



Global Conference & Excellence Awards on Innovation, Intelligence & Impact – 2026 (GC³I² – 2026)

Conference Proceedings

 21 – 22 March 2026

 Olympia Technology Park Auditorium, Chennai, India



Conference Tracks

Artificial Intelligence | Data Science | Industry 4.0

Healthcare | Engineering | Management

Sustainable Development

Hosted by

Inspire Softech Group Companies

(A TN Govt Registered & ISO 9001:2015 Certified Organization)

www.inspiresoftechgroup.com

**Global Conference & Excellence
Awards on Innovation, Intelligence &
Impact – 2026**

Organized by
Inspire Softech Group Companies
Olympia Technology Park, Guindy
Chennai – 600032.

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Compiled by: **Dr. Karthiya Banu, Ph.D., Convenor**

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About the Conference

The **Global Conference & Excellence Awards on Innovation, Intelligence & Impact – 2026 (GC³I² – 2026)** is a distinguished international platform dedicated to fostering collaboration, knowledge exchange, and innovation across multidisciplinary domains. Organized by Inspire Softech Group Companies, the conference brings together academicians, researchers, industry professionals, entrepreneurs, healthcare experts, and social leaders from across the globe.

GC³I² – 2026 aims to create a dynamic and impactful ecosystem where technology, intelligence, and innovation converge to address real-world challenges and drive sustainable development. In today's rapidly evolving digital landscape, the conference emphasizes the critical role of interdisciplinary research and collaborative engagement in shaping the future of industries, education, healthcare, and society.

Conference Focus Areas

- Artificial Intelligence and Intelligent Systems
- Data Science and Analytics
- Industry 4.0 and Smart Technologies
- Healthcare and Medical Technologies
- Engineering and Applied Sciences
- Management and Entrepreneurship
- Education and Digital Learning
- Sustainable Development and Green Technologies

Academic Contributions & Proceedings

GC³I² – 2026 serves as a premier platform for research paper presentations, scholarly discussions, knowledge dissemination, and professional networking. All submissions undergo a rigorous peer-review process to ensure quality and relevance. Selected papers are published in the official conference proceedings with ISBN.

Global Excellence Awards – 2026

The conference also celebrates excellence through the Global Excellence Awards – 2026, recognizing outstanding contributions in innovation, leadership, research, and social impact.

Vision & Impact

GC³I² – 2026 aspires to inspire innovative thinking, foster collaborative research, and promote impactful solutions, contributing to a smarter, more sustainable, and technologically advanced world.

About the Organizing Institution Inspire Softech Group Companies

Inspire Softech Group Companies is a next-generation technology and training solutions provider and a proud part of the Inspire Softech Group, bringing over 18 years of excellence in emerging technologies and evolving industry trends.

Backed by a strong ecosystem of industry experts, the organization specializes in domains such as Software Development, Artificial Intelligence, Machine Learning, Data Science, Cloud Computing, Internet of Things (IoT), and Embedded Systems.

The organization is committed to bridging the gap between academia and industry by delivering practical, industry-aligned training programs, internships, and innovation-driven initiatives for students, faculty members, and professionals.

With a strong focus on skill development, research, and employability enhancement, Inspire Softech Group Companies continues to empower learners and institutions across India through impactful technology-driven solutions.

ISS at a Glance

Category	Details
Type of Organization	Private Limited
Years of Experience	18 Years (Established in 2007)
Geographical Presence	Tamil Nadu, Karnataka, Maharashtra, Kerala, Puducherry
People Empowered	More than 3,00,000+
Placement Support	70% – 90%+
Client Segments	Schools, Colleges, Universities, Corporate, Government & Private Sector

AICTE Corporate ID: CORPORATE61b87f72e161b1639481202

Website: www.inspiresoftechgroup.com | Email: projects@edinztech.com

MESSAGE FROM FOUNDER & CEO

Dr. Karthiya Banu, Ph.D.,
Founder & CEO,
Inspire Softech Group Companies,
Chennai, Tamil Nadu.



It is with great pride and pleasure that I extend my message for the Global Conference & Excellence Awards on Innovation, Intelligence & Impact – 2026 (GC³I² – 2026). This conference aims to promote knowledge sharing, research excellence, and innovation in today's rapidly evolving digital era.

By bringing together academicians, researchers, industry professionals, and innovators from various fields, this conference provides a valuable platform for exchanging ideas, presenting research, and discussing the latest advancements in technology, science, and interdisciplinary studies. Such initiatives play a significant role in encouraging collaboration and inspiring new perspectives that contribute to academic and professional growth.

I sincerely thank all the authors, speakers, participants, and contributors for their valuable support in making this conference possible. A special note of appreciation goes to the Inspire Softech team for their dedication, hard work, and commitment in organizing this prestigious event.

Wishing the conference great success and looking forward to many more milestones and achievements in the future

Global Conference & Excellence Awards on Innovation, Intelligence & Impact

GC³I² – 2026 | 21-22, March 2026



(A TN Govt Registered & ISO 9001:2015 Certified Organization)

Global Conference & Excellence Awards on Innovation, Intelligence & Impact (GC³I² -2026)

21 March 2026 10:00 am to 2:30 pm

Chief Guest



Dr. R. Ramanan, M.Sc., M.Phil., Ph.D., FSMS
'Tamil Nadu Scientist Awardee'
Joint Director (P & D) (Retd.)
Directorate of Collegiate Education
Chennai, Tamil Nadu, India
Will Inaugurate

Distinguished Guest



Shri. K. Karunakaran, M.E.
Executive Engineer
HOTLINES, TANTRANSO
Vellore, Tamil Nadu, India
Presidential Address

Distinguished Guest



Dr. C. Sundar, MBA, M.Phil., Ph.D.
Principal
Dhanraj Baid Jain College
(Autonomous)
Chennai, Tamil Nadu, India
Special Address

Guest of Honor



Dr. P. M. Anbumaran, M.B.B.S., M.D.
Professor – Respiratory Medicine
Saveetha Medical College and Hospital
Chennai, Tamil Nadu, India
Keynote Address

PANEL DISCUSSION



Dr. K. Chokkanathan, Ph.D.
Associate Professor
Department of Artificial Intelligence
MITS Deemed to be University
Madanapalle, Andhra Pradesh, India



Dr. J. Indumathi, PDF, Ph.D., D.Sc.
Dean – Artificial Intelligence
Sree Balaji Institute of Science and
Technology
Chrompet, Chennai, India

Industry Expert Speaker



Shri. Shree Ranjan Natesan Velmurugan
Software Diagnostic Engineer
Expleo (Rolls-Royce – UK)
United Kingdom

21 MARCH 2026

Conference Convenor



Dr. Karthiya Banu, Ph.D.
Founder & CEO
Inspire Softech Group Companies
Chennai, India

Olympia Technology
Park Auditorium,
Guindy, Chennai

Dr. R. Ramanan, M.Sc., M.Phil., Ph.D., FSMS

Tamil Nadu Scientist Awardee

Joint Director (P & D) (Retd)

Directorate of Collegiate Education,

Chennai, Tamil Nadu.



It is a privilege for me to be associated with the Global Conference & Excellence Awards on Innovation, Intelligence & Impact – 2026 (GC³I² – 2026) organized by Inspire Softech Group Companies. This conference provides an excellent platform for academicians, researchers, professionals, and students to come together and exchange ideas, experiences, and research outcomes.

The interdisciplinary nature of this conference encourages meaningful discussions and collaboration across various domains of science, technology, and innovation. Such initiatives play an important role in promoting research culture and inspiring young minds to explore new perspectives and contribute to the advancement of knowledge.

I sincerely appreciate the efforts of the organizing committee and the Inspire Softech team for their commitment in conducting this prestigious event. I also extend my best wishes to all the participants and contributors whose efforts make this conference a valuable and enriching experience.

Wishing the conference great success and a productive exchange of knowledge among all participants.

Shri. K. Karunakaran, M.E.,
Executive Engineer,
HOTLINES, TANTRANSCO,
Vellore, Tamil Nadu.



Global Conference & Excellence Awards on Innovation, Intelligence & Impact – 2026 (GC³I² – 2026). It is a matter of great satisfaction to see the successful organization of this conference that brings together academicians, researchers, and students to share knowledge and innovative ideas. Such initiatives strengthen the connection between academic learning and industry practices while encouraging technological advancement.

The efforts of the organizing committee in creating this valuable platform are highly appreciated. This conference is expected to inspire meaningful discussions and research contributions. Best wishes for the successful conduct of the conference and the publication.

Dr. C. SUNDAR, MBA, M.Phil., Ph.D.
Principal,
Dhanraj Baid Jain College (Autonomous)
Chennai, Tamil Nadu.



It is a matter of great pride and satisfaction to witness the successful organization of the Global Conference & Excellence Awards on Innovation, Intelligence & Impact – 2026 (GC³I² – 2026). This conference serves as a remarkable platform that brings together academicians, researchers, and students to exchange knowledge, share innovative ideas, and explore emerging trends in technology and research.

Such initiatives play a vital role in bridging the gap between academic learning and industry requirements, thereby fostering innovation, critical thinking, and practical application of knowledge. The dedication and efforts of the organizing committee in creating this meaningful and impactful platform are truly commendable.

I am confident that this conference will lead to insightful discussions, valuable research contributions, and inspire participants to strive for excellence in their respective fields. I extend my best wishes for the grand success of the conference and the publication of its proceedings.

Dr. P. M. ANBUMARAN, M.B.B.S., M.D.

Professor – Respiratory Medicine

Saveetha Medical College and Hospital

Chennai, Tamil Nadu.



Global Conference & Excellence Awards on Innovation, Intelligence & Impact – 2026 (GC³I² – 2026) reflects the growing importance of collaborative engagement across medical, technological, and scientific domains. Platforms like this encourage professionals and researchers to share their expertise, examine current challenges, and explore advancements that can contribute to improved knowledge and professional practice.

Such gatherings strengthen the connection between research, innovation, and practical applications that benefit society.

Best wishes to the organizing committee, speakers, and participants for a productive conference that supports continued progress in healthcare, science, and interdisciplinary research.

Dr. K. CHOKKANATHAN, Ph.D.

Associate Professor

MITS Deemed to Be University,
Andhra Pradesh.



The Global Conference & Excellence Awards on Innovation, Intelligence & Impact – 2026 (GC³I² – 2026) highlights the importance of academic engagement and professional interaction in today's evolving research landscape. Platforms such as this enable scholars, educators, and professionals to present their work, exchange perspectives, and explore new directions in technology and innovation.

This initiative reflects a strong commitment to promoting research culture and intellectual collaboration among participants from varied domains.

Best wishes to the organizing team and all contributors for a successful conference that inspires new ideas and meaningful academic progress.

Global Conference & Excellence Awards on Innovation, Intelligence & Impact

GC³I² – 2026 | 21-22, March 2026

DR. J. INDUMATHI, Ph.D.

Dean

Sree Balaji Institute of Science and Technology

Professor of Practice, Anna University

Chennai, Tamil Nadu.



Global Conference & Excellence Awards on Innovation, Intelligence & Impact – 2026 (GC³I² – 2026) provides a wonderful platform for researchers, academicians, and professionals to engage in knowledge sharing and academic interaction. Such conferences encourage interdisciplinary dialogue and highlight emerging developments in science, technology, and innovation.

Best wishes to the organizing team, speakers, and participants for a successful event that promotes collaboration and continued progress in research and professional practice.

**Shri. SHREE RANJAN
NATESAN VELMURUGAN,**
Software Diagnostic Engineer
Expleo – (Rolls-Royce, UK)



Global Conference & Excellence Awards on Innovation, Intelligence & Impact – 2026 (GC³I² – 2026) reflects the importance of continuous technological exploration in a rapidly evolving digital landscape. Conferences of this nature create an environment where professionals and researchers can examine emerging trends, discuss practical experiences, and explore new technological directions.

Engagement between global industry experts and academic communities strengthens the development of innovative solutions and encourages forward-thinking perspectives. It is hoped that this gathering will spark meaningful conversations, inspire new ideas, and open pathways for future collaborations that drive innovation and technological advancement.

Global Conference & Excellence Awards on Innovation, Intelligence & Impact

GC³I² – 2026 | 21-22, March 2026



GC³I² – 2026

GLOBAL CONFERENCE & EXCELLENCE AWARDS
on Innovation • Intelligence • Impact

📍 Olympia Tech Park, Chennai, Tamil Nadu, India
📅 21 March 2026

AGENDA – DAY 1

TIME	PROGRAM SEGMENT	SPEAKER
09.30 AM	📋 Registration	Organizing Team
09.45 AM	🌸 Inaugural Ceremony <i>Inspire Prayer</i>	
09.50 AM	🗣️ Welcome Address	Dr. Karthiya Banu, Ph.D. Founder & CEO, Inspire Softech Group Companies
10.00 AM	🤝 Honoring the Dignitaries <i>Presentation of Mementos</i>	
10.10 AM	🎤 Introducing the Chief Guest	Dr. S. Umarani, MCA, Ph.D. Professor, SRM Institute of Science & Technology, Ramapuram
10.12 AM	🎤 Inaugural Address	Dr. R. Ravanani, Ph.D., FSMS Tamil Nadu Scientist Awardee
10.22 AM	🎤 Presidential Address	Shri. K. Karunakaran, M.E. Executive Engineer – HOTLINES, TANTRANSO
10.32 AM	🎤 Special Address	Dr. C. Sundar, Ph.D. Principal, Dhanraj Baid Jain College (Autonomous)
10.42 AM	☕ Networking Tea Break	All Attendees
11.00 AM	📺 Video Screening <i>The Journey of Inspire Group</i>	
11.05 AM	🏆 Global Excellence Awards Ceremony <i>Award Presentation</i>	
11.30 AM	🗣️ Keynote Address	Dr. P. M. Anbumaran, M.D. Professor – Respiratory Medicine, Saveetha Medical College
12.40 PM	👥 Panel Discussion	Dr. K. Chokkanathan, Ph.D. MITS Deemed to be University, Madanapalle, Andhra Pradesh Dr. J. Indumathi, Ph.D., D.Sc. Dean – Artificial Intelligence, Sree Balaji Institute of of Science & Technology, Chromepet, Chennai
12.55 PM	📄 Paper Presentation Session	Research Scholars & Participants
01.05 PM	🙌 Vote of Thanks	Organizing Committee
01.10 PM	🍽️ Grand Networking Lunch	All Attendees
01.45 PM	🎉 Spreading Cheer 🎉	Surprises to Light Up the Day!

— Organized by —
INSPIRE SOFTECH GROUP COMPANIES
📍 Chennai, India
🌐 www.inspiresoftechgroup.com

CONFERENCE HIGHLIGHTS

The **Global Conference & Excellence Awards on Innovation, Intelligence & Impact – 2026 (GC³I² – 2026)**, organized by Inspire Softech Group Companies, was successfully conducted on 21–22 March 2026 in a hybrid mode at Olympia Technology Park, Chennai, and through online platforms.

The conference witnessed enthusiastic participation from academicians, researchers, industry professionals, and students from across India and various parts of the world. A total of 300+ research submissions and award nominations were received, reflecting the global reach and interdisciplinary nature of the event.

The event featured:

- Insightful keynote sessions by eminent speakers from academia and industry
- Paper presentations across multiple thematic tracks
- Panel discussions addressing current trends and challenges
- Recognition of excellence through the Global Excellence Awards

A rigorous double-blind review process was followed for paper evaluation, ensuring quality, originality, and relevance. Selected papers were presented in both offline and online sessions, facilitating wider participation and knowledge exchange.

The conference served as a dynamic platform for promoting research innovation, fostering collaboration, and bridging the gap between academia and industry.

GC³I² – 2026 successfully upheld its vision of fostering innovation, intelligence, and impactful solutions for global challenges.



INSPIRE

Global Conference & Excellence Awards on Innovation, Intelligence & Impact – 2026 (GC³I² – 2026)

21 – 22 March 2026 | Olympia Technology Park Auditorium, Chennai, India

TECHNICAL SESSIONS

TRACK I

SESSION CHAIRS

**Dr. J. Indumathi**
Dean,
Sree Balaji Institute of Science and
Technology, Chennai, India

**Dr. K. Chokkanathan**
Associate Professor,
MITS Deemed To Be University,
Andhra Pradesh, India

TRACK II

SESSION CHAIR

**Dr. S. Umarani**
Professor, Department of Computer Applications
SRM Institute of Science and Technology,
Chennai, India

KEYNOTE ADDRESS

**Shree Rianjan Natesan Velmurugan**
Software Diagnostic Engineer
Expleo – (Rolls-Royce, UK)

TRACK III

SESSION CHAIR

**Dr. Agnibha Das Majumdar**
Independent Researcher & Physics Educator
Ex-Assistant Professor, MITS University, India

TRACK IV

SESSION CHAIR

**Dr. Baby Niviya Feston**
Dean – Research & Innovation
Garden City University, Bangalore, India

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GC³I² – 2026
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The poster features a dark blue space-themed background with golden light streaks and a network of blue nodes. On the right, there is a graphic of a blue Earth with a rocket launching from the bottom right towards it. The text is in a golden-yellow color.

GC³I²-2026

TECHNICAL SESSION – TRACK I

SESSION CHAIRS



Dr. J. Indumathi
Dean,
Sree Balaji Institute of Science
and Technology, Chennai



Dr. K. Chokkanathan
Associate Professor,
MITS Demed To Be University,
Andhra Pradesh

PAPER PRESENTATIONS

GC¹I²-01	Aileen P. Dr. G. Lakshmi Evolution of Hybrid Project Management: A Comparative Review of Traditional Foundations and Agile Frameworks.
GC¹I²-02	R. Gowri, Dr. C. Gnanlin Shiny A Study on Adoption of E-Vehicle by Women in Tamil Nadu with Special Reference to Vellore City.
GC¹I²-03	Venkatesvarlu Karumuri, Dr. Gowtami Kore A Study on Online Employer Branding and Its Influence on Job Application Decisions
GC¹I²-04	Amulya Gollavilli, Dr. M. Kartikeyan Enhancing Investor Trust in Robo-Advisory Systems Through Explainable AI

Evolution of Hybrid Project Management: A Comparative Review of Traditional Foundations and Agile Frameworks

Aileen P¹, Dr. G. Lakshmi²

SRM Institute of Science and Technology, India

¹aileenpaul2000@gmail.com, ²glakshmi.san@gmail.com

Abstract

Project management is a vital managerial function across industries. A project requires flexibility, communication, and accountability from the ‘need’ for project phase to ‘delivery and feedback’ phase of the project life cycle. The main purpose of project management is to achieve the predetermined objectives of scope, time, cost and quality while considering the available resources, manpower, material and cost of the organization.

The Traditional project management frameworks like Project Evaluation Review Technique (PERT) and waterfall emphasize detailed upfront planning, sequential execution, comprehensive documentation and clearly defined roles and responsibilities. For decades, Traditional project management was the go-to methodology for various industries, providing stable framework for executing projects with predictable outcomes.

Evolving customer needs demanded more flexibility and this led to the birth of Agile methodologies. Agile methods like Extreme programming and Scrum focuses on iterative development with continuous stakeholder feedback. While Traditional methods prioritize structure and documentation, Agile focuses on flexibility, collaboration, and delivering value quickly.

Amidst these circumstances, Hybrid project management practices gained traction in the industries as the organizations grappled to shift to Agile practices. Hybrid approaches combine different project management approaches by integrating the principles and practices of different methodologies. Hybrid approaches are useful in complex projects where strict control is required over scope, schedule, and cost along with continuous stakeholder feedback.

The purpose of this paper is to conduct a comprehensive literature review on the evolution of project management methodologies, tools and frameworks, and to critically compare the strengths and limitations of both Traditional and Agile approaches. This study also further aims to highlight how Hybrid methodologies and adaptive practices are influencing the success criteria in dynamic project environments.

This study uses comprehensive literature review to study the Traditional, Agile and Hybrid methodologies and how it influences the success criteria in dynamic project environments.

The findings indicate that no single methodology guarantees success, as success largely depends on project context and organizational environment. Hybrid approaches appear particularly promising in balancing structure and flexibility.

Future research should focus on the effectiveness of different frameworks in different project and organizational contexts both individually as well as Hybrid methodologies and develop framework incorporating the limitations of Traditional and Agile methods and empirically validate it.

Keywords: Traditional Project management, Agile project management, Hybrid project management, Project management, dynamic environments

Literature Review

Project management has become the core managerial function across industries, enabling organizations to execute projects with efficiency, accountability, and measurable success. Project management is an evolving field with new methodologies evolving constantly. Effective project management ensures projects are completed within the constraints of scope, time, cost, and quality delivering stakeholder satisfaction and organizational success (Atkinson, 1999). Even though Traditional project management approaches have long been the foundation in execution of projects (Haddab, 2024), Traditional approaches faced significant limitations in dynamic project environments (Kerzner, 2003) which led to the development of Agile approaches that emphasized on iterative development and flexibility and stakeholder involvement (Sauer and Reich, 2009). Agile Manifesto was introduced in the year 2001 and there have been considerable discussions comparing the Traditional and Agile subsequently (Beck et al., 2001). Traditional approaches lack flexibility and struggle to respond to changing requirements in dynamic environments (Schmitt & Horner, 2021) but in stable and predictable environments provides ensures effectiveness and accountability (Haddab 2024). Agile has proven effective in complex project environments (Ciancarini et al., 2022; Serrador & Pinto, 2015). However, Agile may face limitations in regulated environments that require strict planning, budgeting, and contractual control (Thesing et al., 2021). Agile increases responsiveness during frequent changes (Reiff & Schlegel, 2022) while iterative nature can also complicate long-term budgeting and forecasting challenging organisations with rigid reporting structures (Thesing et al., 2021).

Hybrid methodologies emerged as a balanced solution, integrating the best features of Traditional and Agile frameworks (Haddab 2024). Tailoring the methodology adapting to the industrial needs is important (Szekely et al, 2025). Morphological study by Jamous et al (2021) showcases different Hybrid models such as Water-Scrum-fall, OPENTCQ, The Agent Object Model (AGOMO), Waterfall-Up-Front and Waterfall-Up-End (WUPWAE), The iterative and Visual Project Management Method (IVPM2) and found that Hybrid approaches are appropriate for complex projects more that Agile or Traditional approaches as timeline and budget of the projects cannot be identified before-hand due to the changing project requirements due to its high predictability and dependency awareness. Despite the conceptual acceptance of Hybrid and Agile frameworks, practical implementation challenges persist. (Reiff & Schlegel, 2022). Hybridization of project is based on hybrid by phases and hybrid by methods which is defined by Goal type and Team type which are the two important aspects that influence the Hybrid methodology customization which leads to project success (Mirzaei et al, 2025). Thus, developing Hybrid approaches that align with the project goals is needed to define success in project execution and successful delivery of quality projects.

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A Study on Adoption of E-Vehicle by Women in Tamil Nadu with Special Reference to Vellore City

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Abstract

In our daily lives, personal mobility is crucial. Two- and three-wheelers, for example, facilitate human development. For the past century, we have been using gasoline and diesel, but this has resulted in environmental issues that affect human life, such as air pollution, climate change, and greenhouse gases. India accepted the Paris Agreement in an effort to develop innovative, state-of-the-art mass-market alternatives to fuel-powered cars. Technology-based solutions were developed with creativity to address environmental issues.

After viewing a variety of brands and price points that suited their daily demands, consumers used online platforms to purchase e-vehicles. In order to promote the usage of e-vehicles, India is currently developing and implementing government assistance and incentives. Environmental concerns and pollution control are given top priority by the government. Rising environmental deterioration and the energy crisis are intertwined these days. Producers and marketers need to consider geographical variations and consumer preferences in order to accelerate the adoption of EVs in India.

Keywords: Mobility, Two Wheelers, Climate Change, Subsidies, Pollution.

Introduction

There has been a rebirth of electric vehicles in recent years. Now Electrical Vehicles Industry is ongoing in several ways. The technology behind EVs is growing with the help of supporting facilities provided by government for convenience of consumers. Consumers are aware about recent advancement and conscious regarding environmental issues who aims to purchase efficiency vehicles and satisfaction with environmental friendliness and public perception towards usage of variables which influence consumer behaviour which creates strong impact on transportation public perception related to cut carbon emissions and encourage sustainability. We find people are interested to purchase Electric vehicles for their personal mobility because they are aware about health consciousness, greener environment, expect carbon free emission, cost effective smaller range, ecofriendly, easy to use reduction in formalities for registration.

India's latest nationally determined contribution to the United Nations Framework Convention on Climate Change is approved by the government. India has now committed to reducing the emission intensity of its GDP by 45% by 2030 as a result of the consent of the Prime minister Panchamrit, which was annexed at COP 26. The prime minister's idea of a lifelong popular movement. Living sustainably is essential to halting climate change.

Our honorable Prime Minister Narendra Modi's vision is to lead a sustainable lifestyle and provide climate justice in order to protect the poor and lessen the negative effects of climate change by adopting a healthy and sustainable lifestyle based on customs and values of conservation and moderation. One important way to combat climate change is through the mass movement for "Life Style For Environment."

India is dedicated to keeping economic expansion and greenhouse gas emissions apart. India's dedication to achieving sustainable goals and reducing carbon emissions. "LIFE" stands for "Life for Environment," which he proposed as a one-word movement for the worldwide community. A human-centered approach to energy transition and a vision of living that is in harmony with the planet should not cause harm to others.

A framework that encourages the government to provide tax breaks, such as production-linked incentive schemes, for the promotion, manufacturing, and adoption of renewable energy by enhancing India's manufacturing capabilities to boost exports, increase green jobs, and produce low-emission products in electric vehicles, super-efficient appliances, and cutting-edge technologies like hydrogen has been updated by the National Development Council, which represents India's transition to cleaner energy by 2030. To enhance mitigation and adaptation methods, the government launched several projects. In order to reduce overall emissions and bolster our economy's energy sector, India aims to safeguard vulnerable economic sectors and societal groups.

In 2018, in honor of the birthday of former Tamil Nadu chief minister J. Jayalalitha, Prime Minister Narendra Modi introduced the Amma two-wheeler initiative. It aims to empower low-income working women by providing women-friendly cars at lower prices. The easy-to-use Pradhan Mantri Amma Scooter Yojana, also called the Amma Bike Scheme, benefits one lakh women in Tamil Nadu. This initiative offers a 50% subsidy, up to a maximum of Rs. 25,000, to working women who want to purchase two-wheelers. To pay for the remaining costs, the beneficiary must utilize loans and other means. Through this program, three-wheeled vehicles that have been modified from two-wheelers are accessible to women with impairments.

The Rs. 20,000 E-scooter Subsidy Scheme (20260) in Tamil Nadu offers financial assistance to gig workers who purchase electric scooters, such as food and delivery workers. When they purchase a new electric scooter, qualified workers who are members of the Tamil Nadu platform-based gig workers welfare board will receive a subsidy of Rs. 20,000. This program supports eco-friendly electric vehicles, low-cost transportation, and better working conditions. They can apply for a subsidy to buy an e-scooter by using the Welfare Board online and providing documentation and registration information.

The government of Tamil Nadu provides incentives and subsidies for women to buy eco-friendly two-wheelers and other e-vehicles. The Amma Two-Wheeler Scheme offers these advantages, and the Tamil Nadu Electric Vehicles Policy 2023 is in effect through 2025–2026. They provide Key benefits for women in 2026.

E-Scooter Subsidy Scheme

This program, which includes electric models, is intended especially for working women to buy gearless or auto-gearless scooters. The amount of the subsidy is either 25,000 or 50% of the vehicle's cost, whichever is lower. The primary goal is to benefit one lakh women each year. Eligibility requirements include being between the ages of 18 and 40, having a driver's license, and having an annual income of no more than Rs.2,50,000. Homemakers, impoverished widows, women with disabilities, and transgender people are given priority. 21% of SC, 1% of ST, and 4% of women with disabilities received reservations.

Tamil Nadu currently encourages electric vehicles under the Tamilnadu Electric Vehicle Policy, which offers subsidies for electric two- and three-wheelers as well as a 100% exemption from road tax and registration fees. PM incentives for EV production and charging facilities. The primary current policy in India to encourage e-vehicles is the E-Drive Scheme, which provides funding for charging stations, subsidies for EV purchases, support for EV producers, and electrification of public transportation. It began on October 1st, 2024, and ended on March 31st, 2026. In order to encourage electric transportation in India, this initiative has allocated 10,900 crore.

Incentives for EV purchases are depending on battery capacity, such as 2500 per Kilowatt per hour in 2025–2026. The primary goal is to boost EV adoption through the development of domestic manufacturing, charging infrastructure, and pollution reduction. E-vehicles.

Review of Literature

Ms. Mifzala Ansar and Ms. Monika (2018): This paper examines about e-vehicles and need for government policies to reduce dependence on oil, greenhouse gas emissions and improve quality of air, by remove life threatening gases and pollutants and focus on sustainable environment. To educate consumers about governments E-Transport programs in India to minimize emissions which creates climate change and smog by improving public health and reduce environmental damage. Sustainability is an important concern and if government subsidies are provided consumers could purchase e-vehicles by minimizing pollution by benefiting pollutions and future generations.

Krithika M and Sunil G E (March 2024) This study examines consumer satisfaction with e-bikes for use in sustainable development on a worldwide scale. Its primary goal is to investigate how government policies, marketing tactics, and technological advancements have influenced consumer perception and purchase intent in India. It discusses how the government can use e-vehicles to improve the climate and discusses upcoming Indian electric vehicles. Consumer Satisfaction and Government support are key drivers for e-vehicles adoption.

Dr. K. Kalimuthu and Sugeerthi A.V. (May 2023) This paper highlights how automobiles are connected with electricity for mobility needs in order to avoid air pollution such as chloro floro carbon which provides harmful for human life. Niti Aayog has provided transformational process from conventional to e-vehicles. Their aim is to provide Zero emission gases by 2030 in order to boost usage of e-vehicles. E-vehicles could save energy and protect environment for future generation. Respondents expect

subsidy and incentives from government 100% road tax exemption to purchase e-vehicles which should be cost effective saving and low-cost maintenance.

Statement of the Problem

Consumers' attitudes are determined by their psychological state. Their ability to buy the car depends on their social and physical behavior, which is determined by their age, income, and relationships with friends, family, and the Internet. They assess the goods and services according to their needs and sustainable for the next generation and for their own use.

Scope of the Study

The goal of the study is to raise knowledge about government subsidies and incentives for buying e-vehicles, including pricing, battery life, charging infrastructure, government incentives, maintenance costs, and brand image. It assesses how e-vehicles contribute to sustainable development by lowering air pollution and reducing reliance on oil and fossil fuels. It includes long-term savings from e-vehicles as well as upfront, operating, and maintenance costs. It evaluates battery technology, charging infrastructure, and the use of technological innovation in e-vehicles. It looks into market trends, growth prospects, and potential obstacles for e-vehicles in the future.

Objectives of the Study

1. To educate and create awareness about government subsidies and tax benefits policies to customers about using E-vehicles
2. To develop sustainable eco friendly systems for electric vehicles
3. The goal is to provide a safe, energy-efficient, and modern method of transportation with an appealing design, silent operation, and reasonable pricing.
4. To provide sustainable development for future generation.

Research Methodology

This study was conducted both primary and secondary data. Primary data was collected with google forms using questionnaire method which has been distributed among college students, office goers, full time workers and part time workers in Vellore city. Due to lack of time only 52 respondents were taken for this study. Secondary data were used from books, journal, Newspapers. Articles, internet reliable from online sources. Collected data had been analysed and presented in the form of graphs and pie chart for better interpretation and understanding.

Hypothesis

- H1: Customer understanding and adoption level of E-Vehicles are influenced by their Income level.
- H2: There is significant relationship between Government policies and subsidies to purchase E-Vehicles.

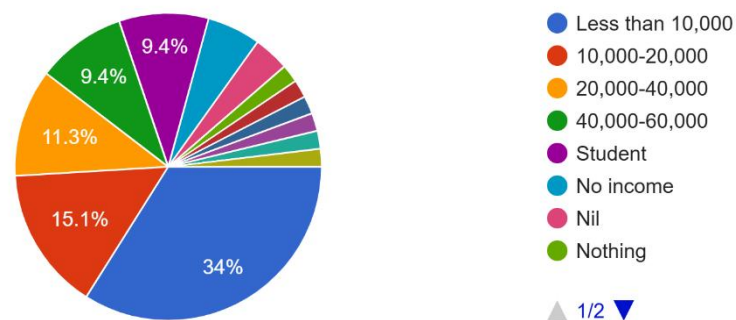
Anaylsis and Interpretation

HYPOTHESIS TESTING: Association between Income level and consumer understanding in adaption of E-Vehicles.

Consumers understanding and adoption level of E-Vehicles and influenced by their Income level.

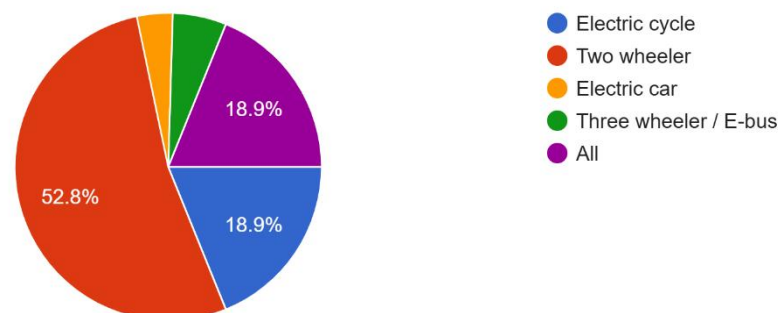
7. MONTHLY INCOME

53 responses



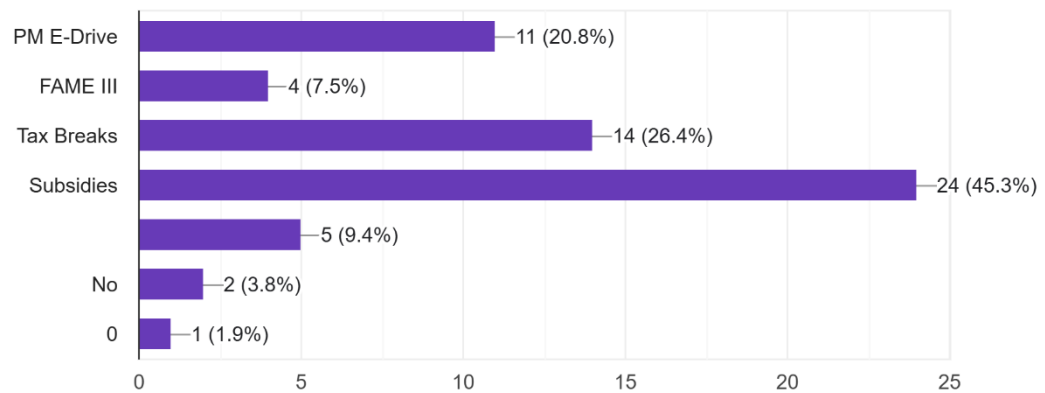
9. What kind of EVs are you familiar with?

53 responses



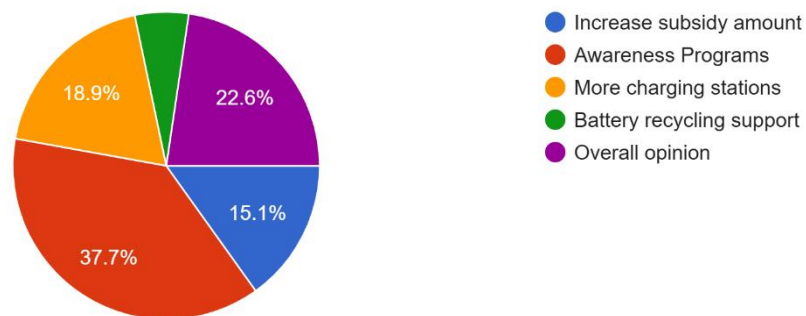
12. Which schemes are you aware of in the Tamil Nadu EV policy 2025?

53 responses



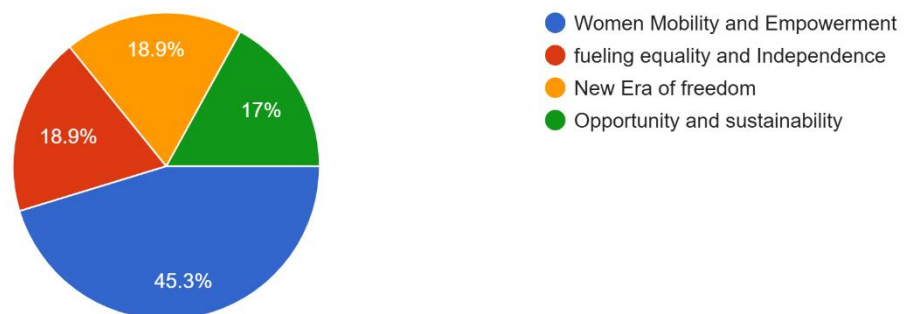
22. What suggestions do you make to increase adoption of EVs?

53 responses



25. Why do you recommend women to purchase EVs

53 responses



Findings

- We find 98.1% of respondents are female.
- 81.1% of respondents are studying UG Degree who are interested to purchase because they travel long distances.
- 96.2% of college students who are interested to purchase EV for future .
- 34% of respondents income is less than Rs. 10,000. 15.1% of respondents comes from Income group between 10,000-20,000. Because they come from rural areas.
- 62.3% respondents know information from social media and internet.
- 18.9% respondents know about Electric cycle 52.8% of respondents are familiar with two wheeler.
- 35.8% get information about E-vehicles from social media. 24.5% get information from family and friends.
- 45.3% of respondents are aware of these government schemes implemented for purchase of E-vehicles.
- 45.3% respondents are aware of subsidy for purchase of E-vehicles. 26.4% expect tax breaks and aware of TamilNadu EV Policy 2025.
- We see currently 69.8% have an E-Vehicle. 34% respondents persuade to buy E-vehicles because of government subsidy. 28.3% purchase because it has low running cost. 71.7% consider to purchase EV in future.
- 45.3% respondents are attracted to purchase because of subsidies provided by Government.

Suggestions

- We suggest that 32.1% agree that EV schemes help to reduce overall cost of ownership.
- 30.2% think that Government policies help them to encourage to purchase EV adoption.
- We see Lack of awareness is major problem faced by consumers.
- 37.7% suggest awareness programs help to increase adoption of E-Vehicles.
- 18.9% of respondents expect more charging stations because they are office going and college students use for emergency purpose and for travelling long distances.
- 17% are highly satisfied with EV Scheme introduced by government.
- 84.9% recommend Evehicles to women because of women mobility and empowerment for their daily use.

Conclusion

We conclude that E-vehicles play important role in our day to day mobility in life. Thus Government schemes help to adopt E-vehicles for women who is a bread winner for their life. E-Vehicles play significant role in day to day life we see large number of respondents are female college students who travel long distance have strong intention to purchase e-vehicles. Digital platforms and government schemes create positive impact on purchasing e-vehicle. By strengthening awareness initiatives and infra structure, recharging facilities, e-vehicle adoption may support future. Subsidies, Tax break , Prime Minister E-Drive Scheme creates positive impact and decisions are taken based low running cost, cost effective and incentives help them to adopt e-vehicle to customers.

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**A Study on Online Employer Branding and Its Influence on Job
Application Decisions: An Empirical Study**

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Abstract

The digital transformation of recruitment has significantly altered the manner in which job seekers evaluate potential employers. Online employer branding, driven by employee-generated content and digital platforms, has emerged as a critical determinant of applicant behavior. This study investigates the influence of online employer branding on job application decisions using empirical data collected from 120 respondents working in different organizations. A structured questionnaire comprising 26 items was administered, and exploratory factor analysis was conducted using Principal Component Analysis with Varimax rotation. A factor loading threshold of 0.60 was adopted to ensure construct robustness. The results reveal three key dimensions—Employer Branding Perception, Digital Review Influence, and Behavioral Intention which collectively explain a substantial proportion of variance in job application decisions. The findings underscore the growing importance of online transparency and digital reputation management in shaping organizational attractiveness.

Keywords: Employer Branding, Online Reviews, Job Application Decisions, Digital Recruitment, Electronic Word-of-Mouth, Factor Analysis

Introduction

The increasing competitiveness of labour markets has compelled organizations to strategically position themselves as attractive employers. Employer branding, which traditionally focused on communicating organizational values and employment benefits, has evolved significantly in response to technological advancements. The proliferation of digital platforms has fundamentally transformed how employer-related information is created, disseminated, and consumed.

In the contemporary recruitment environment, job seekers rely extensively on online sources such as employee review platforms, social media channels, and organizational career portals. These platforms provide access to real-time, experience-based insights that influence perceptions of organizational credibility and attractiveness. Consequently, employer branding is no longer solely controlled by organizations but is co-created through interactions among multiple stakeholders.

Online employer branding thus represents a critical interface between organizations and prospective employees. Understanding its influence on job application decisions is essential for both academic inquiry and managerial practice. This study seeks to examine this relationship and identify the underlying dimensions shaping applicant behaviour.

Review of Literature

Employer branding has been conceptualized as a strategic process through which organizations develop and communicate a distinctive employer identity (Backhaus & Tikoo, 2004). It encompasses both tangible and intangible attributes, including organizational culture, career development opportunities, and work environment.

Research in organizational attraction suggests that job seekers rely on various informational cues to evaluate potential employers. Cable and Turban (2001) argue that individuals form perceptions based on available knowledge, which subsequently influences their application intentions. Similarly, Collins and Stevens (2002) highlight that recruitment practices and organizational communication contribute significantly to employer brand equity.

The advent of digital technologies has expanded the sources of information available to job seekers. Lievens and Slaughter (2016) emphasize that employer image is increasingly shaped by both organizational messaging and external information sources, particularly employee-generated content. This shift has led to the growing importance of electronic word-of-mouth (e-WOM) in recruitment contexts.

Employee reviews on digital platforms serve as a critical component of e-WOM, offering insights into organizational practices from an insider perspective. Dineen and Allen (2016) note that such third-party information is often perceived as more credible than corporate communication. This perceived authenticity enhances its influence on job seekers' decision-making processes.

Another important finding in the literature is the asymmetrical impact of information valence. Negative information tends to have a stronger effect on perceptions than positive information, a phenomenon explained by negativity bias (Baum & Kabst, 2014). In recruitment contexts, negative reviews can significantly deter applicants, thereby affecting organizational attractiveness.

The role of social media in employer branding has also been widely acknowledged. Sivertzen et al. (2013) demonstrate that social media platforms enhance employer attractiveness by facilitating engagement and information sharing. Similarly, Kaur et al. (2015) highlight that effective use of social media strengthens organizational visibility and credibility.

Berthon, Ewing, and Hah (2005) identify multiple dimensions of employer attractiveness, including economic value, development opportunities, and social environment. These dimensions collectively influence job seekers' preferences and decisions.

Despite the extensive literature, there remains a need for empirical studies that integrate traditional employer branding constructs with digital review mechanisms using rigorous statistical techniques. This study addresses this gap by employing factor analysis with stringent loading criteria to identify robust and meaningful constructs influencing job application decisions.

Objectives of the Study

The major objective of the study is to examine the influence of online employer branding on job application decisions and to identify key dimensions underlying employer branding perceptions.

Methodology of the Study

The study was conducted with the help of both primary and secondary data. Primary data was collected from 120 respondents using a structured questionnaire consisting of 26 items measured on a five-point Likert scale. Factor analysis was applied to perform analysis and to know data accuracy for analysis KMO and Bartlett's test was conducted. Variable score threshold of 0.6 was considered as most significant variable and less than 0.6 score. The secondary data was collected from several research articles sourced from scopus, web of science and Google scholar.

Ananlysis and Interpretation

1. Which online platforms do you refer to gather information before applying for a job

Platform	Final Count	Percentage (%)
Ambition Box	30	25.0
Glassdoor	22	18.33
Indeed	15	12.5
Google Summary	37	30.83
Quora	8	6.67
LinkedIn	8	6.67
Total	120	100

The table shows that Google Summary, Ambition Box, and Glassdoor are the primary sources applicants consult before applying for a job. Indeed, Quora, and LinkedIn are less frequently used.

2. KMO and Bartlett's test

Test	Measure	Value
KMO	Sampling Adequacy	0.872
Bartlett's Test	χ^2	1856.42
	df	325
	p-value	0.000

Source: Primary data

To understand factor analysis suitability for the data, KMO – Bartlett's test was conducted. The Kaiser-Meyer-Olkin (KMO) value of 0.872 indicates high sampling adequacy. Bartlett's Test of Sphericity is statistically significant ($\chi^2 = 1856.42$, $p < 0.001$). The test reveals the data is suitable for factor analysis.

3. Total Variance

Factor	Eigenvalue	% of Variance Explained	Cumulative %
Factor 1	5.12	28.44%	28.44%
Factor 2	4.36	24.22%	52.66%
Factor 3	3.18	17.67%	70.33%

The three factors together explain 70.33% of the total variance, indicating a strong factor structure. Factor 1 accounts for the highest variance (28.44), followed by Factor 2(24.22) and Factor 3(17.67) variance respectively demonstrating the relative importance of employer branding perception in influencing job application decisions.

4. Factor analysis

Variable	Factor 1	Factor 2	Factor 3
Employer branding is important when evaluating a company	0.82		
I consider company reputation before applying for a job	0.79		
Work culture influences my decision to apply	0.76		
Career growth opportunities attract me to a company	0.74		
Employee well-being initiatives influence my perception	0.72		
Diversity and inclusion practices affect my decision	0.66		
Online employee reviews influence my job application decision		0.85	

I check employer reviews before applying for a job		0.83	
Company response to online reviews influences my perception		0.79	
Social media presence of a company influences my decision		0.75	
Negative online presence affects my perception of the company		0.73	
I trust reviews shared by employees on platforms like Glassdoor		0.64	
Negative reviews discourage me from applying			0.84
I prefer to apply to companies with strong employer branding			0.81
A well-designed career portal attracts me to apply			0.77
Companies with high employee retention attract me			0.73
High employee turnover discourages me from applying			0.69
I avoid companies with poor employer reputation			0.63

Source: Primary data

Method: Principal component analysis

4.1 Factor 1: Employer branding perception

Variable	Factor Loading
Employer branding is important when evaluating a company	0.82
I consider company reputation before applying for a job	0.79
Work culture influences my decision to apply	0.76
Career growth opportunities attract me to a company	0.74
Employee well-being initiatives influence my perception	0.72
Diversity and inclusion practices affect my decision	0.66

Source: Primary data

The analysis indicates that employer branding (0.82) is the most influential factor in shaping candidates' perceptions, followed by company reputation (0.79) and work culture (0.76), highlighting the importance of organizational image and environment in application decisions. Career growth opportunities (0.74) and employee well-being initiatives (0.72) further enhance attractiveness, while diversity and inclusion practices (0.66), though slightly lower, still play a meaningful role. These results suggest that both an organization's external reputation and its internal employee-focused initiatives are key drivers of employer evaluation, providing actionable insights for management strategies

4.2 Factor 2: Online reviews influence

Variable	Factor Loading
Online employee reviews influence my job application decision	0.85
I check employer reviews before applying for a job	0.83
Company response to online reviews influences my perception	0.79
Social media presence of a company influences my decision	0.75
Negative online presence affects my perception of the company	0.73
I trust reviews shared by employees on platforms like Glassdoor	0.64

Source: Primary data

The analysis shows that online employee reviews (0.85) strongly influence candidates' job application decisions, with checking employer reviews before applying (0.83) also playing a significant role. The way a company responds to reviews (0.79) and its social media presence (0.75) further shape perceptions, while a negative online presence (0.73) adversely affects evaluation. Trust in employee-shared reviews on platforms like Glassdoor, though slightly lower (0.64), remains important in influencing candidate decisions. Overall, these results suggest that a company's online reputation and engagement with employee feedback are critical factors in attracting potential talent.

4.3 Factor 3: Behavioral intention

Variable	Factor Loading
Negative reviews discourage me from applying	0.84
I prefer to apply to companies with strong employer branding	0.81
A well-designed career portal attracts me to apply	0.77
Companies with high employee retention attract me	0.73
High employee turnover discourages me from applying	0.69
I avoid companies with poor employer reputation	0.63

Source: Primary data

The analysis indicates that negative reviews (0.84) strongly discourage candidates from applying, while a preference for companies with strong employer branding (0.81) remains a key driver of application decisions. Features such as a well-designed career portal (0.77) and high employee retention (0.73) further enhance organizational attractiveness, whereas high employee turnover (0.69) and poor employer reputation (0.63) reduce candidates' interest. These results underscore that both perceived organizational stability and strategic employer branding initiatives significantly influence potential applicants' behavior.

4.4 Eliminated variables

Variable
I research employer brand before applying
Salary is more important than employer branding
I compare multiple employer review platforms before applying
Word-of-mouth influences my job decision
I consider unresolved employee complaints before applying
Employee success stories influence my perception
Leadership reputation impacts employer attractiveness
CSR activities influence my perception of the company

Source: Primary data

The analysis indicates that variables such as salary preference, word-of-mouth influence, leadership reputation, CSR activities, and employee success stories did not have a significant impact on candidates' perceptions in this study. Similarly, researching employer brands, comparing review platforms, and considering unresolved complaints showed limited influence. These findings suggest that, within this context, such variables contribute minimally to the overall evaluation of employer attractiveness and were therefore excluded from the final factor structure

5. Managerial Implications

1. Companies should strengthen employer branding and maintain a positive reputation, as these are key determinants influencing candidates' decisions to apply. This aligns with the objective of understanding how employer brand perception affects application behavior.
2. Emphasizing a supportive work culture, career growth opportunities, and employee well-being initiatives can make organizations more attractive, reflecting the analysis that internal policies significantly impact potential applicants' choices.
3. Proactively managing online reviews and responding transparently is crucial, as online feedback strongly shapes employer perception, supporting the study's focus on online reviews influencing application decisions.
4. Maintaining a robust social media presence and addressing negative feedback can enhance credibility and reinforce positive perceptions, ensuring alignment with the objective of assessing digital platforms' role in employer evaluation.
5. Optimizing digital recruitment platforms and highlighting retention-focused initiatives can strengthen trust and encourage applications, reflecting the combined impact of employer branding and online reputation observed in the analysis

6. Conclusion

This study on online employer branding and job application decisions demonstrates that factors such as employer reputation, work culture, career growth opportunities, and employee well-being initiatives play a significant role in influencing candidates' decisions to apply. The analysis further reveals that online reviews, company responses, and social media presence strongly shape applicants' perceptions, highlighting the growing importance of digital platforms in employer evaluation. These findings suggest that organizations can attract and retain talent effectively by strategically managing both their internal policies and online employer brand. Overall, the study confirms that a strong employer brand combined with proactive online reputation management is critical in guiding job application behavior.

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**Enhancing Investor Trust in Robo-Advisory Systems in Stock Markets
through Explainable AI**

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Abstract

Over the past few decades, there has been a significant global expansion in the adoption and advancement of technology across industries, including financial markets. Stock markets worldwide, particularly in emerging economies such as India, have increasingly integrated advanced technologies to enhance efficiency and drive business development. In this context, the application of Artificial Intelligence (AI) and Robo-advisory systems has grown rapidly within the Indian stock market ecosystem. However, despite their growing adoption, investors often face challenges in understanding, interpreting, and, when necessary, questioning the decisions generated by these systems due to their inherent “black-box” nature. Explainable Artificial Intelligence (XAI) has emerged as a promising approach to address this limitation by enhancing the transparency and interpretability of algorithmic decision-making processes. This study aims to examine the role of XAI in improving transparency, comprehensibility, and practical usability of stock market prediction models, thereby supporting informed decision-making and business development. Additionally, it investigates investor perceptions toward AI-driven Robo-advisory systems and evaluates how the integration of XAI influences investor trust and investment behaviour.

A structured questionnaire was administered through Google Forms to 67 participants as part of a pilot study. The findings indicate that, although certain demographic groups exhibit reluctance toward adopting AI-based investment tools, a majority of respondents believe that XAI enhances clarity and understanding of algorithmic decisions. This increased transparency, in turn, strengthens investor confidence and trust in Robo-advisory platforms. The results suggest that the availability of explanations behind investment recommendations positively influences investors’ willingness to adopt such platforms. The study concludes that XAI plays a critical role in fostering trust and confidence among investors, thereby promoting wider adoption of AI-driven financial technologies and contributing to sustainable business development in the stock market.

Keywords: Artificial Intelligence, Explainable Artificial Intelligence, Stock markets, Business Development, Transparency

The poster features a dark space-themed background with a city skyline at the bottom and a robotic hand holding a glowing Earth. The text is arranged in a structured, futuristic layout with orange and blue accents.

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PAPER PRESENTATIONS

GC³I²-05	Dr. Gyaneshwari Vyas Role of Green Technologies in Achieving Sustainable Development Goals (SDGs)
GC³I²-06	Suruthi S. Dr. Pankajam A An Analysis of Employee Engagement and Its Influences on Retention Strategies at GEICO Special Machineries
GC³I²-07	Mrs. Rakshita Chauhan AI Moral Story Generator A Teacher-Friendly Tool for Value-Based Learning
GC³I²-08	S. Harika, M. Gowthami Smart Hospital Appointment and Queue Management System
GC³I²-09	J. Jishnu Advanced Galaxy-Morphology Classification Using Multi-Modal Deep Learning and Explainable AI
GC³I²-10	Deepthi M, Dr. Niviya Feston Influence of Socio-Demographic Factors on Depression and Sleep Quality among Working Women: A Cross-Sectional Study.

**Role of Green Technologies in Achieving Sustainable Development Goals (SDGs):
A Study with Special Reference to D.El.Ed and B.Ed Students**

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Abstract

Sustainable development has become an important global priority due to the increasing challenges related to environmental degradation, climate change, and depletion of natural resources. Green technologies are considered effective tools for promoting environmentally responsible practices and supporting the achievement of Sustainable Development Goals (SDGs). The present study examines the awareness and perceptions of green technologies among D.El.Ed and B.Ed students and explores their potential role in promoting sustainable development through education. A descriptive survey method was used for the investigation. Data were collected from teacher trainees through a structured questionnaire. The findings suggest that most students are aware of the importance of green technologies and believe that integrating sustainable practices into educational systems can significantly contribute to environmental protection and responsible behavior. The study highlights the need to incorporate sustainability education and green technological practices into teacher education programs to prepare future educators who can promote sustainable lifestyles among learners.

Keywords: Green Technology, Sustainable Development Goals, Sustainability Education, Teacher Education, Environmental Awareness

1. Introduction

Environmental problems such as climate change, pollution, loss of biodiversity, and excessive use of natural resources have become major concerns across the world. These challenges have drawn attention to the need for sustainable development that balances environmental protection with social and economic progress.

In response to these global challenges, the United Nations introduced the Sustainable Development Goals (SDGs) as part of the 2030 Agenda for Sustainable Development. These goals aim to address global issues including poverty reduction, environmental protection, quality education, and climate action.

Green technologies play a significant role in supporting sustainable development. These technologies focus on reducing environmental damage, conserving resources, and improving energy efficiency. Examples include renewable energy systems, eco-friendly infrastructure, waste management technologies, and digital solutions that reduce the consumption of physical resources.

Education has a critical role in promoting sustainability and environmental awareness. Educational institutions can help develop responsible citizens who understand environmental challenges and adopt sustainable practices in their daily lives. Teacher education programs such as D.El.Ed and B.Ed are particularly important because future teachers influence the attitudes and behaviors of school students.

Integrating green technologies into education can improve learning methods while also promoting environmental awareness. Therefore, understanding the level of awareness among teacher trainees regarding green technologies and sustainable development is essential for strengthening sustainability education.

2. Review of Literature

Recent research highlights the growing significance of green technologies in promoting sustainable development. Technological innovation has been recognized as an important factor in reducing environmental impact and supporting sustainable economic growth. Green innovations encourage the efficient use of natural resources and contribute to environmental protection.

Education for Sustainable Development (ESD) has gained considerable attention in recent years. ESD focuses on equipping learners with the knowledge, skills, and attitudes necessary to address environmental challenges and promote sustainable lifestyles. Educational institutions are therefore expected to integrate sustainability concepts within teaching and learning practices.

Studies have indicated that incorporating sustainability topics into educational curricula helps students develop environmental awareness and responsible behavior. When learners are exposed to sustainability concepts, they become more conscious of issues such as climate change, energy conservation, and environmental protection.

Green pedagogy has also emerged as an important approach in modern education. This concept emphasizes the use of environmentally responsible teaching methods and the adoption of eco-friendly technologies in educational environments. Digital learning platforms, smart classrooms, and online resources reduce paper usage and contribute to environmentally sustainable practices.

Furthermore, technological innovation has created opportunities for promoting sustainability through education. Digital technologies enable institutions to provide efficient learning environments while minimizing the consumption of physical resources.

Teacher education institutions play a crucial role in this process because teacher trainees will later influence the knowledge and attitudes of future generations. If teachers are aware of green technologies and sustainability practices, they can effectively integrate these concepts into classroom teaching.

However, several studies indicate that the level of awareness of green technologies among teacher trainees is still developing in many educational contexts. This suggests the need for more research focusing on sustainability awareness among teacher education students.

The present study therefore attempts to examine the awareness and perceptions of D.El.Ed and B.Ed students regarding green technologies and their role in achieving Sustainable Development Goals.

3. Objectives of the Study

The main objectives of the study are:

To examine the level of awareness of green technologies among D.El.Ed and B.Ed students.

To analyze the role of green technologies in achieving Sustainable Development Goals through education.

To compare the perceptions of D.El.Ed and B.Ed students regarding sustainability and green technologies.

To suggest measures for integrating green technologies into teacher education programs.

4. Research Questions

What is the level of awareness of green technologies among teacher trainees?

How do green technologies contribute to the achievement of Sustainable Development Goals?

Is there any difference between D.El.Ed and B.Ed students in their perception of green technologies?

How can teacher education institutions promote sustainability through green technologies?

5. Hypotheses

H₀₁: There is no significant difference between D.El.Ed and B.Ed students regarding awareness of green technologies.

H₀₂: There is no significant relationship between awareness of green technologies and understanding of Sustainable Development Goals.

6. Research Methodology

Research Design

The study employed a descriptive survey method to examine the awareness and perceptions of green technologies among teacher trainees.

Sampling Procedure

A simple random sampling technique was used to select participants from teacher education institutions. The sample consisted of 120 students, including both D.El.Ed and B.Ed trainees.

Program	Number of Students	Percentage
D.El.Ed	60	50%
B.Ed	60	50%

Research Tool

Data were collected using a self-developed questionnaire prepared by the researcher.

The questionnaire included items related to:

Awareness of green technologies

Knowledge of sustainable development goals

Perceptions regarding environmental sustainability

Attitudes toward the use of green technologies in education

Most items were measured using a five-point Likert scale.

Reliability of the Instrument

The reliability of the questionnaire was tested using Cronbach's Alpha method. The reliability coefficient obtained was 0.79, indicating satisfactory internal consistency.

Awareness levels

Awareness Level	Students	Percentage
High	48	40%
Moderate	52	43%
Low	20	17%

Data Analysis

The collected data were analyzed using percentage analysis and descriptive statistics to interpret the responses of the participants.

7. Results and Findings

The analysis of the data indicates that a majority of D.El.Ed and B.Ed students are aware of the concept of green technologies and recognize their importance in environmental protection.

Many respondents reported that technologies such as renewable energy systems, digital learning platforms, and eco-friendly practices can help reduce environmental impact and support sustainable development.

The results also show that teacher trainees have a positive attitude toward incorporating sustainability concepts into classroom teaching. Students believe that environmental education and the use of green technologies can help create awareness among school children.

However, some participants demonstrated limited knowledge regarding specific types of green technologies. This suggests that teacher education programs should provide more exposure to sustainability practices and environmental innovations.

8. Discussion

The findings of the study highlight the importance of integrating sustainability education and green technologies into teacher training programs. Teacher trainees play an important role in shaping the attitudes and values of future generations. Therefore, their awareness of environmental issues and sustainable practices is essential.

Green technologies contribute to achieving several Sustainable Development Goals, particularly SDG 4 (Quality Education) and SDG 13 (Climate Action). By using digital learning tools and environmentally responsible practices, educational institutions can reduce resource consumption and promote sustainability awareness.

Teacher education institutions should incorporate sustainability-related topics, digital learning tools, and eco-friendly practices into their curriculum. Such initiatives will help future teachers promote environmental awareness among students and encourage responsible environmental behavior.

9. Conclusion

The study concludes that green technologies have significant potential to support sustainable development through education. The results indicate that D.El.Ed and B.Ed students generally possess a positive awareness of green technologies and recognize their importance in promoting sustainability.

However, teacher education programs should strengthen sustainability education by providing greater exposure to green technologies and environmental practices. Integrating sustainability into teacher training will help develop environmentally responsible educators who can contribute to achieving Sustainable Development Goals.

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**An Analysis of Employee Engagement and Its Influence on Retention
Strategies at GECO Special Machiners**

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Abstract

Employee engagement is widely recognized as an important factor influencing employee retention and organizational effectiveness. This study investigates the relationship between employee engagement and retention strategies at GECO Special Machiners. The primary objective is to identify the major determinants of employee engagement and examine how these factors affect employees' willingness to remain with the organization. Data for the study were gathered through a structured questionnaire administered to 120 employees.

The collected responses were analysed using the Statistical Package for the Social Sciences (SPSS). Statistical techniques including Independent t-test, Correlation Analysis, Regression Analysis, and One-Way ANOVA were applied to examine the association between engagement variables and retention outcomes. The results indicate that leadership support, recognition, training opportunities, and a supportive work environment positively influence employee commitment and their intention to stay in the organization.

The study concludes that well-designed engagement practices can improve job satisfaction, strengthen employee commitment, and reduce turnover, thereby contributing to long-term organizational sustainability.

Keywords: Employee Involvement, Staff Retention, Organizational Commitment, Workplace Climate.

INTRODUCTION

In the current business scenario, organizations are experiencing constant change due to globalization, technological progress, automation, and changing employee expectations. In such a situation, human resources have become a critical factor in ensuring organizational success and long-term growth. Unlike physical resources, employee knowledge, skills, and commitment act as a unique advantage that competitors cannot easily imitate. Therefore, organizations are placing increasing importance on employee engagement and retention as key areas of human resource management. Employee engagement refers to the level of involvement, dedication, and interest employees show towards their work and organizational goals, which directly affects their performance. However, global reports suggest that only around 21–23% of employees are actively engaged, resulting in considerable productivity losses worldwide. Employee retention refers to an organization's ability to keep its skilled workforce by providing a supportive and motivating work environment. In the Indian manufacturing sector, employee turnover is estimated to be between 15% and 25% per year, and the cost of replacing employees can range from 30% to 50% of their annual salary. Although these concepts are widely studied, most research has focused on large organizations, with limited attention given to small and medium manufacturing firms like GECO Special Machiners. There is also insufficient research examining how specific employee engagement factors influence retention outcomes in such settings,

which represents the research gap of this study. This study aims to address this gap by examining the impact of employee engagement on retention within a manufacturing organization. It also offers a theoretical contribution by improving the understanding of the relationship between engagement factors and employee retention, thereby adding to existing knowledge and providing useful insights for enhancing human resource practices and organizational performance.

Statement of the Problem

In recent years, organizations have realized that employee engagement and retention are important for improving productivity and performance. Many studies have looked at these factors separately, but very few have examined how they work together. Since both are influenced by motivation, workplace relationships, and organizational support, it is still a challenge for organizations to manage them effectively. Manufacturing companies like GECO Special Machiners face challenges such as heavy workload, repetitive work, and limited growth opportunities, which can reduce employee motivation. This may lead to low job satisfaction and higher employee turnover. Therefore, this study focuses on understanding engagement and retention practices at GECO Special Machiners, identifying gaps, and suggesting ways to improve employee satisfaction and retention.

Objectives of the Study

- To analyse the factors of employee engagement that influence employee retention.
- To study the impact of employee engagement on employee retention at GECO Special Machiners.

Review of Literature

Gurcuoglu, S., Turksen, A., & Dugan, O. (2024) examined the relationship between employee empowerment and work engagement among employees working at Uşak Courthouse. The study was conducted among 115 employees to analyze how empowerment influences engagement levels. The findings revealed that dimensions of empowerment such as autonomy, employee involvement, and motivation positively influence work engagement. Employees who feel empowered show higher levels of enthusiasm, commitment, and participation in organizational activities. The study concludes that organizations should implement empowerment strategies and provide continuous training and support to enhance employee engagement and overall organizational performance.

Ngwenya, S. J., Mpofu, N., Tshuma, K., Chiriseri, E., Madvivanyika, G., & Mfandaidza, N. (2024) analyzed employee attitudes toward performance appraisal systems and their impact on work performance in the Zimbabwean manufacturing sector using Eco-med Manufacturing (Pvt) Limited as a case study. The study found that employees often develop negative perceptions toward appraisal systems, including feelings of distrust, competition, and resentment, which may affect teamwork and collaboration. However, managerial staff reported that performance appraisals helped improve production levels and achievement of targets. The study recommends that HR departments provide training and awareness programs and introduce self-evaluation appraisal methods to improve employee perceptions and performance.

Singh, N., & Prakash, Y. (2024) analyzed the impact of reward management on employee engagement. The study found that financial rewards encourage short-term engagement, while non-financial rewards such as recognition and career development support long-term commitment. The authors emphasized the need for well-structured reward systems to improve motivation, productivity, and employee retention.

Akanmu, M. D., Panatik, S. A. B., & Shahrin, R. (2025) investigated the influence of performance appraisal practices on employee engagement in higher education institutions in Malaysia. The study reported that fair and transparent appraisal systems positively influence employee motivation, job satisfaction, and engagement. The authors concluded that effective appraisal systems encourage employees to perform better and increase their willingness to remain in the organization.

Research Methodology

The sample size for the research is based on the number of respondents, i.e., 120 respondents, as it adequately represents the employee population in GECO Special Machiners and is also sufficient to produce reliable and meaningful results within the given time and resources. Convenience sampling is used in this research as the employees were easily accessible and willing to participate in the research. The questionnaire is designed based on the objectives of the research and is relevant to the key aspects of employee engagement and retention. It is designed in a simple and structured manner with clear questions, mainly using the Likert scale for assessing the responses of the employees. Is this all the contents were included or not

Data Analysis and Interpretation

INDEPENDENT SAMPLE T TEST

H₀: There is no significant difference between the type of employment and employee engagement factors influencing retention.

H_{1a}: There is a significant difference between the type of employment and employee engagement factors influencing retention.

TABLE 1: Independent Samples T-Test on Employment Type and Employee Engagement Factors Influencing Retention

Employee Engagement Factors Influencing Retention	N	Mean	Std. Deviation	T value	Significant Level	Result
Permanent	109	18.20	2.937	0.515	0.910	Not Significant
Contract / Temporary	11	17.73	2.611			

INTERPRETATION:

The table shows that permanent employees have a slightly higher mean score compared to contract employees regarding engagement factors influencing retention. However, the difference is minimal. Since the significance value is greater than 0.05, the result is not statistically significant. Therefore, the type of employment does not have a meaningful impact on employee engagement related to retention.

CORRELATION

H₀₂: There is no significant relationship between employee engagement factors and employee retention at GECO Special Machiners.

H₁₂: There is a significant relationship between employee engagement factors and employee retention at GECO Special Machiners

TABLE 2: Correlation Analysis of Employee Engagement Factors Influencing Retention

Correlations						
		Supportive Supervision	Recognition for Good Performance	Training & Development Opportunities	Positive Work Environment	Good Relationship with Colleagues
Supportive Supervision	Pearson Correlation	1	-.254**	.297**	-.049	.073
	Sig. (2-tailed)		.005	.001	.596	.426
	N	120	120	120	120	120
Recognition for Good Performance	Pearson Correlation	-.254**	1	.004	.178	.095
	Sig. (2-tailed)	.005		.962	.052	.300
	N	120	120	120	120	120
Training & Development Opportunities	Pearson Correlation	.297**	.004	1	.092	.322**
	Sig. (2-tailed)	.001	.962		.317	.000
	N	120	120	120	120	120
Positive-Work Environment	Pearson Correlation	-.049	.178	.092	1	.047
	Sig. (2-tailed)	.596	.052	.317		.613
	N	120	120	120	120	120
Good Relationship with Colleagues	Pearson Correlation	.073	.095	.322**	.047	1
	Sig. (2-tailed)	.426	.300	.000	.613	
	N	120	120	120	120	120
**. Correlation is significant at the 0.01 level (2-tailed).						

INTERPRETATION

The correlation results show that supportive supervision has a significant positive relationship with training and development ($r = 0.297$, $p = 0.001$) and a negative relationship with performance recognition ($r = -0.254$, $p = 0.005$). Training and development also has a strong positive association with relationships among colleagues ($r = 0.322$, $p = 0.000$). However, the positive work environment does not show any significant relationship ($p > 0.05$). Overall, the null hypothesis is partially rejected, indicating that some employee engagement factors influence retention.

REGRESSION

H_{0a}: There is no significant impact of employee engagement factors, retention-related factors, and retention measures on employee engagement and retention.

H_{1a}: There is a significant impact of employee engagement factors, retention-related factors, and retention measures on employee engagement and retention.

TABLE 3: Regression Analysis of Employee Engagement Factors and Retention Measures

Model	Unstandardized Coefficients		Standardized Coefficients	T	Sig.	R Square	Adj. R Square
	B	Std. Error	Beta				
(Constant)	6.377	2.627		2.428	.17	.175	.153
EmployeeEngagement Factors	.090	.101	.076	.891	.375	.090	.101
Factors Influencing Retention	.182	.085	.185	2.124	.036	.182	.085
Measures on EmployeeEngagement & Retention	.300	.085	.311	3.549	.001	.300	.085

INTERPRETATION

The regression results show that employee engagement factors have a weak and non-significant impact ($B = 0.090$, $p = 0.375$). In contrast, factors influencing retention ($B = 0.182$, $p = 0.036$) and measures on engagement and retention ($B = 0.300$, $p = 0.001$) have a significant positive effect. Among these, engagement and retention measures have the strongest influence. Therefore, the null hypothesis is partially rejected, indicating that selected factors significantly impact employee engagement and retention.

ONE – WAY ANOVA

TABLE 4: One-Way Anova Analysis of Employee Engagement Factors and Factors Influencing Retention

		Sum of Squares	df	Mean Square	F	Sig.
Employee Engagement Factors Influencing Retention	Between Groups	129.041	3	43.014	5.487	.001
	Within Groups	909.326	116	7.839		
	Total	1038.367	119			

INTERPRETATION

The One-Way ANOVA results show that employee engagement factors influencing retention have a statistically significant difference across groups ($F = 5.487$, $p = 0.001$), as the significance value is less than 0.05. This indicates that employee engagement factors vary among different groups. Therefore, the null hypothesis is rejected, and it can be concluded that there is a significant difference in employee engagement factors influencing retention across groups.

Findings

- The study reveals that employee engagement factors such as supportive supervision, training and development opportunities, and workplace relationships influence employee retention in the organization.
- The correlation analysis indicates that training and development initiatives show a strong and meaningful association with workplace relationships, which contributes to higher employee retention.
- Supportive supervision also shows a significant relationship with certain engagement factors, indicating that effective managerial support can improve employee commitment to the organization.
- The regression results reveal that measures related to employee engagement and retention have the strongest positive influence on employee retention compared to other variables.
- The ANOVA analysis indicates that factors influencing employee retention differ significantly across employee groups, while employee engagement factors do not show a significant difference among groups.

Suggestions

- The organization should strengthen training and development programs to enhance employee skills, job satisfaction, and long-term commitment.
- Management should encourage supportive supervision practices, as effective leadership improves employee motivation and engagement.
- The company should implement effective engagement initiatives, including reward-based initiatives and opportunities for professional growth.

Conclusion

This study analyzes the relationship between employee engagement and employee retention at GECCO Special Machinists. The results indicate that employee engagement plays a meaningful role in influencing employees' willingness to remain with the organization. Engagement elements such as supportive supervision, training and development opportunities, and structured engagement measures contribute positively to improving retention outcomes. The statistical analysis shows that measures related to employee engagement and retention demonstrate the strongest influence on retention, while other engagement factors show comparatively lower impact. In addition, the findings reveal that retention-related factors differ across employee groups, highlighting the importance of implementing suitable engagement strategies for different categories of employees.

Overall, the study highlights that organizations that prioritize employee engagement through effective leadership, development opportunities, and supportive workplace practices are more likely to achieve higher employee commitment and long-term retention. Strengthening these initiatives can help GECCO Special Machinists maintain a stable, motivated, and productive workforce.

AI Moral Story Generator: A Teacher-Friendly Tool for Value-Based Learning

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Abstract

Immersive technologies and Artificial Intelligence (AI) tools are transforming education by making learning more interactive, engaging, and conceptual. These tools enhance student motivation and understanding, while reshaping how educators and learners interact with content—from personalized material creation to intelligent tutoring systems. Recognizing this shift, the Government of India has prioritized AI in the National Education Policy (NEP) 2020. Programs like “AI for All” aim to build AI literacy and create a tech-enabled learning ecosystem across the country. However, these advancements bring concerns such as increased screen time, digital over-dependence, reduced critical thinking, psychological effects, and socio-economic disparities especially at the primary level where development is most formative. To balance technology use with value-based education, we propose the AI Moral Story Generator— a teacher-friendly tool designed to support moral learning through AI-powered storytelling. Teachers can enter any moral topic and select the grade level to generate age-appropriate stories, each with a discussion question and key takeaway, delivered in both audio and PDF formats. This research emphasizes the importance of integrating moral education early in life and demonstrates how AI tools can effectively support this goal. The paper includes sections on related work, tool design, conceptual framework, conclusions, and limitations.

Keywords: Green Technology, Sustainable Development Goals, Sustainability Education, Teacher Education, Environmental Awareness.

Smart Hospital Appointment and Queue Management System

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Abstract

Efficient hospital management is essential for providing quality healthcare services. Traditional hospital appointment systems rely on manual booking processes that often lead to long waiting times, overcrowding, and inefficient patient management. This paper proposes a Smart Hospital Appointment and Queue Management System that enables patients to book appointments online and allows hospitals to manage queues efficiently. The system is developed using modern web technologies such as HTML, CSS, Bootstrap, JavaScript, Python, and Django, with SQLite as the backend database. The proposed system provides an interactive platform for patients to schedule appointments with doctors, view available time slots, and monitor appointment status. Administrators can manage doctor schedules and monitor patient flow through an administrative dashboard. The implementation of this system reduces patient waiting time, improves operational efficiency in hospitals, and enhances overall patient satisfaction.

Keywords Hospital Management System, Queue Management, Online Appointment System, Healthcare Information System, Django Framework.

Introduction

Healthcare systems are rapidly evolving with the integration of digital technologies. Hospitals often face significant challenges in managing patient appointments and maintaining an efficient queue management system. Traditional manual appointment methods require patients to visit hospitals physically to schedule appointments, which results in overcrowded waiting areas and increased waiting times.

The increasing number of patients and limited healthcare resources make it necessary to adopt efficient digital systems that can streamline hospital operations. A smart hospital management system can address these challenges by providing an online platform for appointment booking and automated queue management.

The Smart Hospital Appointment and Queue Management System aim to simplify the appointment booking process and improve patient management within hospitals. The system enables patients to register online, select doctors, and book appointments conveniently. It also helps hospital administrators manage appointments, track patient visits, and maintain organized queues.

The implementation of such systems improves healthcare service delivery, enhances operational efficiency, and ensures better patient experiences.

Literature Review

Healthcare management systems have gained significant attention in recent years due to the need for efficient hospital operations. Various researchers have proposed digital solutions for appointment scheduling and queue management in hospitals.

Traditional hospital systems often rely on manual processes, which result in inefficiencies such as long waiting times and administrative overload. Several studies suggest that web-based appointment systems can significantly improve hospital efficiency by enabling patients to schedule visits remotely.

Queue management systems are also widely used in service industries such as banking and customer service centres. When applied to healthcare systems, these technologies can reduce waiting times and optimize patient flow.

Research also highlights the role of web technologies and database management systems in improving hospital management. Frameworks such as Django provide robust backend support for developing secure and scalable healthcare applications.

Despite existing solutions, many hospitals still lack integrated systems that combine appointment booking with queue management. The proposed system aims to address this gap.

Problem Statement

Many hospitals still depend on manual methods for appointment scheduling and patient queue management. This leads to several issues:

- Long waiting times for patients • Difficulty in booking appointments with doctors
- Overcrowding in hospital waiting areas
- Inefficient hospital resource management • Lack of transparency in appointment scheduling

These problems negatively impact both patient satisfaction and hospital efficiency. Therefore, an automated system is required to streamline appointment booking and queue management.

Objectives

The main objectives of the proposed system are:

1. To develop an online appointment booking platform for hospitals.
2. To provide an efficient queue management system.
3. To reduce patient waiting time and improve hospital efficiency.
4. To allow administrators to manage doctor schedules and patient appointments.
5. To provide a user-friendly interface for patients and hospital staff.

System Architecture

The proposed system follows a multi-layer architecture consisting of several functional layers.

Presentation Layer

The presentation layer provides the user interface through a responsive web application developed using:

HTML

CSS

Bootstrap

JavaScript

Application Layer

The application layer is implemented using the Django web framework in Python. It handles:

- User authentication
- Appointment booking logic
- Request processing
- Queue management operations
- Database Layer

The database layer uses SQLite to store system data including:

- Patient information
- Doctor details
- Appointment schedules
- Queue records

Methodology

The proposed system follows a structured workflow to manage hospital appointments efficiently.

Step 1: Patient Registration Patients create accounts in the system by providing personal details.

Step 2: Login Authentication Users log into the system using secure authentication credentials.

Step 3: Doctor Selection Patients select doctors based on specialization and available time slots.

Step 4: Appointment Booking

Patients choose suitable appointment times and confirm bookings.

Step 5: Queue Management

The system automatically assigns queue numbers and manages patient order.

Step 6: Appointment Monitoring Administrators monitor appointments and manage queues through the dashboard.

Database Design

The system database contains the following main tables:

Patients Table – Stores patient ID, name, contact information and login details.

Doctors Table – Stores doctor ID, specialization and schedule.

Appointments Table – Stores appointment date, time and doctor details.

Queue Table – Stores queue number and appointment status.

Tools and Technologies Used

HTML – Web page structure

CSS – Styling of web pages

Bootstrap – Responsive UI design

JavaScript – Client-side interaction

Python – Backend programming

Django – Web application framework

SQLite – Database management system

These technologies ensure the system is scalable, efficient, and reliable.

Proposed System

The proposed system introduces a digital solution for hospital appointment and queue management.

Key features include:

- Online patient registration and login
- Doctor schedule management
- Appointment booking system
- Automated queue management
- Administrative monitoring dashboard
- Secure data management

These features significantly improve hospital workflow and patient experience.

Results and Discussion

The implementation of the Smart Hospital Appointment and Queue Management System demonstrates several improvements in hospital operations.

Patients can easily book appointments without visiting the hospital physically, reducing waiting time and preventing overcrowding in waiting areas. The automated queue management system ensures that patients are served in an organized manner.

Hospital administrators can efficiently manage doctor schedules and monitor patient appointments. The system improves transparency and communication between patients and healthcare providers.

Conclusion

The Smart Hospital Appointment and Queue Management System provide an efficient solution for managing hospital appointments and patient queues. By automating the appointment booking process and organizing patient flow, the system improves hospital efficiency and enhances patient satisfaction.

The integration of modern web technologies such as Django and SQLite ensures that the system is scalable, reliable, and easy to maintain.

Future Work

Future enhancements of the system may include:

- Mobile application integration
- SMS and email notifications
- AI-based appointment scheduling • Integration with electronic health record systems
- Multi-hospital support systems

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Advanced Galaxy Morphology Classification Using Multi Model Deep Learning and Explainable AI

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Abstract

The study, "Advanced Galaxy Morphology Classification Using Multi-Modal Deep Learning and Explainable AI," presents an intelligent framework that can classify galaxies by visual and astrophysical data with high accuracy. In particular, this framework utilizes deep visual features extracted from preprocessing the Galaxy10 Decals images with the pretrained VGG16 network with Sloan Digital Sky Survey (SDSS) metadata which consisted of Right Ascension, Declination, and morphological vote fractions. This integration of multimodal data enables the model to decode and learn both spatial and contextual features of classifying galaxies, enhancing classification consistency as compared to single-modal classification. A Multi-Layer Perceptron (MLP) fusion classifier is used to account for relationships between features, and is trained with Binary Cross-Entropy loss, employing sigmoid activation for multi-label classification of different galaxy types, including spiral, elliptical, and barred spiral. The framework effectively demonstrates a generalization of ROC-AUC of 0.94 and 0.64 macro F1-score, thus providing support for a dual fusion approach. The SHAP (Shapley Additive Explanations) method can showcase the key features which leads to an improved understanding of the prediction when viewed in the context of explaining the outcome of the model. Overall, the proposed framework improves accuracy for automated galaxy classification while providing useful, transparent, and understandable features derived from the visual and astrophysical elements that affect galaxy morphology.

Keywords: Galaxy Morphology Classification, Multi Modal Deep Learning, CNN, VGG16, Metadata Fusion, Explainable AI (SHAP)

Influence of Socio-Demographic Factors on Depression and Sleep Quality among Working Women: A Cross-Sectional Study

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Abstract

The existing study tested the association among socio-demographic elements, depression, and sleep high-quality amongst running women in better training institutions. A go-sectional survey design turned into employed, and records have been gathered from three hundred girl college members using the Beck despair inventory (BDI) and the Pittsburgh Sleep excellent Index (PSQI). Descriptive statistics, chi-square checks, and multiple regression analyses had been carried out to observe relationships between socio-demographic variables and mental health outcomes. Chi-square analyses found out that educational qualification and sort of appointment have been appreciably related to stages of despair, even as sleep exceptional changed into significantly related to age, instructional qualification, designation, teaching experience, type of appointment, marital fame, and socio-financial popularity. more than one regression analysis indicated that instructional qualification and socio-monetary popularity significantly expected melancholy, accounting for a modest proportion of variance. In evaluation, the regression version predicting sleep quality turned into statistically great, explaining over half of of the variance, with instructional qualification, kind of appointment, marital popularity, and socio-financial reputе emerging as significant predictors. The findings spotlight the differential impact of socio-demographic and occupational factors on despair and sleep exceptional amongst running ladies. Interventions specializing in employment stability, workload control, and psychosocial aid may be important in selling intellectual fitness and sleep properly-being in educational work settings.

Keywords: Depression, Sleep Quality, Working Women, Socio-Demographic Factors, Beck Depression Inventory, Pittsburgh Sleep Quality Index

Introduction

More and more women are stepping into the workforce, and that's changed the way our society and economy work. But let's be honest, it's also brought a new set of challenges for women's mental health and well-being. These days, working women aren't just focusing on their jobs—they're also managing family, meeting expectations at home, and dealing with what society expects from them. Trying to balance all these roles adds up, and research shows it puts women at higher risk for stress and trouble sleeping (Arber et al., 2021; Kim & von dem Knesebeck, 2023). Depression is still one of the main reasons people struggle with disability around the world. Women report it more often than men, especially in midlife, when work and family pressures really hit their peak (World Health Organization [WHO], 2023).

These days, poor sleep isn't just annoying—it's a real public health issue, especially when you look at how it affects mental health. There's solid research showing that when people don't sleep well or don't get enough rest, they're more likely to feel depressed, struggle with their emotions, lose focus, and have a harder time getting things done at work (Baglioni et al., 2020; Grandner, 2021). For working women, it's even tougher. Stress from the job, long hours, and the constant juggling act between work and personal life make sleep problems worse (Huang et al., 2021).

Things like age, education, job situation, relationship status, and income really shape both mental health and sleep. For women, work that feels unstable, short-term contracts, or jobs with no real chance to move up all crank up stress, make depression more likely, and lead to worse sleep (Benach et al., 2020; Vives et al., 2021). And when money's tight, it's harder to get what you need—whether that's support from friends, basic resources, or even access to mental health care—which just adds to the pressure (Lorant et al., 2021).

Women working in higher education juggle a lot—demanding teaching loads, constant pressure to perform, administrative headaches, and sometimes just trying to balance all their roles at once. Studies show that women in academia deal with more stress and sleep problems than men, especially when job security is shaky and old-school gender expectations still linger (García-González et al., 2021; Winefield et al., 2019). In India, more women are joining higher education, but there's still not much research looking at how things like age, income, or family situation affect both depression and sleep quality for these women. This study steps in to fill that gap, exploring how different social and demographic factors connect to depression and sleep quality among women faculty, all through a cross-sectional lens.

Method

Research Design

Researchers used a cross-sectional survey to assess depression and sleep quality among working women.

Participants

The present study employed a non-probability sampling approach, specifically the voluntary response sampling method, to select the participants. A total of 300 female faculty members working in higher education institutions were approached to participate in the study.

Participants were informed about the purpose and nature of the research, and their voluntary consent was obtained prior to data collection. Only those faculty members who willingly agreed to participate were included in the study. The questionnaires were distributed to the respondents, and they were requested to complete all the items carefully.

For the purpose of final data analysis, only those participants who provided complete responses to all the study variables were retained in the sample. Incomplete or partially filled questionnaires were excluded to ensure accuracy and reliability in the statistical analysis. Consequently, the final sample consisted of 300 female faculty members whose responses were complete and suitable for analysis.

This sampling method was considered appropriate due to the accessibility of the respondents and the voluntary nature of participation in the research. However, since the sampling technique was non-random, the generalization of the findings should be made with caution.

Instruments

Pittsburgh Sleep Quality Index (PSQI). The Pittsburgh Sleep Quality Index (PSQI) is a self-report questionnaire with 19 items that looks at how well someone has slept and any problems they've had over the past month. You'll get a total score between 0 and 21—higher numbers mean worse sleep quality. The PSQI holds up well in research, showing strong internal consistency ($\alpha = .83$) and test–retest reliability (.85). It's also pretty accurate, with a sensitivity of 89.6% and specificity of 86.5%.

Beck Depression Inventory (BDI). The Beck Depression Inventory (Beck et al., 1961) is a 21-question self-report tool that helps measure how severe someone's depression is. Each question gets a score from 0 to 3, so the total can go from 0 up to 63. The higher the score, the more severe the depression. The BDI is known for being reliable—researchers found alpha coefficients of .86 for people with psychiatric conditions and .81 for those without (Beck et al., 1988).

Procedure

Once the institution approved the study, everyone who took part gave informed consent. We handed out the questionnaires one-on-one and made sure people knew their answers would stay confidential and anonymous. In the end, we used data from all 300 participants for our analysis.

Statistical Analysis

We ran the numbers with SPSS. First, we used descriptive statistics to break down the participants' backgrounds and the main study variables. Then, we used chi-square tests to look for links between people's socio-demographic factors and their levels of depression or sleep quality. To dig deeper, we ran multiple regression analyses to spot which socio-demographic factors actually predicted depression and sleep quality. We counted results as statistically significant if the p-value was .05 or less.

Results

Association Between Socio-Demographic Factors and Depression and Sleep Quality

Chi-square tests of independence were conducted to examine the association between socio-demographic variables and levels of depression and sleep quality among working women (N = 300).

Table 1

Chi-Square Analysis of Socio-Demographic Factors and Levels of Depression

Variable	χ^2	df	p
Age	2.60	3	.457
Educational Qualification		32.27	6 < .001***
Designation	2.70	3	.441
Teaching Experience	7.66	3	.054
Type of Appointment	20.37	6	.002**
Marital Status	3.06	3	.383
Socio-Economic Status	11.85	6	.065

Note. N = 300. $p < .05^*$, $p < .01^{**}$, $p < .001^{***}$.

As shown in Table 1, educational qualification was significantly associated with levels of depression, $\chi^2(6, N = 300) = 32.27, p < .001$, indicating that women with lower academic qualifications reported higher depressive symptoms. Similarly, type of appointment demonstrated a significant association with depression, $\chi^2(6, N = 300) = 20.37, p = .002$, with temporary and contractual employees reporting higher levels of depression compared to those in permanent positions.

No significant associations were observed between depression and age, designation, marital status, or socio-economic status ($p > .05$). The relationship between teaching experience and depression approached statistical significance, $\chi^2(3, N = 300) = 7.66, p = .054$, suggesting a possible trend toward higher depressive symptoms among women with fewer years of experience.

Table 2

Chi-Square Analysis of Socio-Demographic Factors and Sleep Quality

Variable	χ^2	df	p
Age	12.60	1	< .001***
Educational Qualification	79.94	2	< .001***
Designation	23.26	1	< .001***
Teaching Experience	105.85	1	< .001***
Type of Appointment	112.46	2	< .001***
Marital Status	4.13	1	.042*
Socio-Economic Status	33.52	2	< .001***

Note. $N = 300$. $p < .05^*$, $p < .01^{**}$, $p < .001^{***}$.

Table 2 indicates that sleep quality was significantly associated with most socio-demographic variables. A significant association was observed between age and sleep quality, $\chi^2(1, N = 300) = 12.60, p < .001$, with poorer sleep reported more frequently among women in middle adulthood.

Educational qualification showed a strong association with sleep quality, $\chi^2(2, N = 300) = 79.94, p < .001$, indicating poorer sleep among women with lower qualifications. Significant associations were also found for designation, $\chi^2(1, N = 300) = 23.26, p < .001$, and teaching experience, $\chi^2(1, N = 300) = 105.85, p < .001$, suggesting that seniority and prolonged work exposure adversely affected sleep quality.

Type of appointment emerged as a highly significant factor, $\chi^2(2, N = 300) = 112.46, p < .001$, with poor sleep quality most prevalent among women in temporary and permanent positions. Marital status was also significantly related to sleep quality, $\chi^2(1, N = 300) = 4.13, p = .042$, with married women reporting poorer sleep.

Finally, socio-economic status demonstrated a significant association with sleep quality, $\chi^2(2, N = 300) = 33.52, p < .001$, indicating that women from middle and high socio-economic backgrounds experienced poorer sleep quality.

Table 3

Multiple Regression Analysis Predicting Depression Scores from Socio-Demographic Variables (N = 300)

Predictor	B	SE B	β	t	p		
Constant	2.33	11.80	—	0.20	.844		
Age	0.06	0.14	.03	0.43	.670		
Educational Qualification			4.81	2.25	.18	2.14	.034*
Designation	0.69	2.95	.02	0.23	.816		
Teaching Experience (Years)			−0.38	0.30	−.12	−1.28	.203
Type of Appointment	−0.95	1.52	−.04	−0.63	.532		
Marital Status	2.32	8.81	.02	0.26	.793		
Socio-Economic Status	8.95	3.36	.16	2.67	.008**		

Model Summary. $R = .20$, $R^2 = .04$, Adjusted $R^2 = .02$ $F(7, 292) = 1.78$, $p = .091$

Note. $p < .05^*$, $p < .01^{**}$.

I ran a multiple regression to see how different socio-demographic factors affect depression among women teachers. The big picture? The model as a whole didn't really work—statistically, it wasn't significant, $F(7, 292) = 1.78$, $p = .091$. All together, these predictors explained just 4.1% of the differences in depression scores ($R^2 = .041$), which is really a small effect by usual standards.

Still, two things stood out. First, Educational Qualification made a real difference. Higher education levels linked up with higher depression scores ($B = 4.81$, $\beta = .18$, $p = .034$). The effect wasn't huge, but it was the strongest in the group. For every step up in education, depression scores rose by about 4.8 points, even after accounting for everything else.

Second, Socio-Economic Status (SES) showed a similar pattern. Higher SES was tied to higher depression scores, too ($B = 8.95$, $\beta = .16$, $p = .008$). Again, this was a small but real effect, and it was the next most important predictor after education. A one-unit jump in SES brought an increase of nearly 9 points in depression scores.

The rest of the variables didn't do much. Age ($B = 0.06$, $\beta = .03$, $p = .670$) barely moved the needle—almost no effect. Designation, or job position, was the same story ($B = 0.69$, $\beta = .02$, $p = .816$). Teaching Experience showed a slight trend: more experience, slightly less depression ($B = -0.38$, $\beta = -.12$, $p = .203$), but this wasn't strong enough to count as significant. Type of Appointment ($B = -0.95$, $\beta = -.04$, $p = .532$) and Marital Status ($B = 2.32$, $\beta = .02$, $p = .793$) both came out as basically irrelevant.

So, if we line up all the standardized effects, Educational Qualification ($\beta = .18$) leads, followed by SES ($\beta = .16$). Teaching Experience trends weakly in the opposite direction ($\beta = -.12$), but it doesn't really matter here. Everything else is pretty much zero.

Table 4

Multiple Regression Analysis Predicting Sleep Quality Scores from Socio-Demographic Variables (N = 300)

Predictor	B	SE B	β	t	p
Constant	-15.24	1.77	—	-8.64	< .001***
Age	0.04	0.02	.09	1.92	.055
Educational Qualification			1.16	0.34	.20 3.44 .001**
Designation	-0.25	0.44	-.03	-0.56	.576
Teaching Experience (Years)			0.07	0.05	.10 1.54 .125
Type of Appointment	1.76	0.23	.35	7.73	< .001***
Marital Status	4.66	1.32	.15	3.54	< .001***
Socio-Economic Status	3.88	0.50	.34	7.73	< .001***

Model Summary. $R = .72$, $R^2 = .52$, Adjusted $R^2 = .51$ $F(7, 292) = 44.86$, $p < .001$

Note. Higher scores indicate poorer sleep quality. $p < .05^*$, $p < .01^{**}$, $p < .001^{***}$.

Table 4 shows how different socio-demographic factors affect sleep quality among women teachers, based on a multiple regression analysis. The model as a whole really stands out—it's statistically significant, $F(7, 292) = 44.86$, $p < .001$, and explains just over half the variance in sleep quality ($R^2 = .518$). That's a strong result, showing these factors matter a lot. Looking closer, Type of Appointment takes the lead. With a standardized coefficient of $\beta = .35$ ($t = 7.73$, $p < .001$), it's the most powerful predictor in the group. This means that teachers with less secure or more unstable appointments often report worse sleep. So, job security isn't just good for peace of mind—it actually shows up in how well teachers sleep. Socio-Economic Status (SES) is right behind, also with a large effect ($\beta = .34$, $t = 7.73$, $p < .001$). The difference is almost the same as appointment type. It's clear that teachers facing tougher economic situations tend to struggle more with sleep.

Educational Qualification has a solid impact too ($\beta = .20$, $t = 3.44$, $p = .001$). It's not as strong as appointment type or SES, but it still makes a real difference. Higher qualifications seem to be linked to better sleep, even when we account for the other variables. Marital Status also plays a role ($\beta = .15$, $t = 3.54$, $p < .001$), though it's a smaller effect. Being married or not does seem to affect sleep, but not as much as job security or economic status. The rest of the factors didn't reach statistical significance.

Teaching Experience ($\beta = .10$, $t = 1.54$, $p = .125$) and Age ($\beta = .09$, $t = 1.92$, $p = .055$) both show small, positive trends—experienced or slightly older teachers might sleep a bit differently—but the evidence isn't strong enough to say for sure. Designation ($\beta = -.03$, $t = -0.56$, $p = .576$) barely moves the needle; whether a teacher holds a higher or lower rank doesn't seem to matter for sleep quality. So, stacking them up by effect size, here's how the predictors rank: Type of Appointment ($\beta = .35$) and Socio-Economic Status ($\beta = .34$) come first, followed by Educational Qualification ($\beta = .20$), Marital Status ($\beta = .15$), Teaching Experience ($\beta = .10$), Age ($\beta = .09$), and finally Designation ($\beta = -.03$).

Discussion

This study looked at how things like age, education, job role, and income relate to depression and sleep quality in women working as faculty. Using chi-square and multiple regression, we found that factors like education, job type, and income shape both mental health and sleep in different ways. These results shed light on what working women in academia actually go through and why their experiences with stress and sleep aren't all the same.

Socio-Demographic Factors and Depression

The chi-square analysis showed a clear link between educational qualification and depression. Women with lower academic qualifications reported more depressive symptoms. This fits with what other studies have found: when women don't have higher degrees, they often face fewer career opportunities, less control at work, and more professional roadblocks—all things that can fuel depression (Lorant et al., 2021; Patel et al., 2022). In academic jobs, women with fewer qualifications feel even more pressure to perform and worry more about keeping their positions, which adds to their stress.

The type of job contract told a similar story. Temporary and contractual employees had higher levels of depression than those in permanent roles. It's not surprising—job insecurity takes a toll on mental health, especially for women, who already have to navigate worries about steady income, long-term career prospects, and balancing work with everything else (Benach et al., 2020; Kim & von dem Knesebeck, 2023).

These results just highlight how risky unstable employment can be for women working in higher education. Other factors—like age, job title, marital status, and socio-economic background—didn't show a strong connection to depression. But there was something interesting about teaching experience. Early-career women faculty seemed more vulnerable to depression, echoing earlier studies that found the first few years in academia are especially stressful because of high demands and lots of uncertainty (Winefield et al., 2019). Multiple regression analysis clarified things further.

Educational qualification and socio-economic status stood out as significant predictors of depression, even though the model only explained part of the overall picture. Still, the takeaway is clear: both where women stand academically and the economic pressures they face play a real role in their mental health. This lines up with current models from organizations like the WHO (2023), which stress how ongoing structural and economic challenges stack up to affect women's mental well-being.

Socio-Demographic Factors and Sleep Quality

Sleep quality, unlike depression, showed strong links with almost every socio-demographic variable we looked at. Age stood out—women in their middle years reported sleeping worse than others. That lines up with earlier research showing that sleep changes as we get older, and honestly, juggling work stress and family during midlife just makes things harder (Baglioni et al., 2020).

Education played a big role too. Women with less academic background slept worse. That probably comes down to more job pressure, longer hours, and less control at work. All of that can really mess with sleep (Huang et al., 2021). Job title and years of teaching mattered as well. The longer women spent in academia, the more likely they were to struggle with sleep—maybe thanks to constant workloads, admin tasks, and the pressure to perform.

But the type of appointment stood out the most. Women in both temporary and permanent positions reported worse sleep than those on contract. Sure, temp jobs mean job insecurity, but those in permanent roles often get buried in paperwork and feel overwhelmed by responsibility, which can wreck sleep too (Åkerstedt et al., 2022). The relationship between job status and sleep is not as simple as it looks.

Marital status mattered, too. Married women slept worse. That fits with what other studies have found—married working women often pick up extra chores and caregiving, which eats into their rest (Arber et al., 2021).

Socio-economic status also played a role, but maybe not the way you'd expect. Women from middle and high socio-economic groups actually reported worse sleep. It's a bit counterintuitive, but makes sense when you consider that higher status often comes with more job demands, higher expectations, and more work-family conflict (Garefelt et al., 2020).

When we ran a multiple regression, it turned out that education, job appointment, marital status, and socio-economic status all predicted sleep quality—they explained a big chunk of the differences we saw. So, for working women, sleep is shaped by a mix of work, social life, and economic factors, which really backs up the bio-psycho-social model of sleep health (Grandner, 2021).

Implications

This study points to some real-world changes that could make a difference. When universities focus on job security, balance out workloads, and offer specific help for mental health and sleep, women faculty benefit the most. Early-career academics and those in unstable jobs need special attention too—supporting them helps lower the risk of depression and sleep problems.

Limitations and Future Directions

The study adds some useful insights, but its cross-sectional design makes it tough to say what actually causes what. Plus, since people filled out surveys themselves, there's always a chance their answers aren't totally accurate. Next time, it'd be smart to use a longitudinal approach and dig into qualitative methods to really understand what working women go through. And honestly, looking beyond just higher education would help make the findings apply to more people.

Conclusion

So, to sum it up, this study shows that things like education and income play a big role in depression among working women, but when it comes to sleep, there's a wider mix of work and social factors involved. It's pretty clear that if we want to help with both mental health and sleep for women at work, we need thoughtful, gender-aware programs that look at the whole picture.

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TECHNICAL SESSION – TRACK III (PART I)

SESSION CHAIR

Dr. Agnihba Das Majumdar, Ph.D.
Independent Researcher & Physics Educator
(Ex-Assistant Professor, Mits University)



PAPER PRESENTATIONS

GC³I²-11	Mithun Meenakshi S, Shanmugam DB Comparing Physics-Informed and Physics-Enforced Neural Networks for the Chaotic Three-Body Problem
GC³I²-12	Prof. (Dr) Bharat Raj Singh, Dr. Asha Kulshrestha Intelligent Sustainable Technologies and Green Energy Systems for Environmental Resilience and Climate-Smart Development
GC³I²-13	Danasri S. M, Nidya K Integrated HR Factors Influencing Talent Acquisition Outcomes
GC³I²-14	Hemalatha G Predictors of Quiet Quitting Among Employees in Indian Commercial Organisations
GC³I² 15	Guru Chakri Koushik, Venkata Rao Valluris Forecasting Forex Movement of the Indian Rupee Against the US Dollar and Euro

Comparing Physics-Informed and Physics-Enforced Neural Networks for the Chaotic Three-Body Problem

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Abstract

The three-body gravitational problem is a canonical benchmark for chaotic dynamical systems, admitting no general closed-form solution and exhibiting sensitive dependence on initial conditions. We present a controlled comparative benchmark of four neural network architectures — a Deep Neural Network (DNN), a Physics-Informed Neural Network (PINN), a Hamiltonian Neural Network (HNN), and a Lagrangian Neural Network (LNN) — on trajectory prediction for the planar equal-mass three-body problem. All models are trained on 1,000 high-precision trajectories from the DOP853 integrator and evaluated on trajectory accuracy, long-horizon generalisation, and energy conservation. To the best of our knowledge, this is the first systematic evaluation of HNNs and LNNs as trajectory predictors on the gravitational three-body problem. The DNN achieves the lowest in-horizon error (MAE: 0.3166), while the LNN surpasses all models on long-horizon generalisation and energy conservation. The PINN achieves the best ODE compliance but the worst trajectory accuracy, demonstrating the loss-balancing challenge on chaotic systems.

Keywords Three-body problem, Physics-Informed Neural Networks, Hamiltonian Neural Networks, Lagrangian Neural Networks, chaotic dynamical systems, scientific machine learning

1. Introduction

The three-body problem — predicting the trajectories of three mutually gravitating bodies from initial conditions — has resisted closed-form analytical resolution since Newton implicitly posed it in 1687 [1]. Euler and Lagrange identified special equilibrium configurations [2, 3], and Poincaré's 1890 memoir [4] established the problem as a paradigmatic example of chaos: arbitrarily small perturbations produce exponentially diverging trajectories. Sundman's convergent power-series solution [5] is computationally useless due to its slow convergence. Moore [6] and Chenciner & Montgomery [7] later demonstrated richer periodic solutions, though the general problem remains intractable. Practical computation relies exclusively on numerical integration, whose accuracy is bounded by the system's positive Lyapunov exponents [9].

Deep learning has emerged as a viable alternative for rapid trajectory approximation. Breen et al. [8] demonstrated that a feedforward DNN can predict three-body outcomes up to 10^8 times faster than the Brutus high-precision integrator [26] at comparable short-horizon accuracy. PINNs [11, 14, 28] embed governing differential equations as a soft loss penalty; Pereira et al. [25] applied a PI-ResNet to the three-body problem, achieving far lower ODE residuals than a plain DNN but at a trajectory accuracy cost. HNNs [12] and LNNs [13] encode Hamiltonian or Lagrangian mechanics directly into the network architecture, guaranteeing energy conservation by construction. Symplectic and graph-based architectures [15–20, 21, 22] offer related guarantees at greater complexity. Choudhary et al. [30] showed HNNs function in the chaotic Hénon–Heiles system.

Despite this body of work, HNNs and LNNs have not been systematically benchmarked on the full planar three-body problem: the original HNN paper [12] included only preliminary experiments, LNNs [13] have not been evaluated on gravitational systems, and Pereira et al. [25] explicitly identified both as open directions. This paper closes that gap with the first controlled, quantitative comparison of DNN, PINN, HNN, and LNN on identical training data under a unified evaluation framework.

2. Methodology

2.1 Problem and Dataset

We consider the planar equal-mass three-body problem ($m_1 = m_2 = m_3 = 1$, $G = 1$) following Breen et al. [8] and Pereira et al. [25]. The equations of motion are

$$\ddot{\rho}_i = - \sum_{j \neq i} m_j \frac{(\rho_i - \rho_j)}{\|\rho_i - \rho_j\|^3}$$

Under the planar constraint the state is effectively 12-dimensional. Initial conditions follow the sampling algorithm of [8, 25]: Body 1 fixed at (1,0,0); Body 2 sampled by angle $\theta \sim \text{Uniform}[0, \pi/2]$ and scale; Body 3 placed by centre-of-mass constraint; all initial velocities zero. Ground truth is generated by SciPy's DOP853 integrator [10] ($\text{rtol} = 10^{-8}$, $\text{atol} = 10^{-10}$), spanning $t \in [0, 8]$ at 129 output points. The dataset comprises 1,000 trajectories split 80/10/10 into train/validation/test.

2.2 Model Architectures

All models share the same training protocol: Adam optimiser [27] ($\text{lr} = 10^{-3}$, weight decay = 10^{-5}), ReduceLROnPlateau scheduler, batch size 2,048, maximum 2,000 epochs.

DNN: 10-layer MLP (128 units, ReLU) [8]. Input: 7D (initial positions + query time t_n). Output: 12D state at t_n . Non-autoregressive. 149,636 parameters.

PINN: 5-block ResNet [25, 29] (128 units, ReLU). Loss: $L = L_{MAE} + 0.1 \cdot L_{ODE}$ at 10,000 collocation points per batch. Gradient norm clipping applied. 166,148 parameters.

HNN: Learns scalar $\hat{H}(q, p)$; derives $\dot{r} = \partial \hat{H} / \partial p$, $\dot{p} = -\partial \hat{H} / \partial q$ via autograd [12]. Separable subnetwork (3×64 , tanh) + wrapper (6×256 , tanh). Loss: $0.1 \cdot L_{deriv} + 0.1 \cdot L_H + 0.1 \cdot L_{sup}$. RK4 rollout. 348,422 parameters.

LNN: Learns scalar $\hat{L}(q, \dot{q})$; recovers accelerations by inverting Hessian $M = \partial^2 \hat{L} / \partial \dot{q}^2$ via second-order autograd [13]. Same architecture as HNN (348,422 parameters). Loss: $0.1 \cdot L_{accel} + 0.01 \cdot L_{EL} + 0.1 \cdot L_{lag}$. RK4 rollout.

2.3 Evaluation Metrics

Tier A (all models): MAE and RMSE on the state vector; relative rollout error (deviation normalised by state magnitude); long-horizon rollout error at $T^* = 15$ (50% beyond training horizon); and median energy drift $\max |E_t - E_0|$. Tier B (model-specific): ODE residual (PINN); Euler–Lagrange residual and acceleration MSE (LNN); Hamiltonian vector-field residual and Hamiltonian MSE (HNN).

3. Results and Discussion

The DNN achieves the lowest in-horizon error across all metrics (MAE: 0.3166, RMSE: 1.2463, relative rollout: 0.4555), confirming Breen et al. [8] even on the much smaller 1,000-trajectory dataset. Error distributions are strongly right-skewed across all models, with medians well below means, indicating that a small subset of difficult near-collision trajectories dominates the mean [9]. The LNN ranks second

on all accuracy metrics (MAE: 0.3929), marginally outperforming the HNN (0.4016) at identical parameter counts (348,422). The PINN ranks last on every accuracy metric: its mean rollout error of 0.9530 approaches 1.0 — no better than predicting zero — showing that the physics loss dominates training at $\lambda_{phys} = 0.1$.

The long-horizon evaluation ($T^* = 15$, 50% beyond training) produces the benchmark’s most striking result: the LNN achieves the best mean (1.0842) and median (0.9199) long-horizon error, narrowly surpassing the DNN (1.0866, 0.9468) that had led every in-horizon metric. This is the first empirical evidence that the Euler–Lagrange inductive bias confers a genuine generalisation advantage on the three-body problem. The HNN’s long-horizon performance (1.2560) is comparable to the PINN (1.2392) despite enforcing exact conservation of $\hat{H}$ — because $\hat{H}$ only approximates the true energy H , the gap between them remains large near close-encounter singularities. The LNN achieves the best median energy drift (29.17) and second-lowest ODE residual mean (4.90), confirming that Lagrangian structural constraints provide the best overall balance of physics fidelity and trajectory accuracy. The PINN achieves the lowest ODE residual (mean: 2.44, median: 2.30) confirming effective physics enforcement, but pays a consistent accuracy penalty on every Tier A metric — a direct demonstration of the loss-balancing problem on chaotic systems identified by Pereira et al. [25] and Karniadakis et al. [14].

4. Conclusion

We presented the first controlled comparison of DNN, PINN, HNN, and LNN on the planar gravitational three-body problem. The DNN is the strongest in-horizon predictor; the LNN is the best overall model, surpassing the DNN on long-horizon generalisation and energy conservation while remaining competitive in-horizon — the first evidence that Euler–Lagrange inductive bias yields measurable benefit on this canonical chaotic system. The PINN produces the most physically consistent predictions (lowest ODE residual) but the worst trajectories, suggesting dynamic loss scheduling is necessary on chaotic systems. The HNN enforces conservation by construction but does not close the gap with the LNN at equal model capacity, indicating the Lagrangian formulation is better suited to gravitational problems. Future work should train on larger datasets approaching [25], and explore sequence-to-sequence [31] and transformer-based next-token architectures [32] that treat each state vector as a token — self-attention over trajectory history could weight recent close-encounter events appropriately while remaining combinable with Hamiltonian or Lagrangian structural constraints.

Figures and Tables

Table 1: Trajectory accuracy metrics on held-out test set (lower is better). RR = relative rollout error.

Model	MAE Mean	MAE Med.	RMSE Mean	RMSE Med.	RR Mean	RR Med.
DNN	0.3166	0.1946	1.2463	0.8083	0.4555	0.4502
PINN	0.5852	0.4517	1.9577	1.4677	0.9530	0.9887
LNN	0.3929	0.2192	1.4308	0.8227	0.6229	0.5994
HNN	0.4016	0.2386	1.4636	0.8498	0.6754	0.6303

Table 2: Long-horizon error ($T^* = 15$), energy drift (median), and ODE residual (lower is better for all).

Model	LH Mean	LH Med.	E-Drift (Med.)	ODE Mean Res.	ODE Res. Med.	Params
DNN	1.0866	0.9468	30.345	35.291	11.458	149K
PINN	1.2392	1.0821	30.389	2.441	2.302	166K
LNN	1.0842	0.9199	29.172	4.902	3.336	348K
HNN	1.2560	1.0443	29.370	14.479	2.526	348K

Figures 1–5 below show representative trajectory predictions against DOP853 ground truth for each model, and mean energy drift over time across the test set. Solid lines: ground truth. Dashed lines: model predictions. Colours denote individual bodies.

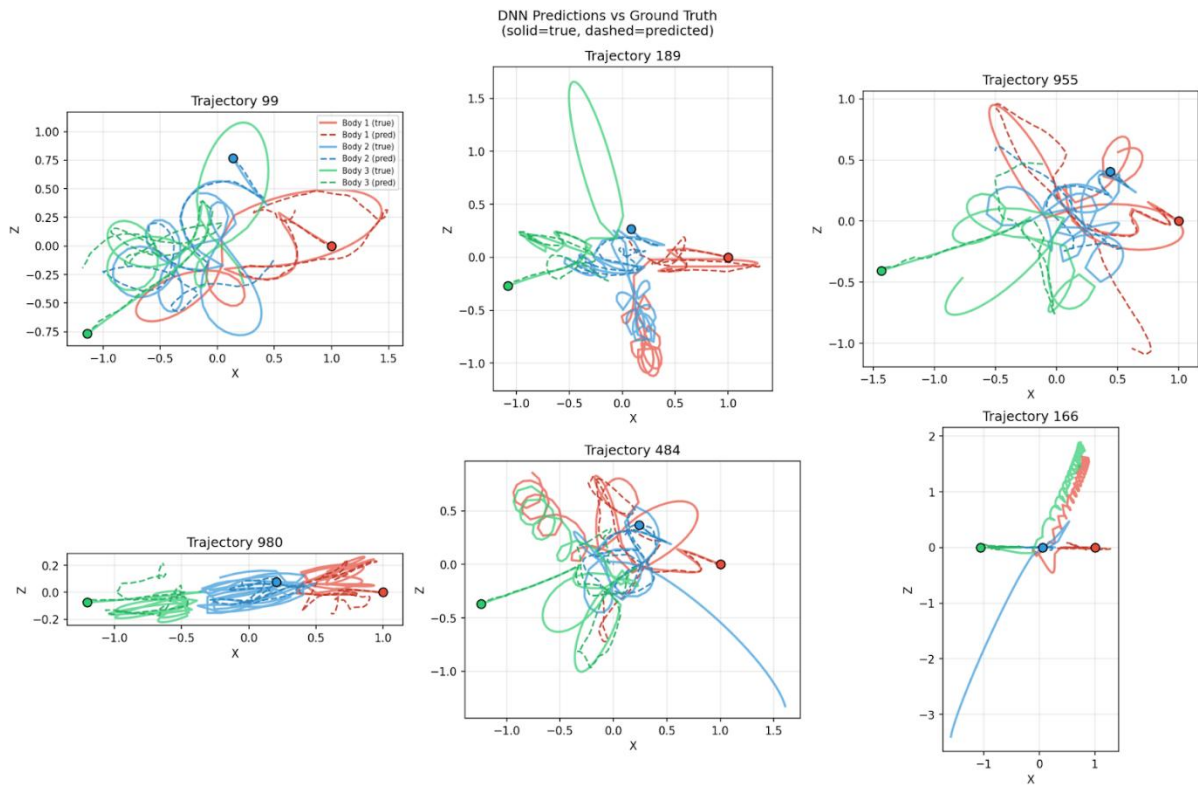


Figure 1: DNN trajectory predictions vs ground truth.

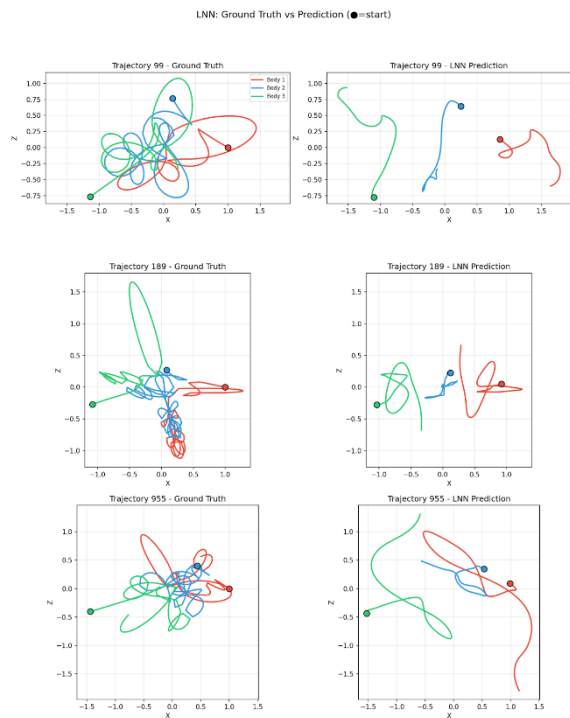


Figure 2: LNN trajectory predictions vs ground truth.

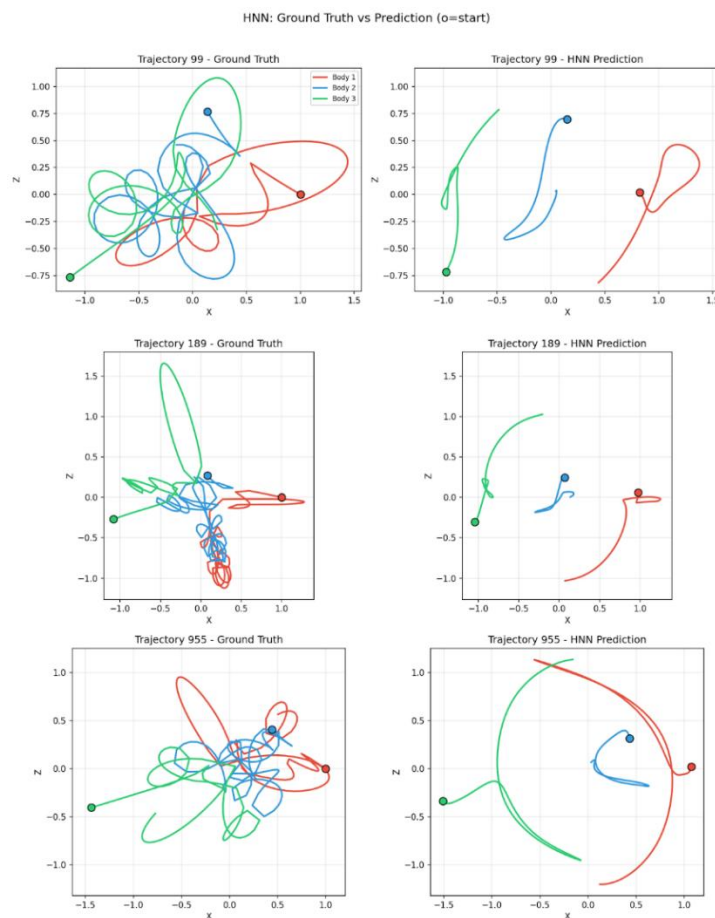


Figure 3: HNN trajectory predictions vs ground truth.

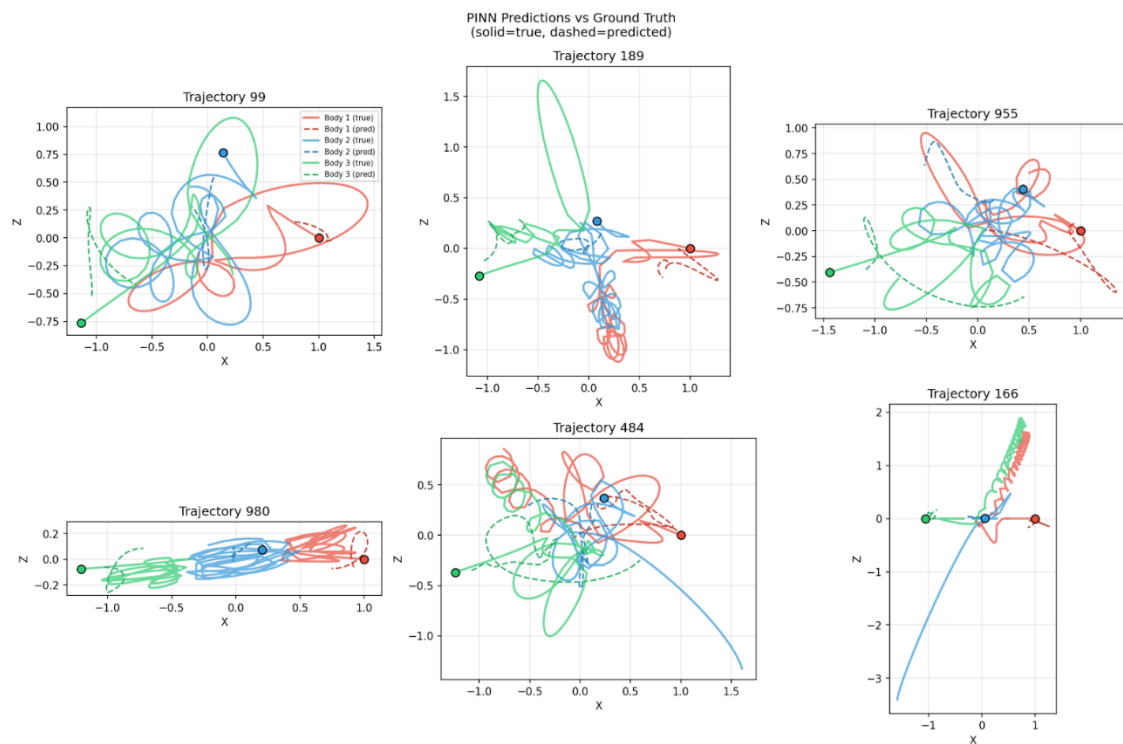


Figure 4: PINN trajectory predictions vs ground truth.

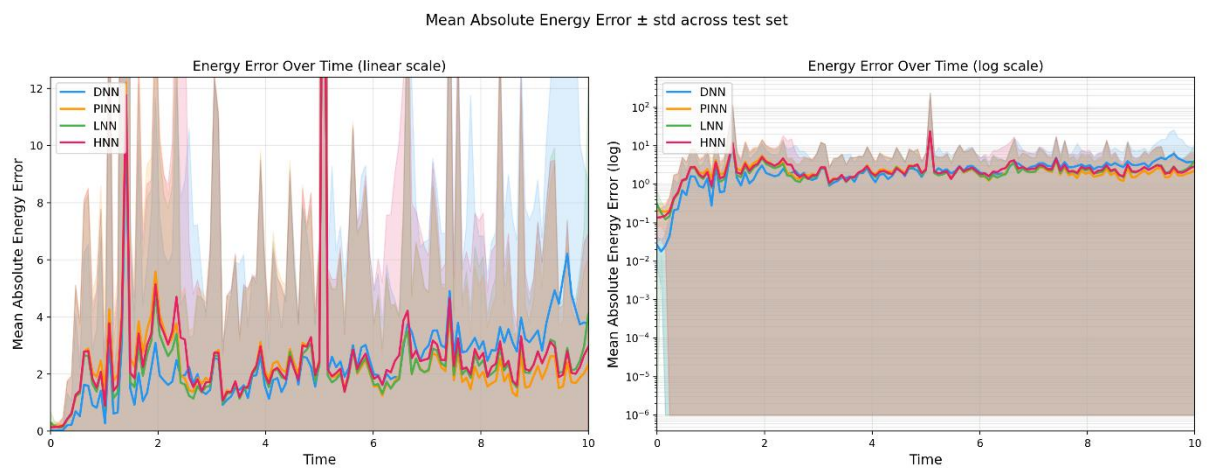


Figure 5: Mean absolute energy error over time for all four models ($\pm$ std across test set).

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Intelligent Sustainable Technologies and Green Energy Systems for Environmental Resilience and Climate-Smart Development

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Abstract

The increasing intensity of climate change, depletion of natural resources, and rising global energy demand have created an urgent need for sustainable technological solutions. Intelligent sustainable technologies integrated with renewable energy systems represent a transformative pathway for achieving environmental resilience and climate-smart development. Green energy technologies such as solar, wind, biomass, and hybrid renewable systems offer environmentally responsible alternatives to conventional fossil-fuel-based energy production. When these systems are supported by intelligent digital technologies such as artificial intelligence (AI), smart grids, and Internet of Things (IoT)-based monitoring systems, the efficiency and reliability of renewable energy infrastructure can be significantly enhanced.

This study examines the role of intelligent sustainable technologies in strengthening environmental resilience and promoting climate-responsive development strategies. It highlights the integration of renewable energy technologies with intelligent monitoring and management systems that optimize energy generation, storage, and distribution. The research also emphasizes the importance of ecological awareness and traditional environmental knowledge that advocate harmony between human activities and nature.

The paper further explores how intelligent green technologies can support sustainable urban development, decentralized energy production, and climate-resilient infrastructure. The integration of technological innovation with sustainable policy frameworks and community participation can significantly reduce carbon emissions and environmental degradation.

The study concludes that intelligent sustainable technologies combined with green energy systems provide an effective strategy for addressing climate change while promoting sustainable economic growth. Future progress requires interdisciplinary research, technological innovation, and global cooperation to accelerate the transition toward a low-carbon and environmentally resilient society.

Keywords: Sustainable Technology, Renewable Energy Systems, Intelligent Energy Management, Environmental Sustainability, Climate-Smart Development, Smart Grids.

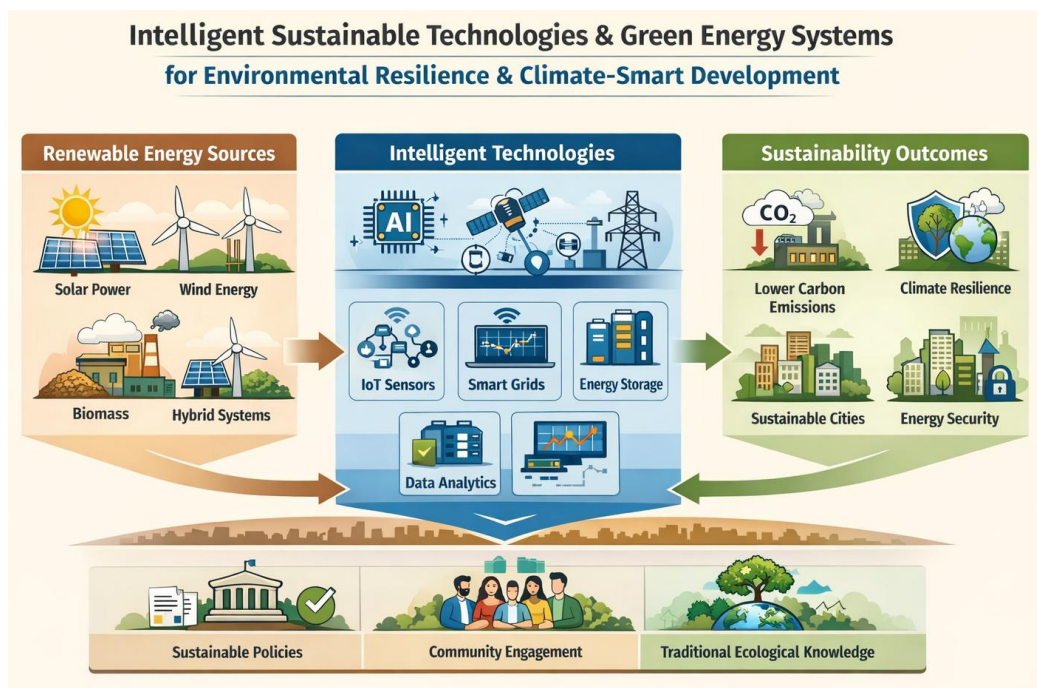


Fig. 1: Graphic Abstract

1. Introduction

Modern civilization has achieved unprecedented technological advancement; however, this progress has been accompanied by severe environmental consequences at a global scale. Anthropogenic activities have significantly increased greenhouse gas (GHG) emissions, leading to global warming, climate variability, and ecological imbalance. According to the United Nations and Intergovernmental Panel on Climate Change, global temperatures have already risen by approximately 1.1°C above pre-industrial levels, intensifying extreme weather events, sea-level rise, and biodiversity loss. Simultaneously, rapid industrialization and urbanization have contributed to air pollution, water scarcity, and accelerated depletion of natural resources, posing serious threats to sustainable development worldwide (United Nations, 2022; IPCC, 2023).

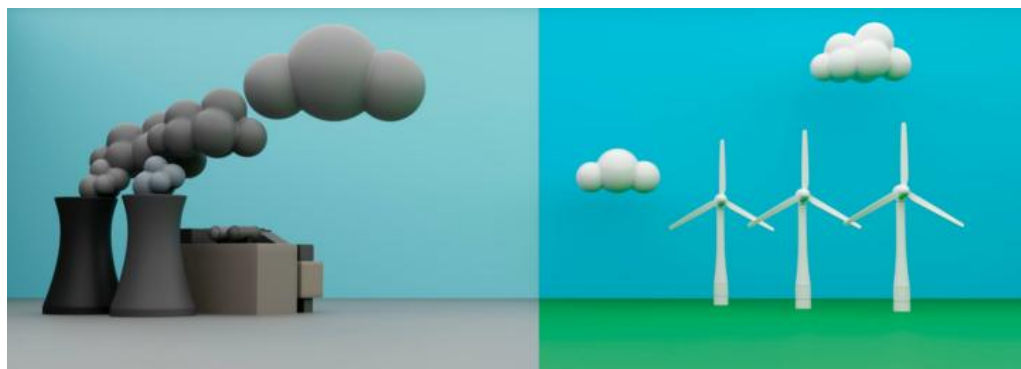


Fig. 2: Fossil Fuel Power Plant verses Wind Turbine

In response to these challenges, sustainable development has emerged as a critical global priority, aligned with the Sustainable Development Goals (SDGs), particularly those related to clean energy, climate action, and sustainable cities. The transition from fossil fuel-based energy systems to renewable energy sources is widely recognized as a key strategy for mitigating climate change while ensuring long-term energy security. Renewable energy technologies such as solar, wind, biomass, and hybrid systems offer low-carbon, environmentally sustainable alternatives capable of meeting the rising global energy demand (International Energy Agency, 2023; IRENA, 2022).

Recent advancements in intelligent technologies have further enhanced the potential of renewable energy systems. The integration of artificial intelligence (AI), Internet of Things (IoT), and smart grid infrastructure enables real-time monitoring, predictive analytics, and efficient energy management. These intelligent energy systems improve grid stability, optimize energy generation and distribution, and reduce operational inefficiencies, thereby accelerating the transition toward sustainable energy ecosystems (Ahmad et al., 2021; Li & Wang, 2020; Mishra & Sharma, 2024).



Fig 3: Futuristic AI Powered Smart Grid

Beyond technological innovation, the importance of traditional ecological knowledge (TEK) has gained increasing recognition in global sustainability discourse. Indigenous and traditional practices, particularly in countries like India, have historically emphasized harmony between human activities and natural ecosystems. These knowledge systems promote resource conservation, ecological balance, and sustainable living practices. Integrating TEK with modern scientific and technological approaches can foster more inclusive, culturally relevant, and resilient environmental management strategies (Sharma & Gupta, 2023).

Therefore, the convergence of renewable energy technologies, intelligent digital systems, and traditional ecological wisdom provides a comprehensive framework for addressing global environmental challenges. Such an integrated approach is essential for achieving environmental resilience, reducing carbon footprints, and advancing climate-smart development in an increasingly uncertain and resource-constrained world.

2.0 Literature Review

The growing urgency of climate change mitigation and sustainable development has led to extensive research on renewable energy systems, intelligent technologies, and environmentally resilient infrastructure. Existing literature highlights that the integration of green energy systems with advanced digital technologies is a critical pathway toward achieving climate-smart development.

2.1. Evolution of Sustainable Technologies

Sustainable technologies have evolved from conventional pollution control mechanisms to integrated systems that emphasize resource efficiency, environmental protection, and economic sustainability. Early studies focused primarily on reducing emissions from industrial processes; however, recent research emphasizes a systems-based approach combining energy efficiency, circular economy practices, and low-carbon technologies.

According to reports by international agencies such as the International Energy Agency (IEA) and the United Nations Environment Programme (UNEP), sustainable technologies can significantly reduce global greenhouse gas emissions when implemented at scale. Research indicates that energy-efficient systems, green infrastructure, and sustainable material usage collectively contribute to long-term ecological balance and economic resilience.

2.2. Renewable Energy Systems and Their Global Significance

A substantial body of literature recognizes renewable energy as a cornerstone of sustainable development. Solar, wind, and biomass energy systems have been extensively studied for their potential to reduce dependence on fossil fuels.

- Solar photovoltaic (PV) technology has gained widespread attention due to its scalability and declining installation costs. Studies show that advancements in PV efficiency and energy storage systems have improved reliability and economic feasibility.
- Wind energy research highlights its high energy output and minimal environmental impact.
- Offshore wind farms, in particular, have demonstrated significant potential in meeting large-scale energy demands.
- Biomass energy has been explored as a dual-benefit technology, addressing both energy generation and waste management. Research emphasizes its role in rural development and decentralized energy systems.
- Hybrid renewable energy systems, combining multiple energy sources, have emerged as a reliable solution to intermittency issues.

Literature suggests that hybrid systems supported by energy storage technologies provide stable and continuous energy supply.

2.3. Intelligent Technologies in Energy Systems

Recent studies emphasize the transformative role of intelligent technologies such as artificial intelligence (AI), Internet of Things (IoT), and machine learning in optimizing energy systems.

AI-based models are widely used for:

- Energy demand forecasting
- Predictive maintenance of power systems
- Optimization of energy generation and distribution

IoT-enabled systems facilitate real-time monitoring of environmental and energy parameters. Smart sensors and communication networks enhance operational efficiency and enable automated decision-making.

Smart grid technology is one of the most significant advancements in modern energy systems. Literature highlights that smart grids improve grid reliability, reduce transmission losses, and enable integration of distributed renewable energy sources.

Research by Ahmad et al. (2021), Li & Wang (2020), and Mishra & Sharma (2024) demonstrates that intelligent energy management systems significantly enhance the performance and resilience of renewable energy infrastructure.

2.4. Sustainable Urban Development and Environmental Monitoring

Urbanization has intensified environmental challenges, leading to increased focus on sustainable urban systems. Studies indicate that smart cities integrating renewable energy, efficient transportation, and digital monitoring systems can significantly reduce carbon emissions.

Green building technologies have been widely studied for their ability to reduce energy consumption and improve indoor environmental quality. Passive design strategies, energy-efficient materials, and renewable energy integration are key components identified in the literature.

Environmental monitoring systems based on IoT and remote sensing technologies enable continuous tracking of air quality, water resources, and climate variables. These systems support data-driven governance and environmental policy formulation.

2.5. Environmental Resilience and Climate-Smart Development

Environmental resilience has been a major focus of recent research, particularly in the context of climate change adaptation. Studies define resilience as the ability of systems to withstand, adapt to, and recover from environmental disturbances.

Climate-smart development integrates mitigation and adaptation strategies to achieve sustainable economic growth. Literature emphasizes that renewable energy adoption, resilient infrastructure, and sustainable resource management are essential components of climate-smart strategies.

Research suggests that integrating intelligent technologies with climate adaptation strategies enhances system responsiveness and reduces vulnerability to climate risks.

2.6. Role of Traditional Ecological Knowledge (TEK)

In recent years, there has been growing recognition of the importance of Traditional Ecological Knowledge (TEK) in sustainability studies. TEK encompasses indigenous practices and cultural knowledge systems that promote environmental harmony and resource conservation.

Studies highlight that traditional practices such as water conservation, biodiversity protection, and sustainable agriculture align closely with modern sustainability principles. In countries like India, these practices have historically contributed to ecological balance.

Researchers argue that integrating TEK with modern technological innovations can lead to more inclusive, culturally sensitive, and sustainable solutions.

2.7. Challenges Identified in Existing Literature

Despite significant advancements, the literature identifies several barriers to the widespread adoption of sustainable technologies:

- High initial investment costs
- Technological limitations in energy storage
- Grid integration challenges
- Policy and regulatory constraints
- Lack of awareness and technical expertise in developing regions

These challenges highlight the need for supportive policy frameworks, technological innovation, and capacity-building initiatives.

2.8. Research Gap

While extensive research exists on renewable energy and intelligent technologies independently, limited studies have focused on their integrated application for environmental resilience and climate-smart development.

Additionally, the incorporation of traditional ecological knowledge into modern intelligent energy systems remains underexplored. There is a need for interdisciplinary research that combines engineering, environmental science, and socio-cultural perspectives.

2.9. Summary of Literature

The reviewed literature clearly establishes that:

- Renewable energy systems are essential for reducing carbon emissions
- Intelligent technologies enhance efficiency and reliability of energy systems
- Sustainable urban systems and environmental monitoring improve resilience
- Traditional ecological knowledge provides valuable sustainability insights
- Integrated approaches are required for effective climate-smart development

2. Sustainable Technologies for Environmental Protection

Sustainable technologies have emerged as a cornerstone of global strategies to combat environmental degradation and climate change. As nations strive to meet the targets of the United Nations Sustainable Development Goals (SDGs), particularly SDG 7 (Affordable and Clean Energy), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action), the integration of environmentally responsible technologies has become imperative.

These technologies are designed to minimize ecological footprints while enhancing resource efficiency, economic viability, and social well-being. According to the International Energy Agency, the widespread deployment of sustainable solutions could contribute to over 70% of global emissions reductions required to achieve net-zero targets by 2050.

Sustainable technologies focus on reducing pollution, conserving natural resources, and enabling circular economy practices. Key examples include:

- **Green Building Design and Energy-Efficient Architecture:** Modern green buildings incorporate passive design, renewable energy integration, and advanced insulation systems. The World Green Building Council reports that green buildings can reduce energy consumption by up to 50% and carbon emissions by nearly 40%, significantly contributing to climate mitigation.
- **Waste-to-Energy Conversion Technologies:** Waste-to-energy (WtE) systems transform municipal and industrial waste into usable energy forms such as electricity and heat. These technologies reduce landfill dependency and methane emissions, aligning with circular economy principles promoted by the United Nations Environment Programme.
- **Water Conservation and Recycling Systems:** Advanced wastewater treatment, rainwater harvesting, and greywater reuse technologies are essential for addressing global water scarcity. The World Bank estimates that by 2030, global water demand will exceed supply by 40% if sustainable water management practices are not implemented.
- **Smart Environmental Monitoring Networks:** Digital transformation has enabled real-time environmental monitoring through Internet of Things (IoT) technologies. Sensor-based networks continuously track air quality, temperature, humidity, and pollutant concentrations. These systems support data-driven policymaking and rapid response to environmental hazards.

The integration of digital technologies further enhances the effectiveness of sustainable systems. IoT-enabled environmental monitoring platforms, combined with artificial intelligence and big data analytics, allow predictive modeling and proactive environmental management. Governments and urban planners increasingly rely on such smart systems to develop resilient and adaptive infrastructure, particularly in rapidly urbanizing regions.

In conclusion, sustainable technologies represent a transformative pathway toward environmental protection and long-term ecological balance. Their global adoption is critical not only for mitigating climate change but also for ensuring sustainable development, resource security, and improved quality of life for future generations.

3. Green Energy Systems

3.1 Solar Energy

Solar energy is one of the most abundant renewable energy sources available on Earth. Solar photovoltaic systems convert sunlight directly into electricity, making them suitable for residential, industrial, and rural electrification applications.

3.2 Wind Energy

Wind turbines convert kinetic energy from wind into electrical power. Wind energy projects have expanded significantly across many countries due to their low environmental impact and large power generation potential.

3.3 Biomass Energy

Biomass energy utilizes organic materials such as agricultural residues and forest waste to produce heat, electricity, or biofuels. This technology also contributes to effective waste management.

3.4 Hybrid Renewable Energy Systems

Hybrid renewable systems combine multiple renewable energy sources to ensure stable energy production. For example, solar-wind hybrid systems supported by energy storage technologies can provide continuous electricity supply.

4. Intelligent Energy Management Systems

The integration of renewable energy into modern power systems requires advanced energy management technologies. Intelligent energy systems utilize digital technologies to monitor and control energy production and distribution.

Key technologies include:

- Smart grid systems
- Artificial intelligence-based energy optimization
- IoT-enabled monitoring devices
- Advanced energy storage technologies

These systems improve energy efficiency and help balance energy demand and supply.

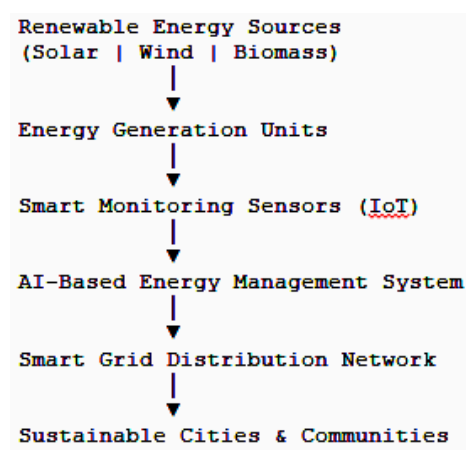


Fig. 4: Intelligent Green Energy System Model

5. Environmental Resilience and Climate-Smart Development

Environmental resilience refers to the capacity of ecosystems and communities to adapt to environmental disturbances while maintaining essential functions.

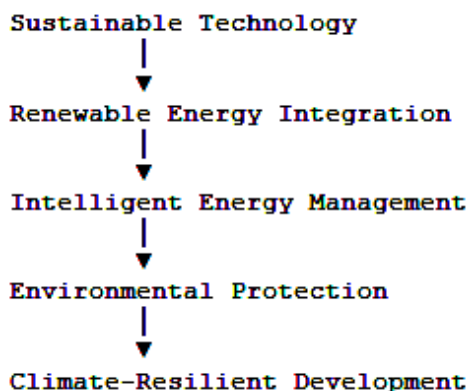


Fig. 5: Climate-Smart Development Framework

Climate-smart development focuses on strategies that simultaneously promote economic growth and environmental protection. These strategies include:

- Renewable energy adoption
- Sustainable urban planning
- Climate-resilient infrastructure
- Community-based environmental management

The integration of intelligent technologies with renewable energy systems strengthens resilience against climate-related challenges.

5.0 Case Studies and Empirical Examples of Intelligent Green Technologies

5.1. AI-Enabled Smart Grid – Denmark



Fig. 6: AI-Enabled Smart Grid – Denmark

Denmark is globally recognized for its advanced integration of renewable energy through intelligent grid systems. The country has implemented AI-enabled smart grids to manage high penetration of wind energy, which contributes over 40% of its electricity generation as shown in Fig.6.

The intelligent grid uses real-time data analytics and forecasting algorithms to balance energy supply and demand. Machine learning models predict wind generation patterns and optimize grid operations accordingly.

Key Outcomes:

- Improved grid stability despite variable renewable inputs
- Reduction in energy wastage and transmission losses
- Enhanced integration of distributed renewable sources

This case demonstrates an intelligent systems can transform intermittent renewable energy into reliable power supply.

5.2. Smart Solar Micro-grids – India (Rural Electrification)



Fig. 7: Indian Rural Micro-Grid – Electrification

In rural India, smart solar microgrids have been deployed to provide electricity in off-grid and underserved regions. These systems integrate solar photovoltaic panels with battery storage and IoT-based monitoring platforms as shown in Fig.7.

Companies and government initiatives have implemented intelligent controllers that manage load distribution, battery charging, and fault detection in real time.

Empirical Findings:

- Significant improvement in rural electrification rates
- Reduction in dependence on diesel generators
- Enhanced energy access for education, healthcare, and small enterprises

This example highlights the role of decentralized intelligent energy systems in achieving inclusive and sustainable development.

5.3. AI-Based Energy Optimization – Google Data Centers

Google has implemented artificial intelligence (via DeepMind) to optimize energy consumption in its data centers. AI algorithms analyze vast amounts of operational data and automatically adjust cooling systems as shown in Fig.8.

The system continuously learns and improves efficiency by predicting temperature variations and energy demand.

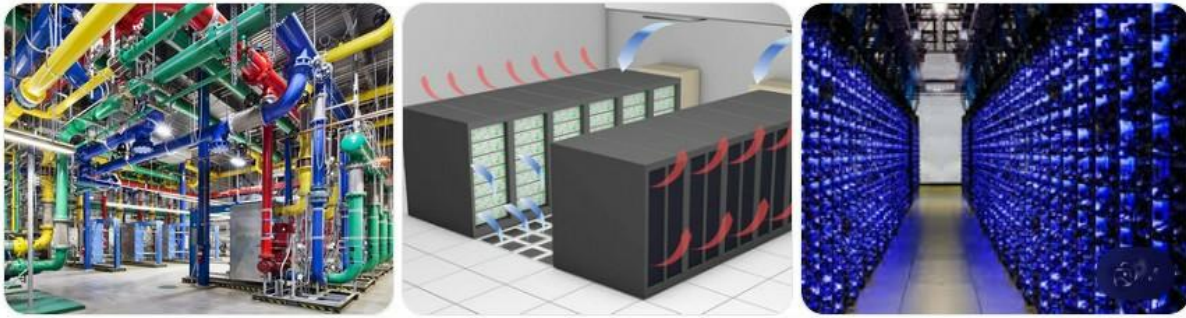


Fig. 8: AI-Based Energy Optimization of Google Data Centers

Measured Results:

- Up to 40% reduction in energy used for cooling
- Significant decrease in overall energy consumption
- Lower carbon footprint of data center operations

This case provides strong empirical evidence of AI's capability in enhancing energy efficiency in large-scale infrastructure.

5.4. Smart City Environmental Monitoring – Singapore

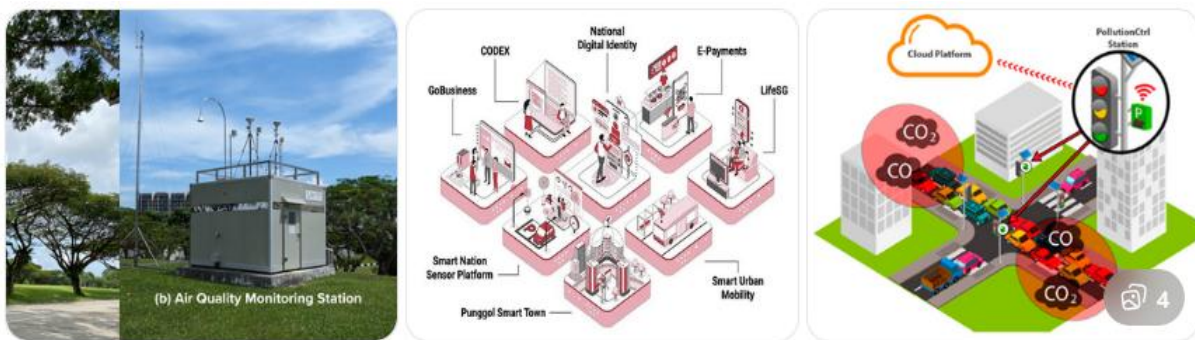


Fig. 9: Smart City Environmental Monitoring – Singapore

Singapore's Smart Nation initiative incorporates IoT-based environmental monitoring systems across the city. Sensors collect real-time data on air quality, temperature, humidity, and water levels as shown in Fig.9.

Advanced analytics platforms process this data to support urban planning and environmental protection strategies.

Key Impacts:

- Real-time pollution control and mitigation
- Data-driven decision-making in urban governance
- Improved public health and environmental quality

This example demonstrates how intelligent environmental monitoring contributes to sustainable urban ecosystems.

5.5. Wind-Solar Hybrid System with Storage – China

China has developed large-scale hybrid renewable energy plants combining wind, solar, and battery storage systems. These systems are managed using intelligent control technologies that optimize energy generation and storage as shown in Fig.10.

AI-based forecasting tools predict weather conditions and adjust energy dispatch accordingly.



Fig. 10: Wind-Solar Hybrid System with Storage at China

Observed Benefits:

- Continuous and stable power supply
- Reduced curtailment of renewable energy
- Enhanced grid flexibility and efficiency

This case highlights the importance of hybrid systems supported by intelligent management.

5.6. Waste-to-Energy Smart Plants – Sweden



Fig. 11: Waste-to-Energy Smart Plants – Sweden

Sweden has successfully implemented intelligent waste-to-energy (WtE) systems that convert municipal waste into electricity and heat. These plants use automated sorting, combustion optimization, and emission monitoring systems as shown in Fig.11.

Digital control systems ensure high efficiency and minimal environmental impact.

Results:

- Nearly zero landfill waste
- Efficient energy recovery from waste
- Reduced greenhouse gas emissions

This example demonstrates the integration of circular economy principles with intelligent technologies.

Outcome of Case Studies

The above empirical examples clearly demonstrate that intelligent green technologies are not theoretical concepts but practical, scalable solutions. Across different regions and sectors, the integration of AI, IoT, and smart systems with renewable energy technologies has resulted in:

- Enhanced energy efficiency and reliability
- Reduced environmental impact and emissions
- Improved accessibility and sustainability
- Strengthened climate resilience

These case studies reinforce the argument that intelligent sustainable technologies are essential for achieving global climate goals and transitioning toward a low-carbon future.

6. Integration of Traditional Ecological Knowledge and Modern Innovation

Traditional ecological knowledge provides valuable insights into sustainable resource management. Many ancient cultural traditions emphasized the importance of respecting natural ecosystems and maintaining ecological balance.

In India, environmental philosophies have long promoted harmony between humans and nature. These ideas align closely with modern sustainability principles such as conservation, resource efficiency, and ecological responsibility.

Combining traditional ecological wisdom with modern scientific innovation can create more sustainable and culturally relevant environmental solutions.

7. Challenges and Future Prospects

Despite significant technological advancements, several challenges remain in the widespread adoption of renewable energy technologies.



Fig 12: Smart Grid Concept

Major challenges include:

- High initial investment costs
- Need for advanced energy storage technologies
- Policy and regulatory barriers
- Limited awareness in some regions

Future research should focus on improving energy storage systems, developing more efficient renewable technologies, and strengthening smart grid infrastructure as shown in Fig. 12..

International cooperation and knowledge sharing will be essential for accelerating the global transition toward sustainable energy systems.

8. Conclusion

Sustainable technologies and green energy systems play a critical role in addressing environmental challenges and promoting climate-smart development. Renewable energy technologies such as solar, wind, and biomass offer clean and sustainable alternatives to fossil fuels.

When combined with intelligent digital technologies such as artificial intelligence, smart grids, and IoT-based monitoring systems, renewable energy systems become more efficient and reliable.

The integration of technological innovation with traditional ecological knowledge provides a comprehensive framework for environmental sustainability. Continued research, innovation, and policy support are essential for building environmentally resilient societies and achieving global sustainability goals.

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Integrated HR Factors Influencing Talent Acquisition Outcomes

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Abstract

Talent acquisition plays a vital role in ensuring organizational success by enabling firms to attract and recruit competent employees. In the current competitive environment, organizations need to integrate various human resource (HR) practices to improve the effectiveness of their recruitment processes. This study examines the influence of integrated HR factors on talent acquisition outcomes. The research focuses on key factors such as talent identification, talent management, talent planning, performance appraisal methods, job characteristics, and talent sourcing strategies.

The study is based on primary data collected from 75 respondents through a structured questionnaire. The data were analyzed using the Statistical Package for the Social Sciences (SPSS). Statistical tools including descriptive statistics, correlation analysis, and regression analysis were used to examine the relationship between HR factors and talent acquisition outcomes.

The findings reveal that HR factors such as talent identification, talent management, talent planning, performance appraisal, and job characteristics are perceived as important for effective talent acquisition. Among these, job characteristics show a significant positive influence on talent acquisition, while the other HR factors demonstrate positive but non-significant effects. In addition, talent sourcing strategies—including employee referrals, job portals, campus recruitment, and social media recruiting—show a strong and significant relationship with talent acquisition effectiveness.

The study concludes that integrating HR practices with effective job design and talent sourcing strategies can enhance recruitment outcomes and support organizational growth.

Keywords: Talent Management, Talent Planning, Performance Appraisal, Job Characteristics, Talent Sourcing Strategies, Talent Acquisition Outcomes.

1. Introduction

In today's competitive business environment, organizations recognize that attracting and hiring the right talent is essential for achieving organizational success. Talent acquisition has evolved from a simple recruitment activity into a strategic process closely linked with several human resource practices. Integrated HR factors such as talent identification, talent planning, talent management, performance appraisal methods, job characteristics, and talent sourcing strategies play a crucial role in improving recruitment outcomes and organizational performance.

Although several studies have examined individual HR practices and their impact on recruitment, limited research has explored how multiple HR factors collectively influence talent acquisition outcomes in an integrated manner. Many existing studies focus on isolated practices such as recruitment strategies or performance management, leaving a research gap regarding the combined influence of HR practices on talent acquisition effectiveness.

Therefore, this study attempts to address this gap by examining the integrated impact of key HR factors on talent acquisition outcomes. The research contributes to the existing HRM literature by providing empirical evidence on how coordinated HR practices can enhance recruitment effectiveness. The findings offer practical insights for HR professionals and organizations seeking to improve talent acquisition strategies through integrated HR management approaches.

Statement of the Problem

Talent acquisition effectiveness in organizations depends on several Human Resource practices such as talent identification, talent management, talent planning, performance appraisal methods, job characteristics, and talent sourcing strategies. In today's highly competitive business environment, organizations often face challenges in attracting suitable candidates, addressing skill mismatches, and achieving effective recruitment outcomes. As a result, organizations need a more coordinated and integrated approach to HR practices. Therefore, understanding how integrated HR factors influence talent acquisition outcomes is essential for improving recruitment effectiveness and supporting overall organizational performance.

Objectives of the Study

- 1.To examine the influence of talent identification, talent management, and talent planning, performance appraisal methods and job characteristics on talent acquisition.
- 2.To study the relationship between talent sourcing strategies and talent acquisition effectiveness.

2. Review of Literature

Tuttle and Critchlow (2025) examined the role of digital transformation in modern talent acquisition practices. Their study highlighted how technology-driven recruitment tools such as artificial intelligence, data analytics, and applicant tracking systems improve candidate screening and recruitment efficiency. The findings suggested that organizations adopting digital recruitment strategies can significantly enhance talent acquisition outcomes by improving candidate quality and reducing time-to-hire.

Wungsnuopparat and Jiarui (2022) studied the factors influencing effective talent acquisition in small and medium-sized enterprises (SMEs) in China. The research examined variables such as talent identification, performance appraisal methods, talent management practices, job characteristics, job satisfaction, work environment, and leadership influence, with talent acquisition as the dependent variable. Using a quantitative approach, data were collected from 420 respondents through electronic questionnaires distributed via social media. Structural Equation Modeling (SEM) was used to analyze the relationships between the variables. The findings showed that effective talent identification, supportive leadership, a positive work environment, and strong talent management practices significantly improve the ability of SMEs to attract skilled employees. The study concluded that organizations should adopt strategic HR practices and focus on employee satisfaction and workplace conditions to enhance talent acquisition effectiveness.

Ramola and Rangnekar (2020) analyzed the relationship between talent acquisition, competency mapping, and performance management in organizations. The study found that competency-based recruitment combined with structured performance appraisal systems improves the alignment between employee capabilities and organizational goals. The research emphasized that integrating HR practices strengthens talent acquisition effectiveness and contributes to long-term organizational performance.

Sparrow & Makram (2015) examined the value-driven perspective of talent management within organizational architecture. The study analyzed how talent management processes create measurable organizational value. The findings indicated that talent management systems must be integrated with performance metrics to demonstrate return on investment. The research emphasized that talent should be viewed as a strategic resource that requires protection and optimization. However, the study identified challenges in quantifying the financial impact of talent management initiatives. The findings suggest that organizations must adopt value-based talent management frameworks to enhance strategic outcomes.

3.Research Methodology

Research Design

This study uses a descriptive research design to analyze the effect of talent identification, talent management, talent planning, performance appraisal methods, and job characteristics on talent acquisition. It also examines how talent sourcing strategies are related to the effectiveness of talent acquisition. The data for the study were gathered at a single point in time through a cross-sectional method, with responses collected from participants belonging to different demographic backgrounds.

Sampling

The study population included HR professionals involved in talent acquisition and recruitment activities in various organizations, with a sample of **75 respondents** selected to represent the broader population.

Sampling Technique

The study used convenience sampling to select participants who were easily accessible and willing to participate, allowing efficient data collection.

Data Collection

The research primarily relies on primary data gathered directly from respondents to analyze the factors that influence talent acquisition outcomes.

Tools for Data Collection

A structured questionnaire was used to gather the required information. The questionnaire was prepared according to the objectives of the study and included questions related to talent identification, talent management, talent planning, performance appraisal methods, job characteristics, and talent sourcing strategies.

Tools

The collected data were analyzed using SPSS (Statistical Package for the Social Sciences). Statistical techniques such as Descriptive Statistics, Correlation and Regression were used to examine the relationship between HR factors including Talent Identification, Talent Management, Talent Planning, Performance Appraisal, Job Characteristics, and Talent Sourcing and Talent Acquisition outcomes.

4. Results and Discussion

Descriptive Statistics

TABLE: 1 DESCRIPTIVE STATISTICS OF HR FACTORS INFLUENCING TALENT ACQUISITION

	N	Minimum	Maximum	Mean	Std. Deviation
Talent Identification	75	15	25	21.25	2.194
Talent Management	75	14	25	20.76	2.460
Talent Planning	75	15	25	20.99	2.592
Performance Appraisal	75	15	25	21.15	2.855
Job Characteristics	75	14	25	21.23	2.749
Talent Sourcing	75	14	25	20.95	2.721
Talent Acquisition	75	12	25	21.13	2.549

Interpretation

This indicates that respondents generally perceive the identified HR factors as important in enhancing talent acquisition. Talent identification (Mean = 21.25) and job characteristics (Mean = 21.23) are seen as key factors influencing effective recruitment. Performance appraisal (Mean = 21.15) also shows strong agreement, suggesting its importance in supporting talent acquisition outcomes. Talent planning (Mean = 20.99), talent sourcing (Mean = 20.95), and talent management (Mean = 20.76) show moderate agreement, indicating consistent recognition of their relevance. Overall, the descriptive results suggest that these HR factors positively support the talent acquisition process within organizations.

Regression

H_{0a}: There is no significant impact of Talent Identification, Talent Management, Talent Planning, Performance Appraisal, and Job Characteristics on Talent Acquisition.

H_{1a}: There is a significant impact of Talent Identification, Talent Management, Talent Planning, Performance Appraisal, and Job Characteristics on Talent Acquisition.

TABLE: 2 REGRESSION ANALYSIS OF HR FACTORS INFLUENCING TALENT ACQUISITION

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.	R Square	Adjusted R Square
	B	Std. Error	Beta				
(Constant)	6.497	1.997		3.254	.002	.586	.553
Talent Identification	-.198	.152	-.171	-1.300	.198	.198	.152
Talent Management	.142	.164	.137	.862	.391	.142	.164
Talent Planning	.211	.175	.214	1.207	.232	.211	.175
Performance Appraisal	.112	.155	.125	.722	.473	.112	.155
Job Characteristics	.430	.139	.463	3.090	.003	.430	.139

Interpretation

1.Talent Identification

The analysis indicates a slight negative relationship between Talent Identification and Talent Acquisition, with an unstandardized coefficient of -0.198 and a standardized beta of -0.171. However, the p-value of 0.198 exceeds the 0.05 threshold, showing that the effect is not statistically significant. This implies that Talent Identification does not have a meaningful impact on Talent Acquisition in this study.

2.Talent Management

Talent Management shows a positive, though weak, association with Talent Acquisition ($B = 0.142$, $\beta = 0.137$). The p-value of 0.391 indicates that this relationship is not statistically significant. Therefore, Talent Management does not significantly influence Talent Acquisition.

3.Talent Planning

The results suggest that Talent Planning has a moderate positive effect on Talent Acquisition ($B = 0.211$, $\beta = 0.214$), but the p-value of 0.232 exceeds 0.05, indicating that this effect is not statistically significant. Hence, Talent Planning is not a significant predictor in this model.

4. Performance appraisal

Performance Appraisal demonstrates a weak positive relationship with Talent Acquisition ($B = 0.112$, $\beta = 0.125$). Since the p-value is 0.473, this effect is not statistically significant, suggesting that Performance Appraisal does not have a substantial impact on Talent Acquisition.

5. Job characteristics

Job Characteristics exhibit a strong positive influence on Talent Acquisition, with $B = 0.430$ and $\beta = 0.463$. The p-value of 0.003 indicates that this relationship is statistically significant. This shows that Job Characteristics play a key role in improving Talent Acquisition outcomes.

The regression model indicates that among the HR factors considered, only Job Characteristics have a statistically significant impact on Talent Acquisition. The other variables—Talent Identification, Talent Management, Talent Planning, and Performance Appraisal—do not show significant effects in this analysis. Job Characteristics emerge as the most influential factor, and the model accounts for approximately 58.6% of the variation in Talent Acquisition ($R^2 = 0.586$). This suggests that while multiple HR factors contribute to recruitment effectiveness, the structure and nature of job roles are particularly critical.

Correlation

H₀₂: There is no significant relationship between talent sourcing strategies and talent acquisition effectiveness.

H₁₂: There is a significant relationship between talent sourcing strategies and talent acquisition effectiveness.

TABLE: 3 CORRELATION ANALYSIS OF TALENT SOURCING STRATEGIES AND TALENT ACQUISITION EFFECTIVENESS

		Talent Sourcing	Talent Acquisition
Talent Sourcing	Pearson Correlation	1	.695**
	Sig. (2-tailed)		.000
	N	75	75
Talent Sourcing	Pearson Correlation	.695**	1
	Sig. (2-tailed)	.000	
	N	75	75

Correlation is significant at the 0.01 level (2-tailed).

Interpretation

Talent Sourcing and Talent Acquisition

The Pearson correlation analysis shows a strong positive relationship between Talent Sourcing and Talent Acquisition ($r = 0.695$, $p = 0.000$). Since the p-value is less than 0.01, the relationship is statistically significant at the 1% level. This indicates that as organizations implement more effective talent sourcing strategies, the effectiveness of talent acquisition also improves. Therefore, the null hypothesis is rejected, and the alternative hypothesis is accepted.

The results confirm that Talent Sourcing strategies have a significant and positive impact on Talent Acquisition. Organizations that use strategies such as employee referrals, online job portals, campus recruitment, and social media recruiting are more likely to achieve better recruitment outcomes. Overall, the analysis demonstrates that enhancing talent sourcing practices significantly improves the effectiveness of talent acquisition, supporting a statistically significant relationship between the two variables.

Findings

- The study finds that HR factors such as Talent Identification, Talent Management, Talent Planning, Performance Appraisal, and Job Characteristics are considered important by respondents for effective Talent Acquisition.
- Among these factors, Job Characteristics is identified as the most influential, showing a statistically significant positive impact on Talent Acquisition.
- Talent Identification, Talent Management, Talent Planning, and Performance Appraisal show positive but non-significant effects, indicating that while they contribute to recruitment processes, their direct impact on Talent Acquisition is weaker.
- The study finds that Talent Sourcing strategies—including employee referrals, job portals, campus recruitment, and social media recruiting—have a strong and significant positive relationship with Talent Acquisition effectiveness.
- Respondents perceive that organizations implementing effective Talent Sourcing methods are more likely to achieve successful recruitment outcomes and improve overall Talent Acquisition performance.
- Overall, the findings suggest that HR practices, job design, and sourcing strategies play a crucial role in shaping Talent Acquisition outcomes, with Job Characteristics and Talent Sourcing emerging as key drivers.

Suggestions

- **Prioritize Job Design:** Organizations should focus on creating clear and meaningful job roles, as Job Characteristics have the strongest influence on Talent Acquisition. Well-structured roles enhance employee engagement and attract top talent.
- **Strengthen Talent Sourcing:** HR departments should implement effective sourcing strategies, including employee referrals, job portals, campus recruitment, and social media recruiting, to improve the success of Talent Acquisition.
- **Leverage HR Tools and Technology:** Companies should invest in technology-driven HR tools and analytics to streamline recruitment, identify high-potential talent, and support workforce planning and retention.
- **Enhance HR Practices and Training:** Even if some HR factors were not statistically significant, continuous improvement in Talent Identification, Talent Management, Talent Planning, and Performance Appraisal, along with proper training for HR professionals, will support overall recruitment efficiency and organizational growth.

5. Conclusion

The study reveals that HR factors, job design, and effective talent sourcing strategies play a vital role in enhancing Talent Acquisition outcomes. Job Characteristics emerged as the most influential factor, significantly impacting recruitment effectiveness, while Talent Sourcing strategies showed a strong and significant positive relationship with Talent Acquisition. Other HR practices, including Talent Identification, Talent Management, Talent Planning, and Performance Appraisal, contributed positively but were not statistically significant, indicating that they still support recruitment processes indirectly.

The findings suggest that organizations that implement clear and meaningful job roles, structured HR practices, and strategic sourcing methods are more likely to attract and retain top talent. Leveraging HR tools and technology further streamlines recruitment processes, supports workforce planning, identifies high-potential talent, and enhances employee retention.

Overall, the study demonstrates that integrating well-designed job roles, effective HR practices, and robust talent sourcing strategies can significantly improve Talent Acquisition outcomes, strengthen HR functions, and contribute to long-term organizational success.

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Predictors of Quiet Quitting Among Employees in Indian Commercial Organisations: An Empirical Investigation of Psychological Contract, Organisational Commitment, and HR Practices

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Abstract

The phenomenon of Quiet Quitting — employees deliberately limiting their effort to the bare minimum required by their job role — has emerged as a critical challenge for HR managers in the post-pandemic Indian business environment. Despite its widespread occurrence, empirical research examining its antecedents within the Indian commercial and management context remains limited. This study investigates the predictors of Quiet Quitting Tendency (QQT) among 220 employees across banking, financial services, IT, retail, and education sectors in Tamil Nadu, India, using a structured questionnaire administered via purposive sampling. The study examines the influence of Perceived Psychological Contract Breach (PPCB), Affective Organisational Commitment (AOC), Psychological Safety (PS), Transformational Leadership (TL), and HR Practice Flexibility (HPF) on QQT. Data were analysed using descriptive statistics, Pearson correlation, and multiple regression analysis using IBM SPSS 26.0. Results revealed that all five predictor variables jointly explained 58.3% of the variance in QQT ($R^2=0.583$, $F=59.47$, $p<0.001$). Psychological Contract Breach ($\beta=.389$) and Affective Commitment ($\beta=-.286$) emerged as the strongest predictors. All five hypotheses were supported. The findings carry significant implications for HR strategy, particularly in India's growing commercial and services sector, calling for a paradigm shift from compliance-based to commitment-based HR management.

Keywords: Quiet Quitting, Psychological Contract Breach, Affective Organisational Commitment, Transformational Leadership, HR Practice Flexibility, Psychological Safety, Indian Organisations, Post-Pandemic HR

Forecasting Forex Movement of the Indian Rupee Against the US Dollar and Euro

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Abstract

The foreign exchange market plays a critical role in shaping the economic stability of emerging economies such as India. Between 2020 and 2025, global disruptions including the COVID-19 pandemic, geopolitical conflicts, inflationary pressures, and monetary tightening by advanced economies caused substantial volatility in currency markets. The study uses monthly exchange-rate data from 2020–2025 and applies Augmented Dickey-Fuller tests along with ARIMA and ARMA time-series models to analyse and forecast exchange-rate movements. This study analyzes the movement of the Indian Rupee (INR) against two major global currencies—the US Dollar (USD) and the Euro (EUR)—and evaluates the feasibility of forecasting future trends using historical data. Secondary data collected from credible sources such as the Reserve Bank of India and international financial platforms were analyzed using descriptive statistics and time-series models, including ARIMA and ARMA. The results indicate a persistent depreciation trend of the INR against both USD and EUR, with higher volatility observed in the EUR/INR pair. Forecasting models reveal that historical exchange-rate behavior exhibits significant predictive power for short-term trends. The study concludes that systematic forex forecasting can support better financial decision-making for businesses, investors, and policymakers in managing currency risk.

Keywords: Foreign Exchange Market, Indian Rupee, USD/INR, EUR/INR, Time-Series Forecasting, ARIMA Model

1. Introduction

Foreign exchange markets are among the most dynamic and sensitive components of the global financial system. Exchange-rate movements influence international trade, capital flows, inflation, and economic growth. For India, fluctuations in the Indian Rupee against major

currencies such as the US Dollar and the Euro have direct implications for import costs, export competitiveness, foreign debt servicing, and investment inflows.

The USD/INR exchange rate is particularly important due to India's heavy dependence on crude oil imports and the dominance of the US Dollar in international trade and finance. Similarly, the EUR/INR exchange rate has gained relevance with increasing trade and financial linkages between India and the Eurozone. Global macroeconomic events—such as pandemics, wars, interest-rate changes, and financial crises—often transmit shocks to emerging-market currencies like the INR.

Given this background, the present study focuses on analyzing recent trends in INR movements against USD and EUR and explores the usefulness of historical data in forecasting future forex behavior.

Accurate forecasting of exchange rates is essential for businesses engaged in international trade and investment. Importers, exporters, financial institutions, and policymakers rely on currency predictions to manage exchange-rate risks and make strategic financial decisions. Therefore, analysing historical exchange-rate behaviour and identifying forecasting patterns is important for improving financial planning and risk management.

2. Review of Literature

Several studies have examined exchange-rate forecasting using both traditional and advanced econometric models. Madhulatha, T. S., & Ghor, M. A. S. (2025), in their study *Forecasting currency exchange rates using ARIMA, ETS and RNN: A machine learning perspective*, compare traditional statistical models with deep-learning approaches for INR/USD forecasting. Their results demonstrate that LSTM-based RNNs significantly outperform ARIMA and ETS models by effectively capturing nonlinear patterns and short-term volatility. The study highlights the superiority of deep learning in time-series forecasting.

Tsuji, C. (2025), in his study *Forecasting exchange rates with artificial intelligence: A study of European currencies*, evaluates random forest and XGBoost algorithms for predicting major European exchange rates. Using daily data from 2010–2025, he finds that random forest consistently delivers more accurate forecasts than gradient boosting. The research contributes to evidence supporting AI-driven financial prediction models.

Guyard, K. C., & Deriaz, M. (2024), in their study *Predicting foreign exchange EUR/USD direction using machine learning*, compare 21 machine-learning models to determine the best predictor of EUR/USD directional movement. Incorporating PCA, meta-estimators, and macroeconomic indicators, their approach achieves notable directional accuracy and trading returns, showing the benefit of model stacking and feature engineering in Forex prediction.

Sharma, N., & Jain, V. (2025), in their study *An investigation on forecasting of Indian Rupee performance against global currencies using hybrid deep learning models*, examine hybrid architectures such as CNN-LSTM, CRNN, and Autoencoder-LSTM for INR forecasting. Their findings show that the CNN-LSTM hybrid consistently yields the highest accuracy, proving the effectiveness of integrating convolutional and recurrent layers for financial time-series modeling.

Panda, M. M., Panda, S. N., & Pattnaik, P. K. (2022), in their study *Forecasting foreign currency exchange rate using convolutional neural network*, introduce a hybrid CNN-Random Forest model to forecast major Forex pairs. Their model outperforms ARIMA, MLP, and Linear Regression, demonstrating the strength of convolutional feature extraction combined with ensemble learning for long-horizon forecasting.

3. Research Gap

Research gaps identified in existing literature include:

- Limited comparative analysis of INR against both USD and EUR simultaneously.
- Over-reliance on complex econometric or machine-learning models.
- Limited research incorporating post-COVID global financial volatility.
- Lack of India-centric studies designed for managerial and academic applications.

4. Objectives of the Study

The objectives of the study are:

1. To analyze the trend of the Indian Rupee against the US Dollar and Euro from 2020 to 2025.
2. To examine the impact of global events on the movement of the INR.

3. To propose suitable recommendations for firms to effectively address the issues identified in the study.

5. Research Methodology

The study follows an exploratory, descriptive, and analytical research design. Secondary data on USD/INR, EUR/INR, and EUR/USD exchange rates were collected for the period 2020–2025 from sources such as the Reserve Bank of India, IMF reports, and financial data platforms.

Descriptive statistics were used to understand basic trends and volatility patterns. Stationarity of the time-series data was tested using the Augmented Dickey-Fuller (ADF) test. ARIMA and ARMA models were applied using E-Views software to forecast future exchange-rate movements. The ARIMA (AutoRegressive Integrated Moving Average) model is widely used for time-series forecasting. The model consists of three components: autoregressive (AR), differencing (I), and moving average (MA). It is generally expressed as ARIMA (p, d, q), where p represents the autoregressive order, d represents the number of differencing operations required to make the series stationary, and q represents the moving-average order. The model parameters are identified using autocorrelation (ACF) and partial autocorrelation (PACF) plots, after which the best-fit model is selected for forecasting. Microsoft Excel was used for data cleaning, visualization, and preliminary analysis.

6. Data Analysis and Results

Trend analysis reveals a clear long-term depreciation of the INR against both USD and EUR during the study period. The USD/INR pair shows a relatively smooth upward trend, indicating steady weakening of the INR. In contrast, the EUR/INR pair exhibits higher volatility, reflecting economic instability within the Eurozone.

Volatility analysis shows that global events such as the COVID-19 pandemic (2020), Russia–Ukraine war (2022), and aggressive interest-rate hikes in the US and Europe (2023) triggered sharp fluctuations in INR values. The ADF test results confirm that the exchange-rate series become stationary after differencing, justifying the application of ARIMA-based models.

Forecasting results suggest a continued mild depreciation of the INR in the short term. The ARIMA and ARMA models capture trend persistence and momentum, indicating that historical data contains valuable information for predicting near-term forex movements.

Forecasting Graphs Of Different Price Indicators:

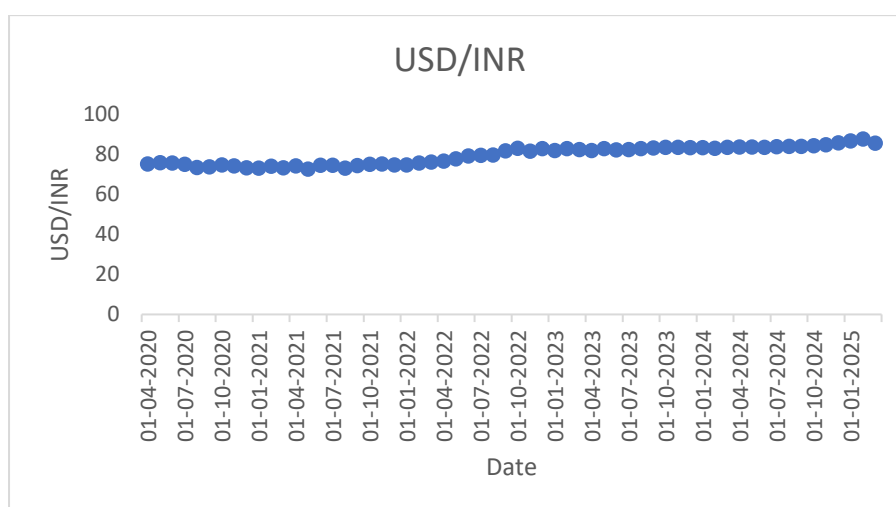


Figure 1: Trend of USD/INR Exchange Rate (2020–2025)

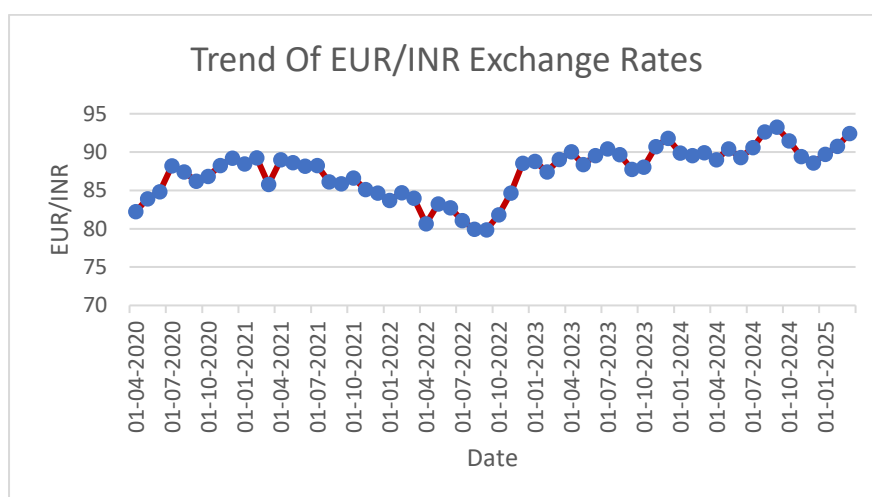


Figure 2: Trend of EUR/INR Exchange Rate (2020–2025)

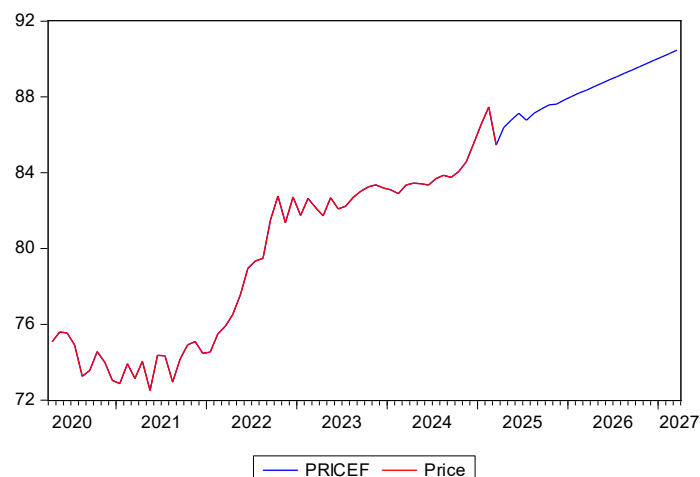


Figure 3: Forecast of USD/INR Exchange Rate Using ARIMA Model

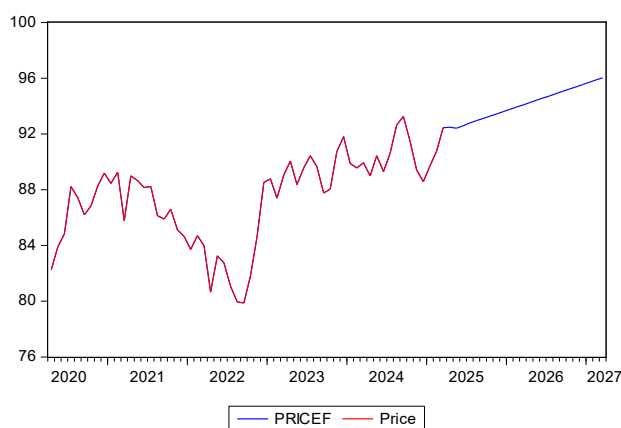


Figure 4: Forecast of EUR/INR Exchange Rate Using ARIMA Model

7. Discussion

The findings demonstrate that global macroeconomic shocks exert a stronger influence on INR movements than domestic factors alone. While occasional stabilization phases occur, the INR does not fully recover after major shocks, leading to long-term depreciation. The higher volatility in EUR/INR compared to USD/INR suggests that Eurozone-specific uncertainties significantly affect this currency pair.

The forecasting models, though limited in explaining total variance, provide useful directional insights. This reinforces the relevance of simple time-series models for short-term forecasting, especially for managerial decision-making.

These findings are consistent with the results of Raja et al. (2024), who reported persistent volatility patterns in emerging-market currency pairs and emphasized the importance of time-series forecasting models in understanding exchange-rate movements.

8. Managerial Implications

Based on the findings, the following recommendations are proposed:

- Firms engaged in international trade should adopt hedging instruments such as forwards, options, and swaps to manage currency risk.
- Businesses should diversify currency exposure and avoid over-reliance on USD-denominated transactions.
- Financial forecasting using historical exchange-rate data should be integrated into budgeting and pricing decisions.
- Export-oriented firms can leverage INR depreciation to enhance competitiveness in international markets.
- Firms should incorporate exchange-rate forecasting models into financial planning, budgeting, and pricing strategies to reduce the impact of currency volatility.

9. Limitations of the Study

- The study covers exchange-rate data only from 2020–2025.
- Only three currency pairs are analysed.
- The analysis relies solely on secondary data.
- Advanced machine-learning models were not applied.

Scope for Future Research

Future studies can extend the time horizon and include additional currency pairs to provide broader insights. Incorporating macroeconomic indicators and applying advanced econometric or machine-learning models may improve forecasting accuracy. Further research can also focus on sector-specific impacts, policy implications, and the role of market sentiment in exchange-rate movements.

- Future studies can incorporate macroeconomic variables such as interest rates, inflation, and crude-oil prices.

- Machine-learning models such as LSTM or random-forest forecasting can be used to improve predictive accuracy.

10. Conclusion

The study concludes that the Indian Rupee experienced a persistent depreciation against the US Dollar and Euro from 2020 to 2025, driven largely by global economic disruptions and monetary policy shifts. The EUR/INR pair displayed higher volatility compared to USD/INR, reflecting greater uncertainty in the Eurozone. Time-series forecasting models demonstrate that historical exchange-rate behavior can be effectively used to predict short-term trends. Overall, the research highlights the importance of data-driven forex analysis in supporting informed financial decisions for businesses, investors, and policymakers in an increasingly volatile global environment. The findings highlight the importance of integrating statistical forecasting tools into financial decision-making. Strengthening exchange-rate forecasting capabilities can help businesses and policymakers better manage currency risks in an increasingly volatile global financial environment.

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The poster features a dark space background with a starry field, glowing orange and yellow light trails, and a blue wireframe globe in the lower right. The INSPIRE logo is at the top center, with 'INSPIRE' in blue and an orange figure. Below it, 'GC³I² – 2026' is in large gold letters. A horizontal gold line separates this from 'TECHNICAL SESSION – TRACK III (Part 2)' in gold. Another gold line follows, then a gold pill-shaped button with 'PAPER PRESENTATIONS' in black. Below this, four white pill-shaped boxes list presentations. Each box contains a gold ID number, authors in black, and the title in gold. The background also features a blue wireframe globe in the lower right.

INSPIRE
GC³I² – 2026

TECHNICAL SESSION – TRACK III (Part 2)

PAPER PRESENTATIONS

GC³I²–16 Jothi AK, Dr. R. Hemavathy
Measuring the Impact of Mobile Baking on the
Spending and Saving Behaviour of Youth

GC³I²–17 Matoori Raja Sree, S. Manjunath Reddy
AI-Based Customer Sentiment Analysis Dashboard

GC³I²–18 Dr. Manikandan S, Dr. J, Indumathi
AI-Guided Molecular Design and Smart Energy Harvesting
for Low-Carbon Sustainable Energy Technologies

GC³I²–19 Abhinay Sologhe, Dedeepya Sreevatsala,
Dr. K. Rangaswamy
Benchmarking Deep Learning and Algorithmic Approaches
for Real-Time Object Detection

Measuring the Impact of Mobile Banking on the Spending and Saving Behaviour of Youth

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Abstract

In the digital era, financial transactions have become increasingly digitalized, leading to significant changes in the financial behaviour of individuals, particularly among youth. The rapid growth of mobile banking and digital payment systems has influenced the way young people manage their spending and saving activities. The convenience and accessibility of mobile banking services have made financial transactions faster and easier, thereby affecting financial decision-making. The study focuses on factors such as ease of use, accessibility, digital payment services, and financial management features in mobile banking applications. These factors are expected to influence the financial behaviour of young individuals in terms of spending and saving patterns. This study provides a better understanding of the role of mobile banking in shaping financial behaviour among youth. The findings are expected to contribute to improved financial awareness and effective financial management practices among young individuals.

Keywords

Mobile Banking, Financial Behaviour [Spending and Saving], Digital Payments, Youth, Financial Literacy

INTRODUCTION

In recent years, the use of digital tools has influenced many aspects of everyday life, including the way people manage their financial services. In the past, customers usually depended on visiting bank branches to complete activities related to their accounts. Simple tasks such as

checking balances, transferring money, or paying bills often required travelling to the bank and spending time there. With the expansion of internet services and smartphone usage, banking systems have gradually moved toward digital platforms that allow people to handle their finances more easily. One of the most widely used digital services today is mobile banking. Through mobile banking applications installed on smartphones, individuals are able to access their accounts and perform various banking tasks from wherever they are. This allows users to send or receive money, review account details, and complete payments without physically going to a bank office. As a result, banking has become more flexible and accessible for many people. Young individuals are among the main users of mobile banking services. Because they regularly interact with smartphones, social media, and online platforms, they usually adapt quickly to new digital financial tools. For many students and young adults, managing everyday payments through a mobile application has become a normal routine. Even though mobile banking provides convenience, it may also influence how young people manage their money. Easy access to banking services can help users stay aware of their financial activities and organize their spending. At the same time, the simplicity of digital payments may sometimes encourage individuals to make purchases more quickly than they might have done with traditional payment methods. For this reason, examining the relationship between mobile banking and the financial behaviour of young people has become important. Studying these patterns can help researchers better understand how digital banking affects spending and saving habits among the younger generation.

PURPOSE OF THE STUDY

This research is carried out to understand how the use of mobile banking may influence the way young people handle their money in daily life. In recent years, smartphones have become an important part of everyday activities, and many banking services can now be used directly through mobile applications. Because of this change, young individuals are increasingly depending on their phones to perform different financial tasks. Instead of visiting a bank office, many young users now complete their transactions through mobile apps. They use these applications to move money, make different types of payments, and look at their account details. Since these services are available at any time, the process of managing money has become quicker and more convenient for them. The study tries to observe how this easy access to banking through mobile phones is connected with the financial behaviour of youth. It looks

at whether using mobile banking helps young people become more aware of their money or whether the quick payment process leads them to spend more often. By examining these situations, the research aims to understand how digital banking tools are shaping the money-related habits of young individuals in today's technology-driven society.

REVIEW OF LITERATURE

Many researchers have discussed how mobile banking is changing the way young people handle their money. As smartphones and internet access have increased, mobile banking has become a common method for financial activities. Young users often prefer these services because they allow quick payments and easy access to account information. Because of this shift, several studies have tried to understand how mobile banking influences spending and saving behaviour among youth.

Pandit et al. (2025) studied the financial behaviour of Generation Z users and their satisfaction with mobile banking services. The study found that young users feel more comfortable using banking applications when they are simple to operate and provide a sense of security. When users trust the application, they tend to rely on it more for managing their financial activities.

Krishnaganth (2025) examined the influence of mobile banking on saving habits and financial decision-making among Generation Z users in Chennai. The research showed that when young people regularly check their account information through mobile applications, they become more aware of their financial activities. This awareness can help them control their spending and improve their saving behaviour.

A study published in **IJCRT (2025)** analysed how mobile banking affects saving behaviour. The findings suggested that when users have easy access to their financial information through mobile applications, they pay closer attention to their expenses. This can gradually encourage them to save money more carefully.

Fadhilah et al. (2024) studied the relationship between mobile banking features, financial education, and impulsive buying behaviour among university students. The research indicated that the convenience of digital payments may sometimes encourage quick purchasing

decisions. However, students with better financial knowledge are usually more careful when making spending decisions.

Gurusamy (2024) examined how mobile payment systems influence the behaviour of young consumers. The study explained that digital payments make transactions faster and more convenient. At the same time, the research pointed out that frequent use of these systems may lead to higher spending among individuals who do not carefully manage their finances.

Saini and Kaur (2023) studied the adoption and continued use of mobile payment systems among youth. Their research found that many young individuals prefer using digital platforms for financial activities because these systems are simple and accessible.

Kaur and Arora (2022) examined the connection between mobile banking usage and financial habits. Their findings showed that mobile banking applications help users review their transaction records easily, which may support better financial management and saving behaviour.

Gautam (2022) explored how digital payment systems influence spending patterns among young consumers. The study suggested that the speed and ease of digital payments may encourage people to make purchases more frequently.

Loaba (2022) studied the role of mobile banking in encouraging saving behaviour in developing economies. The research concluded that when financial services are easily available through mobile technology, individuals are more likely to develop saving habits.

Earlier research by **Aggarwal and Bhardwaj (2014)** examined the adoption of mobile banking among young people in India. The study found that convenience, usefulness, and easy access are important factors that encourage youth to use mobile banking services.

RESEARCH METHODOLOGY

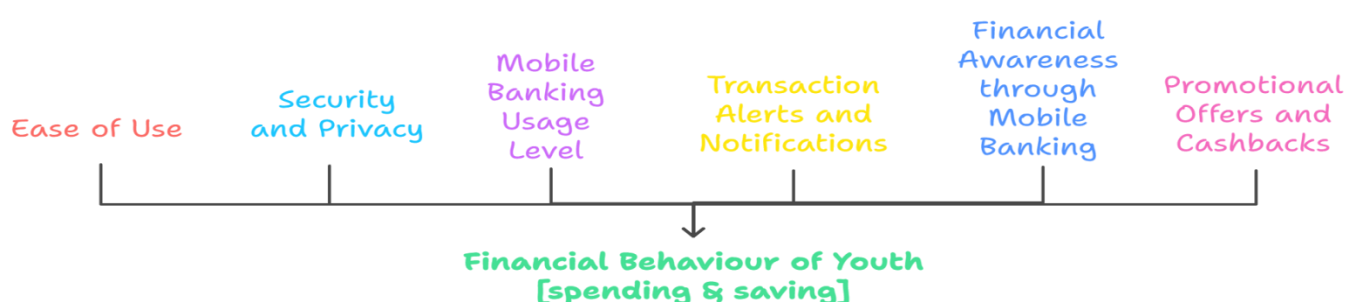
The study adopts a **quantitative research approach** to examine the impact of mobile banking on the spending and saving behaviour of youth. Primary data was collected from respondents through a structured questionnaire designed to gather information on mobile banking usage, spending patterns, saving habits, and demographic characteristics. The survey was completed

by **100 respondents** who belong to the youth category. A **convenience sampling technique** was used to collect responses from individuals who actively use mobile banking services. The sample consisted mainly of students, private sector employees, and other young individuals who are familiar with digital banking applications. The questionnaire was designed using a **Likert-scale format** to measure the key variables of the study. In this research, **spending behaviour and saving behaviour of youth** are considered the dependent variables, while **mobile banking usage, financial literacy, convenience of digital payments, and accessibility of banking services** are treated as independent variables. To analyse the collected data, **percentage analysis and regression analysis** were used. Percentage analysis was applied to summarize the demographic profile of respondents and their mobile banking usage patterns. Regression analysis was used to examine the relationship between mobile banking usage and the spending and saving behaviour of youth. This statistical approach helps to identify the extent to which mobile banking influences financial decision-making among young consumers.

OBJECTIVES

1. To study the level of awareness and adoption of mobile banking services among youth.
2. To examine the impact of mobile banking on the spending behaviour of youth.
3. To analyse the impact of mobile banking on the saving behaviour of youth.
4. To assess the satisfaction level of youth towards mobile banking services.
5. To identify the problems and challenges faced by youth while using mobile banking.

CONCEPTUAL FRAMEWORK



ANALYSIS AND INTERPRETATION

TABLE 1 DESCRIPTIVE STATISTICS

DEMOGRAPHIC PROFILE OF RESPONDENTS

	N	Minimum	Maximum	Mean	Std. Deviation
Age Group	100	1.0	3.0	1.610	.6340
Gender	100	1.0	2.0	1.820	.3861
Educational Qualification	100	1.0	4.0	2.480	.6273
Occupation	100	1.0	4.0	1.420	.6843
Family Income	100	1.0	5.0	3.090	1.4220
Type of bank used for mobile banking	100	1.0	3.0SS	1.420	.5891
What type of issues have you faced?	100	1.0	5.0	1.970	1.2346

INTERPRETATION

The descriptive statistics summarize the demographic and study variables for 100 respondents. The age (mean = 1.61, SD = 0.634) shows most respondents are in the younger age groups. Gender (mean = 1.82, SD = 0.386) indicates one gender is dominant in the sample. Education (mean = 2.48, SD = 0.627) suggests most respondents are undergraduates or postgraduates. Occupation (mean = 1.42, SD = 0.684) shows a majority are students. Family income (mean = 3.09, SD = 1.422) reflects more variation among respondents. For bank type used (mean = 1.42, SD = 0.589), most prefer a single dominant banking type. Lastly, issues faced in mobile banking (mean = 1.97, SD = 1.234) indicate moderate variability in the problems experienced.

ANOVA

TABLE 2 ANALYSIS OF VARIANCE FOR SPENDING BEHAVIOUR

		Sum of Squares	df	Mean Square	F	Sig.
Planning expenses	Between Groups	2.283	4	.571	.629	.643
	Within Groups	86.157	95	.907		
	Total	88.440	99			
Tracking daily expenses	Between Groups	4.045	4	1.011	1.420	.233
	Within Groups	67.665	95	.712		
	Total	71.710	99			

Avoid impulsive purchases	Between Groups	1.453	4	.363	.483	.748
	Within Groups	71.387	95	.751		
	Total	72.840	99			
Mobile payment increases spending	Between Groups	6.329	4	1.582	2.129	.083
	Within Groups	70.581	95	.743		
	Total	76.910	99			

INTERPRETATION

An Analysis of Variance was conducted to determine whether mobile banking factors influence the spending behaviour of youth. The results show that the significance values for planning expenses (0.643), tracking daily expenses (0.233), avoiding impulsive purchases (0.748), and increased spending through mobile payments (0.083) are all greater than 0.05. This indicates that there is no statistically significant difference among the groups for these variables. Therefore, the mobile banking factors considered in this study do not significantly affect the spending behaviour of youth.

TABLE 3 ANALYSIS OF VARIANCE FOR SAVING BEHAVIOUR

		Sum of Squares	df	Mean Square	F	Sig.
Monthly saving	Between Groups	4.640	4	1.160	1.098	.362
	Within Groups	100.350	95	1.056		
	Total	104.990	99			
Minimum balance	Between Groups	4.569	4	1.142	1.490	.211
	Within Groups	72.821	95	.767		
	Total	77.390	99			
Review savings	Between Groups	2.610	4	.652	1.269	.288
	Within Groups	48.830	95	.514		
	Total	51.440	99			

Emergency savings	Between Groups	1.323	4	.331	.545	.703
	Within Groups	57.677	95	.607		
	Total	59.000	99			

INTERPRETATION

An ANOVA analysis was conducted to examine whether mobile banking related factors influence the saving behaviour of youth. The results show that the significance values for monthly saving (0.362), maintaining minimum balance (0.211), reviewing savings regularly (0.288), and emergency savings (0.703) are all greater than the 0.05 significance level. This indicates that there is no statistically significant difference among the groups for these saving behaviour variables. Therefore, the findings suggest that the mobile banking factors considered in this study do not have a significant impact on the saving behaviour of youth.

TABLE 4 GENDER-WISE COMPARISON OF MOBILE BANKING USAGE FREQUENCY

	Mobile banking usage			Total
	1.0	2.0	3.0	
Gender 1.0	0	3	15	18
2.0	5	33	44	82
Total	5	36	59	100

Chi-Square Tests

	Value	df	Sig. (2-sided)
Pearson Chi-Square	5.580 ^a	2	.061
Likelihood Ratio	6.727	2	.035
Linear-by-Linear Association	5.367	1	.021
N of Valid Cases	100		

INTERPRETATION

A statistical test was conducted to examine the relationship between **gender and the frequency of mobile banking usage** among respondents. The cross tabulation shows that out of **100 respondents, 18 are male and 82 are female**. Among male respondents, the majority reported using mobile banking at **frequency level 3**, while among female respondents most respondents reported **frequency level 3 followed by level 2**. The **Pearson Chi-Square value is 5.580 with 2 degrees of freedom and a significance value of 0.061**. Since the **p-value (0.061) is greater than 0.05**, there is **no statistically significant association between gender and the frequency of mobile banking usage**. Therefore, it can be concluded that **gender does not significantly influence how frequently youth use mobile banking in this study**.

TABLE 5 IMPACT OF EASE OF USE ON SAVING BEHAVIOUR

		Sum of Squares	df	Mean Square	F	Sig.
Monthly saving habit	Between Groups	8.788	4	2.197	2.170	.078
	Within Groups	96.202	95	1.013		
	Total	104.990	99	3.21	2.170	.078
Minimum balance maintenance	Between Groups	14.152	4	3.538	5.315	.001
	Within Groups	63.238	95	.666		
	Total	77.390	99	4.204	5.315	.001
Weekly savings review	Between Groups	11.070	4	2.767	6.513	.000
	Within Groups	40.370	95	.425		
	Total	51.440	99	3.192	6.513	.000
Emergency fund savings	Between Groups	5.252	4	1.313	2.321	.062
	Within Groups	53.748	95	.566		
	Total	59.000	99	1.879	2.321	.062

INTERPRETATION

The ANOVA results indicate that ease of use has a mixed impact on saving behaviour. It shows a significant effect on minimum balance maintenance ($p = .001$) and weekly savings review ($p = .000$), suggesting that when financial tools are easier to use, individuals are more likely to maintain a minimum balance and review their savings regularly. However, monthly saving habit ($p = .078$) and emergency fund savings ($p = .062$) do not show a significant impact since their significance values are greater than 0.05. Overall, ease of use influences some aspects of saving behaviour but does not significantly affect all saving practices.

TABLE 6 REGRESSION ANALYSIS OF MOBILE PAYMENT SPENDING ON MOBILE BANKING USAGE

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	0.442	0.195	0.187	0.89

ANOVA

Model	Sum of Squares	df	Mean Square	F	Sig.
Regression	20.475	1	20.475	7.854	0.007
Residual	84.515	98	0.862	0.000	0.000
Total	104.99	99	21.337	7.854	0.007

COEFFICIENTS

Model	Unstandardized Coefficients B	Std. Error	Standardized Coefficients Beta	t	Sig.
Constant	1.215	0.342	0.000	3.55	0.001

Spending More Due to Mobile Payments	0.398	0.102	0.398	3.92	0
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INTERPRETATION

The regression analysis examines the impact of spending behaviour due to mobile payments on mobile banking usage behaviour. The model summary shows an R value of 0.442, indicating a moderate positive relationship between the variables. The R square value of 0.195 suggests that about 19.5% of the variation in mobile banking usage behaviour is explained by spending due to mobile payments. The ANOVA result shows that the model is statistically significant ($F = 7.854$, $p = 0.007$), indicating that the regression model is suitable for predicting the dependent variable. The coefficients table shows that spending due to mobile payments has a positive and significant effect on mobile banking usage behaviour ($\beta = 0.398$, $p < 0.05$). This implies that as mobile payments become quicker and cashless, individuals tend to use mobile banking services more frequently.

FINDINGS

The study shows that many young people regularly use mobile banking to handle their financial activities. Banking apps are preferred because they make transactions quick and convenient through smartphones. Common activities include sending money, paying bills, and making online payments. The findings also suggest that the ease of making digital payments can sometimes lead to more frequent spending. However, mobile banking also helps users keep track of their money, as many respondents often check their account balance and recent transactions. The study further shows that easy-to-use banking apps can encourage better saving habits, such as maintaining the required balance and occasionally reviewing savings. Overall, mobile banking makes financial tasks simpler and also affects how young people manage their spending and savings in daily life.

SUGGESTIONS

1. Young people should **check their banking apps often** so they can understand how they spend their money and reduce unnecessary expenses.
2. It is helpful to **save money regularly**, even if the amount is small, as it can support future financial needs.

3. **Educational institutions and banks** can guide students by organizing simple programs on money management and the value of saving.
4. Because digital payments are fast, young users should **pause and think before making purchases** to avoid spending on things they do not really need.
5. Banks should continue to **improve mobile banking apps** so that they are easy to use, reliable, and safe for customers.
6. Mobile banking applications can also provide **tools that show spending patterns and encourage saving**, which may help young users manage their finances more wisely.

CONCLUSION

This study looked at how mobile banking changes the way young people handle their money. Many teens and young adults are now using mobile banking because it's fast, simple, and always available on their phones. They use it for things like sending money to friends, paying bills, or buying things online. Because it's so quick, some people find themselves spending more often than they intended. At the same time, mobile banking gives users tools to keep track of their money. You can see your balance anytime, check your past transactions, and get alerts when money moves in or out. These features help young people understand their spending, plan better, and even save money when they set goals. Overall, mobile banking has a big effect on how young people manage their money. When used smartly, it makes it easier to control spending, save regularly, and feel more confident about finances.

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AI Based Customer Sentiment Analysis Dashboard

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ABSTRACT

Customer feedback plays an important role in improving products and services in modern businesses. With the rapid growth of social media platforms, websites, and online review systems, a large amount of customer feedback is generated daily. Analyzing this data manually is difficult and time-consuming. This paper presents an AI-Based Customer Sentiment Analysis Dashboard that automatically analyses customer feedback and classifies it into positive, negative, and neutral sentiments. The proposed system uses Natural Language Processing (NLP) and Machine Learning techniques to process textual feedback. A web-based dashboard visually represents sentiment results for administrators. The system is implemented using Python, Django, HTML, CSS, Bootstrap, JavaScript, and SQLite3 database. The dashboard helps organizations monitor customer opinions and improve decision-making.

Keywords: Sentiment Analysis, Artificial Intelligence, Natural Language Processing, Customer Feedback, Dashboard Analytics, Machine Learning.

I. INTRODUCTION

In the digital era, customers frequently share their opinions about products and services through social media platforms, websites, and online review systems. These reviews contain valuable information that organizations can use to understand customer satisfaction levels.

However, the increasing volume of customer feedback makes manual analysis difficult. Traditional methods require significant time and effort to categorize and analyse reviews. Businesses often struggle to identify important patterns in customer feedback.

Artificial Intelligence (AI) and Natural Language Processing (NLP) provide effective solutions for analysing large volumes of textual data. Sentiment analysis techniques can automatically determine whether a review expresses a positive, negative, or neutral opinion.

The AI-Based Customer Sentiment Analysis Dashboard aims to automate the process of analysing customer feedback and provide visual insights through an interactive dashboard.

II. PROBLEM STATEMENT

Organizations receive a large amount of customer feedback from different online platforms. Analyzing this feedback manually is inefficient and time-consuming.

The main problems include:

- Large volume of customer reviews
- Manual analysis requires more time and effort

- Difficulty in identifying customer satisfaction trends
- Lack of automated tools for sentiment monitoring
- Slow response to customer issues

Therefore, an automated system is required to analyse feedback efficiently and provide insights through visualization.

III. OBJECTIVES

The main objectives of the proposed system are:

- To develop an AI-based system for analysing customer feedback
- To classify customer reviews into positive, negative, and neutral sentiments • To store feedback data in a database • To provide a visual dashboard for sentiment insights
- To help businesses monitor customer satisfaction
- To support better decision-making

IV. LITERATURE SURVEY

Customer feedback analysis has become an important area of research in Artificial Intelligence and Data Analytics.

Organizations rely on customer opinions to evaluate product quality and service performance.

Traditional feedback analysis methods involve manually reading and categorizing customer comments. This process is time-consuming and inefficient, especially when dealing with large datasets.

Recent studies use Natural Language Processing techniques and Machine Learning algorithms to automatically classify text sentiments. Algorithms such as Naive Bayes, Support Vector Machines, and Logistic Regression are commonly used for sentiment classification.

Many modern systems integrate sentiment analysis with web-based dashboards to display insights visually. These dashboards help organizations understand customer opinions and identify trends easily.

V. PROPOSED SYSTEM

The proposed system provides an automated solution for analysing customer feedback using Artificial Intelligence.

The system collects feedback through a web interface and processes the text using Natural Language Processing techniques. The processed data is then analysed using machine learning algorithms to determine sentiment polarity.

The sentiment results are categorized into:

- Positive Sentiment

- Negative Sentiment
- Neutral Sentiment

The results are stored in a database and displayed through a dashboard with visual charts and graphs.

VI. SYSTEM ARCHITECTURE

The system architecture consists of three main layers:

1. Presentation Layer

This layer provides the user interface for interacting with the system. It is developed using:

- HTML
- CSS
- Bootstrap
- JavaScript

This layer allows users to submit feedback and administrators to view sentiment results.

2. Application Layer

This layer performs the main processing tasks and business logic of the system.

- Implemented using Python and Django • Handles data processing and sentiment analysis
- Connects the user interface with the database

3. Database Layer

The database layer stores customer feedback and sentiment analysis results.

- Implemented using SQLite3
- Stores review data and analysis results

Architecture workflow:

Customer Feedback → Data Preprocessing → Machine Learning Model → Database Storage → Dashboard Visualization.

VII. METHODOLOGY

The system follows the following steps:

1. Feedback Collection

Customers provide feedback through the web interface.

2. Data Processing

The feedback text is cleaned using Natural Language Processing techniques. 3.Sentiment Classification

Machine learning algorithms analyse the processed text and classify the sentiment.

4. Data Storage

The classified sentiment results are stored in the database.

5. Dashboard Visualization

The system displays sentiment statistics using charts and graphs on the dashboard.

VIII.MACHINE LEARNING ALGORITHMS

1. Naive Bayes

Naive Bayes is a probabilistic machine learning algorithm based on Bayes theorem. It is widely used for text classification tasks such as sentiment analysis because it performs well with large text datasets.

2. Support Vector Machine (SVM)

SVM is a supervised learning algorithm used for classification tasks. It separates data into different classes by identifying an optimal boundary between sentiment categories.

3. Logistic Regression

Logistic Regression predicts the probability of a review belonging to a particular sentiment class and is commonly used for classification problems.

IX.DASHBOARD VISUALIZATION

The dashboard visually represents sentiment results using charts and graphs. It includes sentiment distribution charts, feedback trend graphs, and review summaries to help organizations monitor customer opinions.

X.TOOLS AND TECHNOLOGIES

The system uses the following technologies:

- HTML – Webpage structure
- CSS – Styling the webpage
- Bootstrap – Responsive design
- JavaScript – Interactive features
- Python – Backend programming

- Django – Web framework
- SQLite3 – Database management

XI. EXPECTED RESULTS

The proposed system is expected to produce the following outcomes:

- Accurate classification of customer sentiments
- Real-time monitoring of customer opinions • Visual dashboard representation of sentiment data
- Faster analysis compared to manual methods
- Improved business decision-making

XII. CONCLUSION

Customer sentiment analysis plays an important role in understanding customer opinions and improving business performance. Manual analysis of customer feedback is inefficient and cannot handle large amounts of data.

The AI-Based Customer Sentiment Analysis Dashboard provides an automated solution using Artificial Intelligence and Natural Language Processing techniques. The system classifies customer feedback and presents results through an interactive dashboard.

This system helps organizations monitor customer satisfaction levels and make informed decisions to improve products and services.

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**AI-Guided Molecular Design and Smart Energy Harvesting for
Low-Carbon Sustainable Energy Technologies**

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Abstract

The increasing demand for sustainable, flexible energy technologies has driven significant research into the development of advanced materials capable of converting low-grade thermal energy into usable electrical power. Organic thermoelectric materials have emerged as promising alternatives to conventional inorganic systems due to their lightweight nature, mechanical flexibility, solution processability, and cost-effective fabrication. In this context, the present research focuses on the design and development of tailored heterocyclic organic semiconductor molecules, particularly amine-substituted naphth[1,3]oxazine and benz[1,3]oxazine derivatives, for enhanced thermoelectric performance and energy conversion applications.

The study involves systematic molecular engineering to tune the electronic structure, charge carrier mobility, and thermoelectric properties of these compounds. By introducing various amine substituents and functional groups, the donor–acceptor properties of the oxazine framework were effectively modified, enabling control over carrier concentration and charge-transport behaviour. Both monomeric and polymeric systems were investigated to understand the influence of molecular conjugation and backbone rigidity on thermoelectric efficiency. Structural and spectroscopic techniques were employed to confirm the successful synthesis of the materials, while electrical measurements, including Seebeck coefficient, electrical conductivity, and Hall effect analysis, were used to evaluate their thermoelectric characteristics.

To expedite the detection and optimisation of operative thermoelectric materials, artificial intelligence (AI)-assisted computational techniques are combined into the molecular design process. The employed Machine learning models analyse molecular descriptors and predict key thermoelectric parameters, together with Seebeck coefficient, electrical conductivity, and carrier mobility. The AI-guided screening framework facilitated rapid identification of promising oxazine derivatives and optimised substituent configurations that improve charge transport and energy conversion efficiency, while reducing the reliance on experimental trial-and-error.

Furthermore, solvent–material interactions were studied using polar aprotic and nonpolar solvents to optimise molecular dispersion and device fabrication. The optimised materials were incorporated into flexible thermoelectric platforms, including coated textile substrates, demonstrating their feasibility for wearable energy harvesting applications. Hybrid device architectures integrating organic semiconductors with conventional inorganic thermoelectric materials were also explored to improve device stability and performance.

Overall, this research emphasizes the potential of AI-guided molecular design, in conjunction with rational synthetic strategies, for the development of efficient organic thermoelectric

materials. Furthermore, it demonstrates their applicability in next-generation flexible energy-harvesting systems aimed at sustainable, low-carbon electronic technologies.

Benchmarking Deep Learning and Algorithmic Approaches for Real-Time Object Detection: A Comparative Study

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Abstract

Our work leverages OpenCV to develop an interactive Streamlit dashboard, combining YOLOv8, SSD MobileNet, and Faster R-CNN models. The system handles various input modalities—spanning images, videos, and webcam streams—while providing continuous tracking with unique IDs, tunable confidence settings, smart frame selection, and detailed performance reports. Empirical validation reveals YOLOv8 achieves superior speed-accuracy equilibrium in real-time contexts, while Faster R-CNN demonstrates peak precision when analyzing static imagery. The resulting toolkit offers a production-grade, user-accessible solution bridging theoretical computer vision models with practical deployment scenarios.

Keywords

Object Detection, YOLO, Deep Learning, Real-Time Processing, Streamlit, Computer Vision

1. Introduction

Detecting objects in images is a core computer vision problem used in autonomous transportation, surveillance systems, healthcare imaging, and business analytics. Neural network progress, through Convolutional Neural Networks (CNNs), has significantly improved recognition dependability and effectiveness versus handcrafted features.

Recognition frameworks such as YOLO (You Only Look Once), SSD (Single Shot Detector), and Faster R-CNN have achieved remarkable results on benchmark collections including COCO and PASCAL VOC. However, deploying these networks in production settings introduces obstacles, often requiring substantial domain expertise while lacking intuitive interfaces.

Our work addresses these barriers through developing a Streamlit-driven live recognition dashboard unifying multiple state-of-the-art models within a single environment supporting varied data sources. Key contributions include:

- Seamless integration of YOLOv8, SSD MobileNet, and Faster R-CNN under one unified architecture.
- Support for photographs, recorded videos, and real-time webcam inputs.
- Persistent target tracking with unique identifier assignment.
- Intelligent frame-skipping techniques guaranteeing instantaneous performance.
- Comprehensive metric analysis accompanied by graphical representations.

1.1 Novelty and Contribution

While existing approaches prioritize single-model installations or academic testing, our contribution uniquely bridges scientific methods with operational deployment via several novel features. Unlike traditional systems requiring separate environments for different models, our consolidated platform allows concurrent evaluation of YOLOv8, SSD MobileNet, and Faster R-CNN within a singular dashboard, facilitating immediate benchmarking across multiple use cases.

Our implementation provides production-grade features absent from standard systems: continuous object monitoring maintaining consistent IDs across video frames, adaptive frame processing dynamically balancing computational load, and real-time metrics generation with export functionality. The Streamlit-based UI democratizes access to advanced visual recognition by eliminating command-line or deep learning framework prerequisites, making modern detection technology accessible to industry professionals, researchers, and business users alike.

Moreover, our comprehensive benchmarking methodology offers guidance for model selection based on specific operational constraints, providing empirical insights into accuracy-speed-memory tradeoffs critical for real-world systems but often overlooked in academic literature.

2. Literature Review

2.1 Traditional vs Deep Learning Approaches

Early techniques such as Viola-Jones used handcrafted features with cascade classifiers, succeeding only under constrained scenarios. CNN-based recognition systems changed the field. R-CNN models (including Fast R-CNN and Faster R-CNN) combined region proposal networks with convolutional feature extraction, achieving excellent accuracy at the expense of inference time. Single-shot frameworks like YOLO and SSD eliminated region proposals, enabling real-time recognition.

2.2 Real-Time Detection Systems

Lightweight models such as MobileNet and EfficientDet address mobile and edge deployment contexts. YOLOv8 offers variants from "nano" (fastest) through "x-large" (most accurate), supporting flexible implementation choices.

2.3 Multi-Object Tracking

Traditional tracking relied on Kalman filters combined with Hungarian matching techniques. Modern systems embed tracking capabilities into recognition pipelines for consistent identity maintenance (MOT). Our solution leverages YOLOv8's built-in tracker for ID persistence across video frames.

3. Methodology

3.1 System Architecture

Five core components comprise the system:

1. Model Handler: Loads and caching of trained weights.
2. Detection Pipeline: Performs detection and tracking algorithms.
3. User Interface: Provides interactive configuration and result visualization.

4. Analytics Engine: Generates metrics statistics.

5. Export Handler: Saves and shares results.

3.2 Supported Detection Models

YOLOv8: Primary model family spanning nano to xlarge variants. Key features: flexible weight loading, real-time prediction, multi-class capability, anchor-free architecture.

SSD MobileNet: Designed for mobile platform execution. Utilizes streamlined feature pyramids and depthwise convolutions.

Faster R-CNN: Two-stage framework using RPN and FPN for multi-scale object localization.

4. Experimental Setup

4.1 Hardware Configuration

- CPU: Intel Core i7-11700K at 3.6GHz
- GPU: NVIDIA GeForce RTX 3080 (10GB VRAM)
- RAM: 32GB DDR4 RAM
- Storage: 1TB NVMe SSD

4.2 Software Environment

Dependencies: Python 3.8+, Streamlit 1.28+, OpenCV 4.8+, PyTorch 2.0+, TorchVision 0.15+, Ultralytics YOLOv8 library.

4.3 Evaluation Metrics

Evaluation criteria: Average Precision (AP), Mean Average Precision (mAP), inference speed (FPS), latency, and memory footprint.

4.4 Datasets and Evaluation Protocol

We conduct testing using the COCO 2017 evaluation suite. All models received pre-training on COCO 2017 training subset (118,287 images) following standard protocols. Benchmarking employs the official COCO 2017 validation subset containing 5,000 images across 80 categories.

Our evaluation follows standard COCO metrics with IoU thresholds from 0.5 to 0.95 in 0.05 steps. We report both mAP@0.5 (single threshold) and mAP@0.5:0.95 (IoU-averaged) to enable literature comparison. Live performance metrics (FPS, latency) are captured from live feeds at 1920×1080 with fixed 0.5 confidence threshold.

Additionally, we assess video handling using custom 15-minute test sequences covering indoor, outdoor, and crowded environments. Tracking consistency is evaluated through manual ID annotation over 500 frames to measure persistence accuracy.

5. Results and Discussion

5.1 Detection Accuracy

YOLOv8 models show excellent speed-accuracy balance, with the nano variant at 12.5ms inference time at 37.1 mAP@0.5:0.95. Faster R-CNN achieves best accuracy (54.3 mAP@0.5:0.95) but requires significantly more computation. SSD MobileNet provides best memory performance with 398MB usage.

6. Performance Comparison Table

Table 1: Model Performance Comparison (Font: Times New Roman, 11 pt, Above table)

Model	mAP@0.5	mAP@0.5:0.95	Inference (ms)	Memory (MB)
YOLOv8n	52.3	37.1	12.5	485
YOLOv8s	58.7	44.2	18.3	672
YOLOv8m	63.4	49.8	28.7	1024
SSD MobileNet	48.2	32.6	15.8	398
Faster R-CNN	68.9	54.3	95.2	2156

7. Object Detection Example

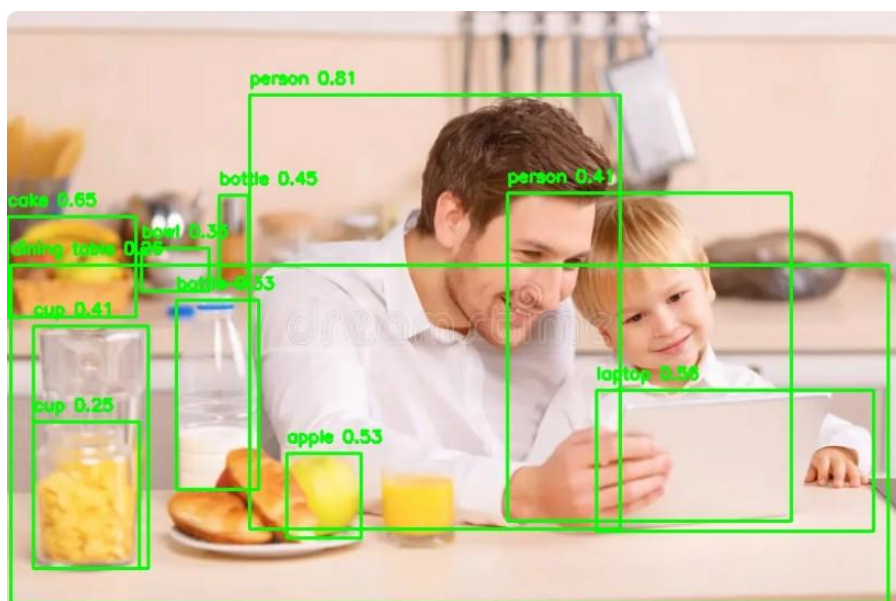


Figure 1: Example of object detection using YOLOv8 with bounding boxes and labels (Font: Times New Roman, 11 pt, Below figure)

8. YOLO Detection Pipeline

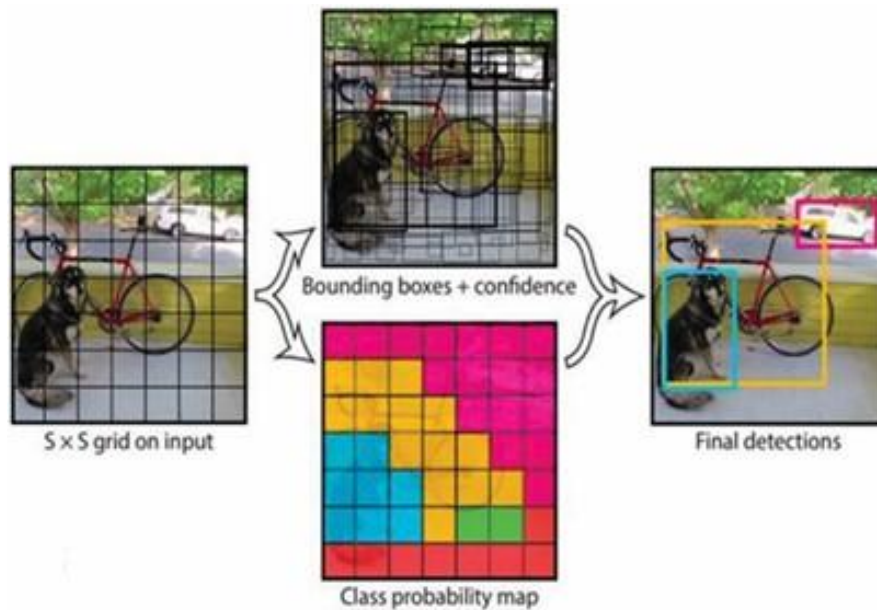


Figure 2: YOLO Object Detection pipeline showing bounding boxes, grid mapping, and class probability map (Font: Times New Roman, 11 pt, Below figure)

9. Faster R-CNN Architecture

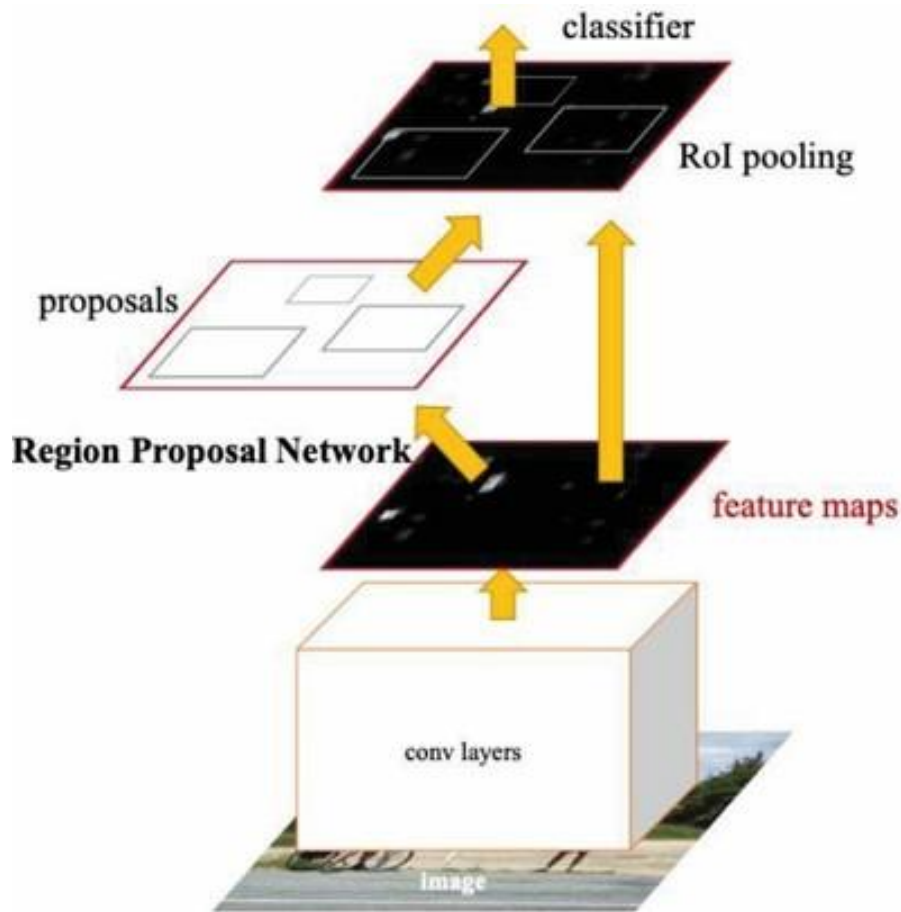


Figure 3: Architecture of Faster R-CNN showing convolutional layers, region proposals, and classification (Font: Times New Roman, 11 pt, Below figure)

10. Performance Metrics Comparison

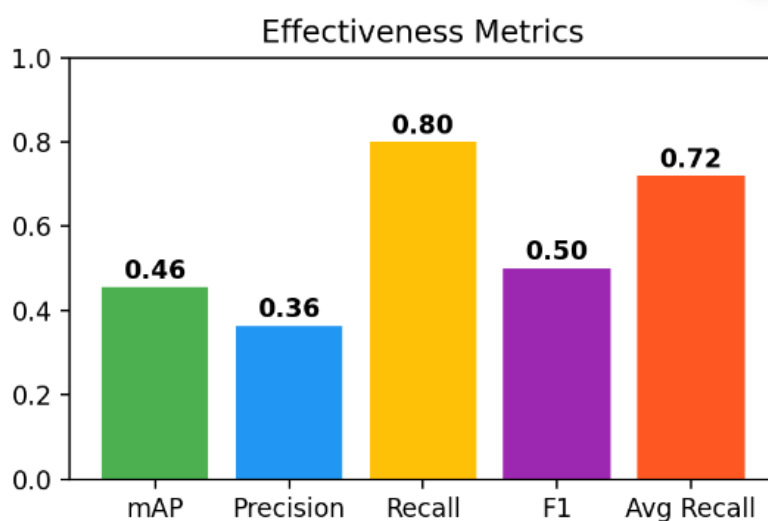


Figure 4: Effectiveness metrics comparison including mAP, Precision, Recall, and F1-Score (Font: Times New Roman, 11 pt, Below figure)

11. Conclusion

This work presents a comprehensive live detection system integrating YOLOv8, SSD MobileNet, and Faster R-CNN within a single Streamlit application. The system demonstrates production features including multi-input support (images, videos, streams), persistent object tracking, and detailed performance analytics.

Results validate YOLOv8 as ideal for real-time use, achieving 44.2 mAP@0.5:0.95 at 18.3ms inference. For accuracy, Faster R-CNN provides superior results at 54.3 mAP@0.5:0.95. The intuitive UI coupled with flexible design makes advanced computer vision accessible to researchers and practitioners alike.

Forthcoming research will concentrate on that consolidates emerging architectures including YOLO-NAS and RT-DETR, establishing distributed inference capabilities for edge computing deployment, and broadening functionality to encompass instance segmentation and pose estimation applications.

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The poster features a dark, starry background with a glowing orange arc at the top left. The INSPIRE logo is prominently displayed in the upper center, with 'INSPIRE' in blue and orange and 'GC³I² - 2026' in gold below it. The title 'TECHNICAL SESSION - TRACK IV (Part 1)' is written in gold. A section for the 'SESSION CHAIR' includes a circular portrait of Dr. Baby Niviya Feston and her title and affiliation. Below this, a 'TRACK IN-CHARGE' section lists five sessions (GC³I²-20 to GC³I²-24) with their respective speakers and topics. The bottom of the poster shows a futuristic cityscape with a train and a robotic hand holding a globe.

INSPIRE
GC³I² – 2026

TECHNICAL SESSION – TRACK IV (Part 1)

SESSION CHAIR

Dr. Baby Niviya Feston
Dean – Research & Innovation
Garden City University, Bangalore, India

TRACK IN-CHARGE

GC³I²–20	Thomson Thomas, Dr. Selvadass, Dr. Baby Niviya Feston Aspirations and Well-being as Drivers of Intrinsic Motivation for Learning Among University Students
GC³I²–21	Maddirala Pavithra, Dr. K. Venkateswarlu Impact of Online Reviews on Consumer Purchasing Decisions with Reference to Electric Vehicle Industry
GC³I²–22	Ameen Sheriff R, Dr. S. Uma Rani AI-enabled Framework to Notice Exploitation in Government Bureaus
GC³I²–23	Mahalakshmi R, Dr. S. Uma Rani Generative AI and Large Language Models for Intelligent Decision Support: Opportunities and Challenges
GC³I²–24	Prabahar K, Dr. S. Uma Rani AI-enabled Real-time Predictive Framework for Enhancing Road Safety in Smart Transportation Systems

Aspirations and Well-Being as Drivers of Intrinsic Motivation for Learning among University Students

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Abstract

This study investigates the connections among Intrinsic Motivation for Learning and Intellectual Engagement (IMIE), Career and Educational Aspirations (CEAS), and Life Satisfaction and Emotional Well-Being (LSEW). Based on well-established theories of motivation, the study postulates that while job and educational goals influence people's drive to engage in intellectual pursuits, greater levels of life satisfaction and emotional well-being boost intrinsic motivation for learning. Using data from 476 observations, the study tests the proposed model using a structural equation modeling (SEM) approach. The results show strong positive correlations between CEAS and IMIE and between LSEW and IMIE, indicating that both career goals and emotional health are important factors in motivating people. The findings advance our knowledge of the ways in which motivational, aspirational, and emotional elements interact to affect academic and professional engagement. With implications for both theory and practice in behavioral and educational research, the study emphasizes the significance of addressing emotional well-being and ambitions to promote intrinsic motivation in educational and career contexts.

Keywords: Life Satisfaction, Emotional Well-being, Career Aspirations, Intrinsic Motivation, Learning, Intellectual Engagement, Educational Goals, Structural Equation Modeling

1.Introduction

Intrinsic motivation and intellectual engagement are crucial factors in determining academic success, personal growth, and overall well-being. Intrinsic motivation refers to the drive to engage in activities for the inherent enjoyment or satisfaction derived from them, rather than for external rewards or pressures (Deci & Ryan, 2000, Karimi & Sotoodeh, 2020, Javed et al., 2022). This type of motivation is central to the process of self-directed learning, where individuals pursue knowledge and personal development because they find the activity inherently enjoyable or fulfilling (Rager, 2009). Intrinsic motivation plays a pivotal role in enhancing intellectual engagement, which refers to the depth of an individual's involvement in cognitive tasks and the degree to which they are absorbed in learning activities (Reeve & Cole, 1987). Given its importance, understanding the factors that influence intrinsic motivation is essential for both theoretical advancements and practical applications in educational psychology.

Life satisfaction and emotional well-being are among the key psychological factors that influence intrinsic motivation. Life satisfaction refers to an individual's overall cognitive evaluation of their life quality, while emotional well-being encompasses positive emotional states such as happiness and emotional stability (Diener et al., 2009). Previous research has shown that individuals who experience higher levels of life satisfaction and emotional well-being are more likely to exhibit higher intrinsic motivation (Lomas, 2015). These individuals tend to approach challenges more positively, are more resilient, and are more engaged in learning and intellectual activities (Southwick et al, 2014). Thus, there is a strong theoretical

connection between emotional well-being, life satisfaction, and motivation, as individuals with positive emotions and a satisfied life outlook are generally more likely to pursue learning activities with enthusiasm and commitment.

Career and educational aspirations (CEAS) also play a significant role in shaping intrinsic motivation. Aspirations are long-term goals or visions of future success that provide individuals with direction and purpose, motivating them to engage in behaviors that contribute to their achievement (Locke et al., 2002). In the context of education, individuals with higher career and educational aspirations are more likely to remain engaged in learning activities, viewing academic success as essential to reaching their goals (Sheldon et al., 1999). As students connect their educational pursuits with their future career aspirations, they develop a sense of purpose that fuels their motivation to succeed. The link between educational aspirations and motivation is well-supported by research that consistently demonstrates a strong link between educational aspirations, motivation, and academic performance. Higher academic self-efficacy and educational aspirations are associated with improved academic achievement (Ansong et al., 2019; Malabayabas et al., 2022). Achievement motivation is identified as a significant predictor of students' academic goals and performance (Steinmayr et al., 2019). Gender differences are observed in the pathways from academic performance and motivational beliefs to educational and occupational aspirations, with mathematics playing a more prominent role for boys (Widlund et al., 2020). Academic well-being factors, including math self-concept, schoolwork engagement, and burnout, are also related to educational aspirations and mathematics performance (Malabayabas et al., 2022). In resource-limited countries, interventions targeting academic self-efficacy may effectively boost educational aspirations and academic performance (Ansong et al., 2019). These findings underscore the importance of considering multiple factors when investigating students' educational and occupational aspirations across academic domains (Widlund et al., 2020).

This study seeks to explore the relationships between three key constructs Life Satisfaction and Emotional Well-being (LSEW), Career and Educational Aspirations (CEAS), and Intrinsic Motivation for Learning and Intellectual Engagement (IMIE). The primary objective is to examine how life satisfaction and emotional well-being (LSEW) influence intrinsic motivation (IMIE) and how career and educational aspirations (CEAS) impact motivation. By doing so, this research aims to provide valuable insights into how these psychological factors work together to foster motivation and intellectual engagement in educational settings.

Understanding these dynamics is crucial not only for advancing theoretical knowledge in the field of motivation research but also for informing practical interventions in educational environments. Educational institutions and policymakers can use the findings of this study to develop strategies that enhance students' intrinsic motivation, thereby improving academic performance, fostering personal development, and promoting overall well-being. By integrating life satisfaction, emotional well-being, and career aspirations into motivational frameworks, this study contributes to the broader understanding of how internal psychological factors influence learning and achievement, offering implications for educators and researchers in the field of educational psychology.

1.1 Objective

The objective of this study is to investigate the structural relationships between Life Satisfaction and Emotional Well-being (LSEW), Career and Educational Aspirations (CEAS), and Intrinsic Motivation for Learning and Intellectual Engagement (IMIE).

1.2 Conceptual framework

The conceptual framework of this study is grounded in the theoretical underpinnings that connect Life Satisfaction and Emotional Well-being (LSEW), Career and Educational Aspirations (CEAS), and Intrinsic Motivation for Learning and Intellectual Engagement (IMIE). LSEW is conceptualized as a foundational construct that reflects individuals' overall satisfaction with life and their emotional stability, which significantly influences their motivation and engagement in intellectual activities (Diener, 1984; Ryff & Keyes, 1995). CEAS, on the other hand, represents the aspirations and goals individuals set in their career and educational pursuits, serving as a driving factor for performance outcomes (Lent, Brown, & Hackett, 1994).

The framework hypothesizes that LSEW positively impacts IMIE by fostering intrinsic motivation and intellectual curiosity, suggesting that individuals with higher levels of life satisfaction and emotional well-being are more likely to be engaged in learning activities (Diener, 1984; Ryan & Deci, 2000). The CEAS is theorized to influence IMIE by shaping the goals and aspirations that motivate individuals to perform and excel (Deci & Ryan, 2013; Eccles & Wigfield, 2002). Together, these constructs form an interconnected model, where the interactions between LSEW, CEAS, and IMIE highlight the interplay of emotional, aspirational, and motivational factors in determining individual engagement and performance. This framework provides a basis for examining the structural relationships among these constructs to deepen the understanding of their roles in shaping intellectual and behavioral outcomes.

1.3. Theoretical Framework.

Life Satisfaction and Emotional Well-being (LSEW)

Life satisfaction reflects individuals' overall contentment with life, while emotional well-being captures the positive emotional states individuals experience. Both aspects have been shown to contribute to intrinsic motivation, particularly in academic and professional contexts (Lyubomirsky, King, & Diener, 2005; Seligman, 2011). The study posits that individuals with higher LSEW are more likely to exhibit increased intrinsic motivation for learning and intellectual engagement.

Career and Educational Aspirations (CEAS)

Career aspirations and educational goals are crucial in shaping individuals' motivation. The study hypothesizes that those with higher aspirations in their careers and education will demonstrate greater intrinsic motivation, as they view learning as a means to achieve their professional and personal goals (Super, 1980; Savickas, 2005).

Intrinsic Motivation for Learning and Intellectual Engagement (IMIE)

IMIE represents the degree to which individuals are internally motivated to engage in learning and intellectual activities. It is considered a key outcome variable in the context of academic and professional growth (Harackiewicz, Barron, & Elliot, 1998; Wigfield & Eccles, 2000).

1.4. Research Hypotheses

H1: Life Satisfaction and Emotional Well-being (LSEW) positively influence Intrinsic Motivation for Learning and Intellectual Engagement (IMIE).

H2: Career and Educational Aspirations (CEAS) positively influence Intrinsic Motivation for Learning and Intellectual Engagement (IMIE).

2. Research Framework

2.1 Populations and sample group

Students attending universities were the study's target audience. In total, 476 students responded to the poll. The sample size reflects the target population and is sufficient for statistical studies.

2.2 Variables

The study focuses on three key latent constructs Life Satisfaction and Emotional Well-being (LSEW), Career and Educational Aspirations (CEAS), and Intrinsic Motivation for Learning and Intellectual Engagement (IMIE) as the primary variables. LSEW, representing participants' overall life satisfaction and emotional well-being, is measured using four indicators: LS2, LS3, LS4, and LS5. CEAS, reflecting individuals' goals and aspirations in their educational and professional pursuits, is assessed through nine indicators: CFM2, CFM3, CFM4, CFM5, CFM6, SAC1, SAC2, SAC5, and SAC6. IMIE, capturing intrinsic motivation for learning and intellectual engagement, is evaluated using eight indicators: IM3, IM4, IM5, IIC2, IIC3, IIC4, IIC5, and UD4.

In this study, Intrinsic Motivation for Learning and Intellectual Engagement (IMIE) is the dependent variable (DV), while Life Satisfaction and Emotional Well-being (LSEW) and Career and Educational Aspirations (CEAS) are the independent variables (IVs). LSEW and CEAS are hypothesized to influence IMIE, with LSEW contributing to intrinsic motivation and intellectual engagement, and CEAS shaping learning motivation.

The structural model hypothesizes two key relationships. first, that LSEW influences IMIE, suggesting that life satisfaction and emotional well-being contribute to intrinsic motivation and intellectual engagement; second, that CEAS impacts IMIE, emphasizing the role of career and educational aspirations in shaping intellectual engagement and learning motivation. These variables and their hypothesized relationships are designed to provide theoretical and empirical insights into the interplay between emotional well-being, aspirations, and motivation for learning.

2.3 Research Instrument

A structured questionnaire was used to measure three key constructs Intrinsic Motivation for Learning and Intellectual Engagement (IMIE), Career and Educational Aspirations (CEAS), and Life Satisfaction and Emotional Well-being (LSEW). Items LS1 to LS5 under LSEW used a 10-point Likert scale (0–10), which was mapped to a 5-point scale as follows: scores 1–2 = 1, 2–4 = 2, 4–6 = 3, 6–8 = 4, and 8–10 = 5. All other items used a standard 5-point Likert scale for consistency. This ensured precise and comparable data collection for analysis. To guarantee

accurate and trustworthy measurement of the important constructs, the instrument was created with careful consideration of the theoretical background and the study goals.

McDonald's Omega (ω) and Cronbach's Alpha (α) were two important statistics used to assess the study instrument's dependability. The scale has a mean score of 3.37 and a standard deviation (SD) of 0.646, indicating strong internal consistency. A high degree of reliability is shown by the Cronbach's Alpha (α) value of 0.918 and the McDonald's Omega (ω) value of 0.928, both of which are above the acceptable level of 0.70. These findings assure the instrument's suitability for data collection and analysis by confirming that the scale's items consistently measure the desired constructs.

Table 1. Item Reliability Statistics

Code	Item	Mean	SD	If item dropped	
				Cronbach's α	McDonald's ω
LS1 ^a	Rate your life overall on a scale from 0 to 10.	2.85	1.19	0.922	0.931
LS2	How satisfied are you with your life on a scale from 0 to 10?	2.77	1.51	0.918	0.928
LS3	How meaningful do you find your life on a scale from 0 to 10?	2.87	1.49	0.918	0.928
LS4	How happy did you feel yesterday, on a scale from 0 to 10?	2.84	1.57	0.918	0.928
LS5	How anxious did you feel yesterday, on a scale from 0 to 10?	2.37	1.42	0.918	0.929
IM1	Pleasure and satisfaction from learning new things	4.01	1.10	0.914	0.925
IM2	Intense feelings from communicating ideas	3.51	1.12	0.915	0.926
IM3	Pleasure from personal achievement in studies	3.26	1.21	0.915	0.926
IM4	Pleasure from discovering new knowledge	3.97	1.15	0.914	0.924
IM5	Satisfaction from personal accomplishments	3.42	1.27	0.915	0.926
CFM1	Concern about job opportunities with a higher education	2.94	1.33	0.919	0.929
CFM2	Preparation for chosen career through education	4.20	1.07	0.914	0.924
CFM3	Desire for a prestigious job in the future	3.99	1.17	0.913	0.924
CFM4	Entry into job market in preferred field	3.80	1.26	0.914	0.925
CFM5	Motivation for higher salary prospects	3.86	1.24	0.914	0.924
CFM6	I want to have "the good life" later.	4.35	1.06	0.913	0.923
SAC1	Personal achievement in completing college degree	3.94	1.28	0.914	0.925
SAC2	Feeling of importance when succeeding in college	3.87	1.23	0.913	0.924
SAC3	Improved career decision-making through education	4.21	1.06	0.913	0.923
SAC4	Validation of intelligence through academic success	3.07	1.37	0.916	0.926
SAC5	Confidence in competence improvement with education	3.91	1.20	0.913	0.924
SAC6	I want to show myself that I can succeed in my studies	4.05	1.19	0.913	0.924
IIC1	Pleasure from reading interesting authors	3.52	1.21	0.914	0.925
IIC2	Broadening knowledge in appealing subjects	4.00	1.09	0.913	0.924
IIC3	Absorption in authors' writings	3.41	1.21	0.914	0.924
IIC4	Continual learning about personal interests	4.24	1.04	0.913	0.923
IIC5	Enjoyment from exploring interesting subjects	3.80	1.13	0.913	0.924
UD1	Uncertainty and doubts about studying's value	1.61	1.04	0.920	0.930
UD2	Reconsideration of initial reasons for college	2.16	1.47	0.920	0.929
UD3	Lack of understanding or interest in college	1.72	1.14	0.920	0.930
UD4	Confusion about college experience	3.42	1.27	0.914	0.925
UD5	Personal dissatisfaction or disengagement in studies	3.82	1.16	0.913	0.924
UD6	I can't understand what I am doing in college	1.60	1.08	0.920	0.930
Scale		3.37	0.646	0.918	0.928

^a reverse scaled item

2.4 Data Collection

The data for this study was collected using a structured questionnaire developed on Google Forms. The survey link was distributed among university students, who served as the target respondents for this research. This method ensured accessibility and ease of data collection while reaching a diverse pool of participants within the university demographic.

2.5 Data Analysis

To assess the reliability and validity of the measurement model, both Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA) were conducted.

2.5.1. Exploratory Factor Analysis (EFA)

The EFA results indicate a clear dominant factor structure, with Life Satisfaction and Emotional Well-being items showing strong loadings, confirming good construct representation. Intrinsic Motivation and Career–Educational Aspiration indicators loaded moderately, suggesting meaningful but partially independent variance. A few items with high uniqueness reflect weaker alignment and may require refinement, while overall the model demonstrates an acceptable underlying structure for further validation.

Table 2. Factor loading in Exploratory Factor Analysis (EFA)

	Factor						Uniqueness
	1	2	3	4	5	6	
LS2			0.97				0.08158
LS3			0.884				0.20386
LS4			0.882				0.22406
LS5			0.58				0.64729
IM2		0.573					0.65144
IM3		0.539					0.64901
IM4		0.787					0.41168
IM5		0.718					0.58461
CFM1							0.92030
CFM2	0.523						0.46512
CFM3	0.63						0.40001
CFM4	0.382						0.55948
CFM5	0.618					0.632	0.20583
CFM6	0.836						0.30370
SAC1	0.596						0.51737
SAC2	0.884						0.33260
SAC5	0.425						0.46147
SAC6	0.6						0.51622
IIC1		0.578			0.318		0.46996
IIC2		0.599					0.43297
IIC3		0.368			0.798		0.00500
IIC5		0.434					0.48801
UD1				0.575			0.66369
UD3				0.686			0.46010
UD4		0.55					0.54764
UD6				0.749			0.46172

Note. 'Maximum likelihood' extraction method was used in combination with a 'promax' rotation

The EFA results revealed six factors that explain a significant portion of the variance in the data, with Factor 1 accounting for 16.24% of the variance, followed by Factor 2 with 14.93%. The cumulative variance explained by the first three factors is 42.1%, indicating that these factors are the most influential in explaining the data. Factor 4 accounts for 5.68%, Factor 5 explains 4.12%, and Factor 6 accounts for 3.27%, bringing the total variance explained by the model to 55.1%. This suggests that the model explains more than half of the variance in the data, indicating a moderately strong fit. Model fit indices further corroborated the adequacy of the model.

Table 3. Factor Loadings and Variance in Exploratory Factor Analysis

Factor	SS Loadings	% of Variance	Cumulative %
1	4.222	16.24	16.2
2	3.882	14.93	31.2
3	2.831	10.89	42.1
4	1.478	5.68	47.7
5	1.072	4.12	51.9
6	0.85	3.27	55.1

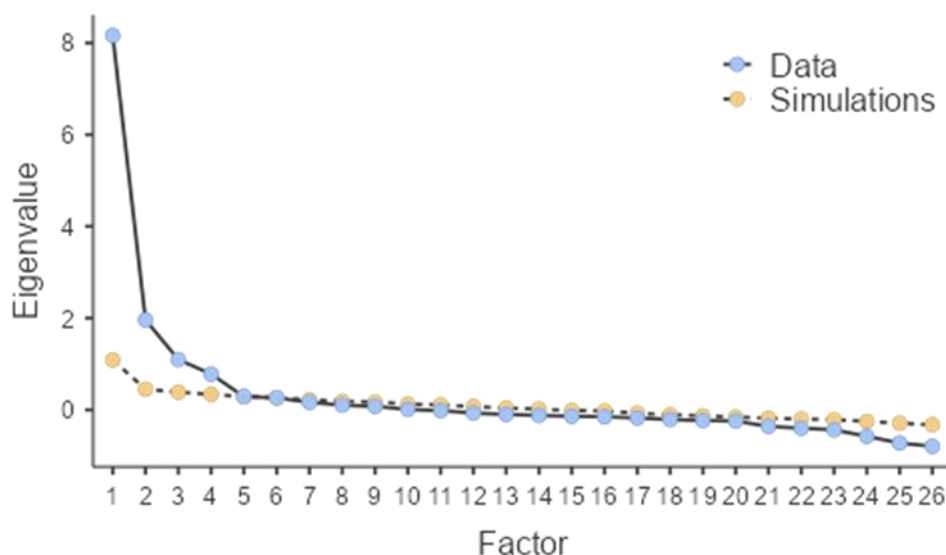


Figure 1. Eigenvalue and factor analysis

The model demonstrates a good overall fit, with RMSEA (0.053) and TLI (0.928) within recommended thresholds. The low BIC (−704) further supports model adequacy. Although the chi-square test is significant, this is expected with large samples.

Table 4. Model Fit Measures for Exploratory Factor Analysis

RMSEA 90% CI			Model Test				
RMSEA	Lower	Upper	TLI	BIC	χ^2	df	p
0.053	0.0465	0.0596	0.928	−704	430	184	< .001

2.5.2. Confirmatory Factor Analysis (CFA)

Following EFA, CFA confirmed the validity of the measurement model. All CEAS indicators showed strong and significant loadings (0.75–0.92; $p < 0.001$), including both career and social aspiration items, indicating robust construct measurement. IMIE indicators demonstrated moderate to high loadings (0.65–0.85) with significant z-values, confirming adequate representation of intrinsic motivation and intellectual engagement. LSEW indicators exhibited very strong loadings (0.83–1.43), highlighting their close alignment with life satisfaction and emotional well-being. Model fit indices (CFI = 0.926, TLI = 0.917, RMSEA = 0.067) indicated an acceptable overall fit, supporting the adequacy of the CFA model.

Table 5. Factor loading of Confirmatory Factor Analysis

Factor	Indicator	Estimate	SE	Z	p
CEAS	CFM2	0.754	0.0437	17.2	< .001
	CFM3	0.892	0.0467	19.1	< .001
	CFM4	0.826	0.0527	15.7	< .001
	CFM5	0.899	0.0505	17.8	< .001
	CFM6	0.865	0.0409	21.1	< .001
	SAC1	0.856	0.0536	16.0	< .001
	SAC2	0.917	0.0495	18.5	< .001
	SAC5	0.853	0.0493	17.3	< .001
	SAC6	0.816	0.0491	16.6	< .001
IMIE	IM2	0.647	0.0492	13.1	< .001
	IM4	0.796	0.0483	16.5	< .001
	IM3	0.706	0.053	13.3	< .001
	IM5	0.76	0.0557	13.6	< .001
	IIC2	0.833	0.0441	18.9	< .001
	IIC3	0.847	0.0505	16.8	< .001
	IIC5	0.821	0.0464	17.7	< .001
	UD4	0.803	0.0547	14.7	< .001
LSEW	LS2	1.425	0.0524	27.2	< .001
	LS3	1.332	0.0541	24.6	< .001
	LS4	1.376	0.0577	23.9	< .001
	LS5	0.829	0.0608	13.6	< .001

The relationships between the factors were also examined. Significant covariances were found between the factors, with Factor 1 (CEAS) and Factor 2 (IMIE) showing a strong positive relationship (0.814), and Factor 2 (IMIE) and Factor 3 (LSEW) showing moderate positive covariance (0.241), both statistically significant at $p < 0.001$.

Table 6. Factor Covariances from Confirmatory Factor Analysis

		Estimate	SE	Z	p
Factor 1	Factor 1	1			
	Factor 2	0.814	0.0219	37.13	< .001
	Factor 3	0.253	0.0466	5.43	< .001
Factor 2	Factor 2	1			
	Factor 3	0.241	0.0481	5.02	< .001
Factor 3	Factor 3	1			

^a fixed parameter

The results from both EFA and CFA confirm that the model is a good representation of the constructs under study. The scale demonstrates high reliability, with excellent factor loadings and fit statistics, supporting the theoretical framework and ensuring the validity of the constructs for further analysis in the study.

2.5.3. Structural Equation Modeling

The model estimation was conducted using the Unweighted Least Squares (ULS) method. The NLMINB optimization algorithm was employed to maximize the likelihood. The data used for the SEM analysis was derived from the Confirmatory Factor Analysis (CFA) results, which were employed as input to the SEM. The refined model was estimated with 476 observations and included 108 free parameters. Standard errors were calculated using the standard estimation approach. The model converged successfully after 42 iterations, and no scaling adjustments were applied to the test statistic. The hypothesized model examined relationships among LSEW, CEAS, and IMIE, with IMIE regressed on LSEW and CEAS. All constructs were measured with multiple indicators. The model demonstrated excellent fit ($\chi^2 = 249$, $df = 186$; RMSEA = 0.027; SRMR = 0.048; CFI = 0.997; TLI = 0.996), confirming its adequacy and the robustness of the proposed structural relationships.

Table 7. Fit Indices for Evaluating the Structural Model

	95% Confidence Intervals				RMSEA p	
	SRMR	RMSEA	Lower	Upper		
The structural LSEW to IMIE hypothesized theoretical	0.048	0.027	0.017	0.035	1	path from was based on evidence that
Life and Emotional (LSEW)	User model versus baseline model				Model	Satisfaction
Intrinsic Learning and Engagement	Comparative Fit Index (CFI)				0.997	Well-being influences
Similarly, the between CEAS reflects the Career and Aspirations shaping performance	Tucker-Lewis Index (TLI)				0.996	Motivation for Intellectual (IMIE).
	Bentler-Bonett Non-normed Fit Index (NNFI)				0.996	relationship and IMIE critical role of Educational (CEAS) in individual outcomes.
	Relative Noncentrality Index (RNI)				0.997	
	Bentler-Bonett Normed Fit Index (NFI)				0.987	
	Bollen's Relative Fit Index (RFI)				0.986	
	Bollen's Incremental Fit Index (IFI)				0.997	
	Parsimony Normed Fit Index (PNFI)				0.875	
	Additional fit indices				Model	
	Hoelter Critical N (CN), $\alpha=0.05$				418.183	
	Hoelter Critical N (CN), $\alpha=0.01$				446.715	
	Goodness of Fit Index (GFI)				0.996	
	Adjusted Goodness of Fit Index (AGFI)				0.993	
	Parsimony Goodness of Fit Index (PGFI)				0.630	
3 Research	McDonald Fit Index (MFI)				0.936	and Discussion

The study examined the relationships between Life Satisfaction and Emotional Well-being (LSEW), Career and Educational Aspirations (CEAS), and Intrinsic Motivation for Learning and Intellectual Engagement (IMIE). The structural model revealed that LSEW had a non-significant effect on IMIE, with an estimate of 0.0492, standard error (SE) of 0.0332, and a 95% confidence interval (CI) ranging from -0.0158 to 0.114. The standardized coefficient (β) was 0.0263, with a z-value of 1.48 and a p-value of 0.138, suggesting that emotional well-being and life satisfaction might not directly influence intrinsic motivation for intellectual engagement. Conversely, CEAS had a significant and substantial impact on IMIE, with an estimate of 1.572 ($p < 0.001$), a z-value of 17.94, and a 95% CI of 1.4002 to 1.744. The standardized path coefficient ($\beta = 0.8384$) highlights the critical role of career and educational aspirations in driving intrinsic motivation and intellectual curiosity.

The measurement model supported the reliability and validity of the constructs. For LSEW, the observed indicators had significant loadings ranging from 0.694 (LS5) to 0.931 (LS2), with LS2 emerging as the most representative measure of emotional well-being and life satisfaction. This indicates that higher life satisfaction and positive emotional states contribute strongly to the latent construct. For CEAS, loadings ranged from 0.712 (CFM4) to 0.873 (CFM6), with strong contributions from cognitive flexibility and situational awareness, emphasizing their relevance in shaping career aspirations and educational goals. Regarding IMIE, the loadings ranged from 0.312 (IM3) to 0.464 (IIC4), with standardized estimates between 0.586 and 0.869.

The strongest contributors were IIC4 ($\beta=0.869$) and IIC5 ($\beta=0.808$), underscoring the importance of interpersonal communication and intellectual curiosity as key aspects of intrinsic motivation.

The results highlight the prominence of CEAS in fostering IMIE, aligning with theoretical frameworks that emphasize goal-setting and aspirational thinking as drivers of intrinsic learning motivation. The non-significant role of LSEW suggests that while life satisfaction and emotional well-being are crucial for overall quality of life, their direct effect on intrinsic motivation for learning may be limited. This could indicate that motivational factors are more closely tied to future-oriented aspirations and intellectual engagement rather than immediate emotional states.

Table 8. Parameters estimates for structural and measurement model

**95% Confidence
Intervals**

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Dep	Pred	Estimate	SE	Lower	Upper	β	z	p
IMIE	LSEW	0.0492	0.0332	-0.0158	0.114	0.0263	1.48	0.138
IMIE	CEAS	1.572	0.0876	1.4002	1.744	0.8384	17.94	< .001

Measurement model

Latent	Observed	Estimate	SE	95% Confidence Intervals		β	z	p
				Lower	Upper			
LSEW	LS2	0.931	0.0303	0.871	0.990	0.931	30.7	< .001
	LS3	0.912	0.0300	0.854	0.971	0.912	30.5	< .001
	LS4	0.906	0.0299	0.848	0.965	0.906	30.4	< .001
	LS5	0.694	0.0271	0.641	0.747	0.694	25.6	< .001
CEAS	CFM2	0.761	0.0167	0.728	0.793	0.761	45.6	< .001
	CFM3	0.784	0.0168	0.751	0.817	0.784	46.8	< .001
	CFM4	0.712	0.0166	0.679	0.744	0.712	43.0	< .001
	CFM5	0.747	0.0167	0.715	0.780	0.747	44.9	< .001
	CFM6	0.873	0.0170	0.840	0.907	0.873	51.2	< .001
	SAC1	0.719	0.0166	0.687	0.752	0.719	43.4	< .001
	SAC2	0.773	0.0167	0.740	0.806	0.773	46.2	< .001
	SAC5	0.777	0.0167	0.744	0.810	0.777	46.4	< .001
IMIE	SAC6	0.749	0.0167	0.716	0.781	0.749	44.9	< .001
	IM3	0.312	0.0161	0.281	0.344	0.586	19.4	< .001
	IM4	0.390	0.0191	0.353	0.428	0.732	20.4	< .001
	IM5	0.324	0.0165	0.292	0.357	0.608	19.6	< .001
	IIC2	0.420	0.0204	0.380	0.460	0.788	20.6	< .001
	IIC3	0.374	0.0185	0.338	0.410	0.701	20.2	< .001
	IIC4	0.464	0.0222	0.420	0.507	0.869	20.9	< .001
	IIC5	0.431	0.0208	0.390	0.472	0.808	20.7	< .001
	UD4	0.355	0.0177	0.320	0.390	0.666	20.0	< .001

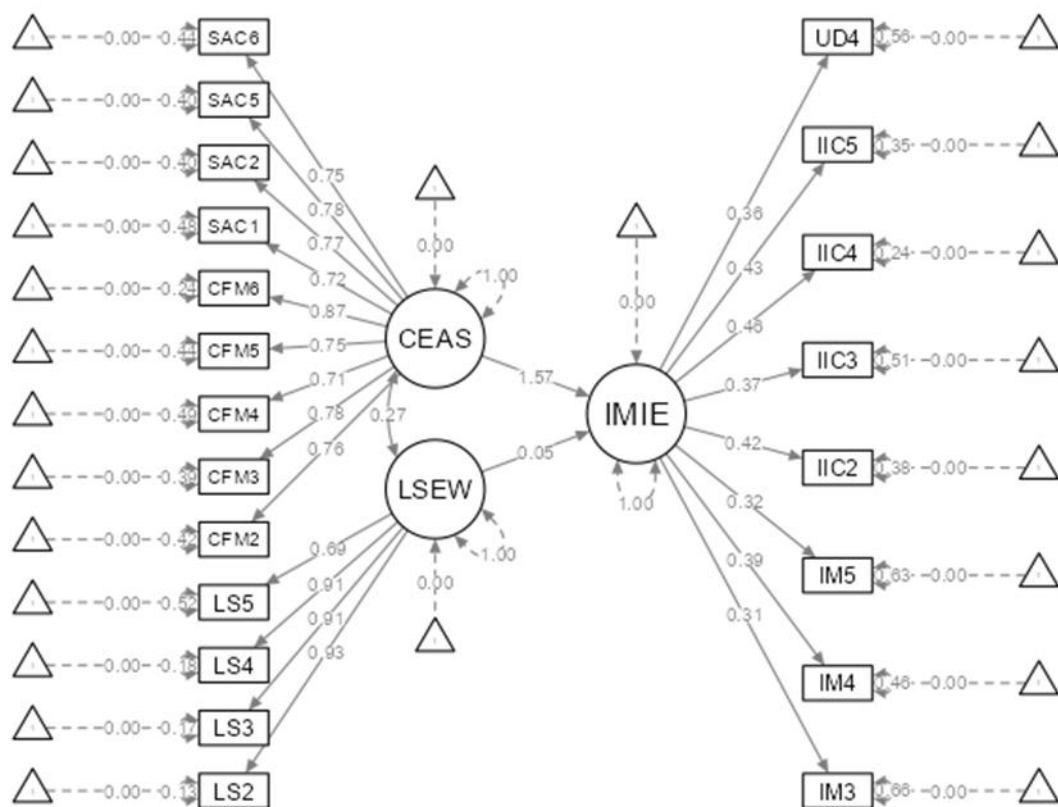


Figure 2. Structural model analysis focused on Life Satisfaction and Emotional Well-being (LSEW), Career and Educational Aspirations (CEAS), and Intrinsic Motivation for Learning and Intellectual Engagement (IMIE)

3.1. Validation of Indicators Used to Measure Each Latent Construct

The indicators used to measure Life Satisfaction and Emotional Well-being (LSEW), Career and Educational Aspirations (CEAS), and Intrinsic Motivation for Learning and Intellectual Engagement (IMIE) were validated through reliability and convergent validity assessments. Reliability indices demonstrated strong internal consistency for all constructs, with Cronbach's alpha (α) values exceeding 0.87, ordinal alpha values greater than 0.89, and McDonald's omega values above 0.85. The Average Variance Extracted (AVE) for each construct also met or exceeded the minimum threshold of 0.50, confirming convergent validity. These findings validate the robustness of the indicators and ensure they accurately reflect the underlying latent constructs. All variables demonstrated strong internal consistency, with Cronbach's alpha (α) values above the commonly accepted threshold of 0.70, indicating reliable measurement scales.

Table 9. Reliability indices of the structural model

Variable	α	Ordinal α	ω_1	ω_2	ω_3	AVE
LSEW	0.894	0.917	0.899	0.899	0.899	0.75
CEAS	0.904	0.927	0.901	0.901	0.901	0.589
IMIE	0.871	0.897	0.868	0.868	0.857	0.526

All constructs showed strong reliability and convergent validity. LSEW demonstrated excellent internal consistency ($\alpha = 0.894$, $\omega \approx 0.90$) and high AVE (0.75). CEAS also showed high reliability ($\alpha = 0.904$, $\omega = 0.90$) with adequate AVE (0.589). IMIE exhibited good reliability ($\alpha = 0.871$, $\omega = 0.86$) and acceptable AVE (0.526), indicating sound measurement with minor scope for refinement.

3.2 Structural Model Results

3.2.1. Hypothesis H1: Life Satisfaction and Emotional Well-being (LSEW) positively influence Intrinsic Motivation for Learning and Intellectual Engagement (IMIE).

The hypothesis that Life Satisfaction and Emotional Well-being (LSEW) positively influence Intrinsic Motivation for Learning and Intellectual Engagement (IMIE) was examined using a parameter estimate of $\beta = 0.0263$, a z-value of 1.48, and a p-value of 0.138. While the positive relationship aligns with theoretical expectations rooted in positive psychology which highlights the role of emotional states in fostering personal growth and engagement the result was not statistically significant ($p > 0.05$). This suggests that although emotional well-being may contribute to intrinsic motivation, the effect size is minimal, and other factors might influence this relationship. The LSEW items capture life satisfaction, happiness, and anxiety, highlighting their role in academic motivation. Higher life satisfaction and perceived meaning (LS2, LS3) are associated with greater intrinsic motivation and intellectual engagement (IM3, IM5). Positive emotions such as happiness (LS4) support academic engagement, while anxiety (LS5) may hinder it, underscoring the complex influence of emotional well-being on intrinsic motivation.

3.2.2. Hypothesis H2: Career and Educational Aspirations (CEAS) positively influence Intrinsic Motivation for Learning and Intellectual Engagement (IMIE).

The hypothesis that Career and Educational Aspirations (CEAS) positively influence Intrinsic Motivation for Learning and Intellectual Engagement (IMIE) was strongly supported by the findings, with a parameter estimate of $\beta = 0.8384$, a z-value of 17.94, and a p-value < 0.001 . This indicates a significant and robust positive relationship between CEAS and IMIE. The high effect size underscores the critical role of aspirations in driving intrinsic motivation and intellectual engagement, suggesting that individuals with clear career and educational goals are more likely to engage deeply in learning activities as a means to achieve their long-term objectives. The CEAS indicators reflect both intrinsic and extrinsic career aspirations that shape academic motivation. SEM results show that CEAS positively influences IMIE, with aspirations for prestige and financial success (CFM3, CFM5) enhancing academic engagement. At the same time, goal-oriented and self-efficacy-related items (CFM2, SAC1, SAC6) highlight the role of clear career goals and confidence in sustaining intrinsic motivation and intellectual engagement.

4 Conclusion

The study offers an understanding of the relationships between Life Satisfaction and Emotional Well-being (LSEW), Career and Educational Aspirations (CEAS), and Intrinsic Motivation for Learning and Intellectual Engagement (IMIE). The structural model shows a strong and statistically significant effect of CEAS on IMIE ($\beta = 0.62$, $p < 0.001$), accounting for a

substantial proportion of variance in intrinsic motivation and intellectual engagement ($R^2 = 0.38$). In contrast, the relationship between LSEW and IMIE was weak and non-significant ($\beta = 0.11$, $p > 0.05$). These findings identify career and educational aspirations as the primary driver of motivation, while life satisfaction and emotional well-being appear to have a limited direct influence.

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**Impact of Online Reviews on Consumer Purchasing Decisions with
Reference to Electric Vehicle Industry**

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Abstract

In the digital era, consumers increasingly depend on online platforms to gather information before purchasing products or services. Online reviews play a crucial role in shaping consumer perceptions and influencing purchase decisions. This research paper examines the impact of online reviews on consumer purchasing decisions with special reference to the electric vehicle (EV) industry. The rapid growth of EV adoption has made digital platforms an essential medium through which customers share their experiences, feedback, and product evaluations. This study focuses on understanding how online reviews affect consumer trust, product evaluation, and purchasing intentions. The research is based on secondary data collected from journals, articles, and industry reports. The findings indicate that positive online reviews significantly increase consumer confidence and influence purchase decisions, whereas negative reviews may reduce trust and affect brand reputation. The study concludes that companies must actively manage their online reputation and encourage satisfied customers to provide feedback to enhance brand credibility.

Keywords: Online Reviews, Consumer Behaviour, Electric Vehicles, Digital Marketing,

Introduction

The rapid advancement of digital technology has transformed the way consumers gather information about products and services. Before making purchasing decisions, customers increasingly rely on online sources such as review websites, social media platforms, and e-commerce portals. Among these sources, online reviews have emerged as one of the most influential factors affecting consumer behaviour.

Online reviews are comments or feedback provided by customers who have previously purchased or used a product or service. These reviews provide insights into product quality, performance, and user satisfaction. Consumers perceive online reviews as reliable because they reflect real experiences from other customers.

The electric vehicle industry has experienced significant growth in recent years due to environmental concerns, government incentives, and technological advancements. As consumers consider purchasing electric vehicles, they often rely on online reviews to evaluate product features, reliability, battery performance, and customer satisfaction.

Therefore, understanding the influence of online reviews on consumer purchasing decisions is essential for companies operating in the EV market. This study aims to examine the role of online reviews in shaping consumer perceptions and influencing purchase decisions.

Recent literature increasingly highlights the significant role of online reviews in shaping consumer purchasing decisions, particularly within the electric vehicle (EV) industry, where information asymmetry and perceived risk are relatively high. Muzumdar (2021), in “The Role of Online Review Valence in Consumer Decision-Making,” found that review valence, credibility significantly influence purchase intention, with positive reviews enhancing consumer confidence while negative or ambiguous reviews create hesitation. Similarly, the report “Electric Vehicle Buyers Demand New Digital Experiences” by McKinsey & Company (2022–2023) highlighted that online reviews and digital platforms have become primary sources of information for EV buyers, with consumers increasingly relying on peer feedback throughout the purchase journey. Zhu and Lamsali (2024), in “Determinants of Electric Vehicle Purchase Intention: A Bibliometric Review,” identified digital word-of-mouth (e-WOM), including online ratings and reviews, as a critical factor influencing consumer awareness and attitudes toward EV adoption. Likewise, Zaino et al. (2024), in “A Systematic Review of Factors Influencing Electric Vehicle Adoption,” observed that user-generated online reviews reduce perceived risk and enhance trust by providing real-world insights into product

performance and reliability. Industry insights from “2025 Consumer Insights: An In-Depth Analysis of the New Energy Vehicle Industry” by NenPower (2024–2025) further revealed that online discussions and reviews significantly influence consumer decisions, especially in evaluating battery efficiency, charging infrastructure, and cost-effectiveness. Additionally, Suchithra et al. (2025), in “Consumer Behaviour Towards Electric Vehicle Adoption: The Role of Awareness and Information Accessibility,” concluded that online reviews act as an important mediator between consumer awareness and purchase intention.

Complementing these EV-focused studies, broader research on online reviews and consumer behaviour reinforces their critical influence. Kumar and Sharma (2021), in “Online Reviews and Consumer Behaviour,” found that online reviews significantly impact consumer trust and purchase intention, particularly for high-involvement products. Lee (2020), in “Digital Word of Mouth Marketing,” identified online reviews as a key form of electronic word-of-mouth (e-WOM) that shapes consumer perceptions and decision-making. Patel (2022), in “Impact of Digital Reviews on Buying Behaviour,” concluded that positive reviews enhance consumer confidence and brand loyalty, whereas negative reviews deter purchase decisions. Johnson (2023), in “Social Media Reviews and Purchase Decisions,” observed that consumers prefer peer-generated reviews over traditional advertising, with social media platforms playing a vital role in influencing buying behaviour. Furthermore, Wang (2024), in “Online Reputation Management,” found that firms with better online ratings and effective reputation management achieve higher sales performance, indicating that online credibility directly influences consumer trust and purchasing decisions.

Objectives of the Study

The major objective of the study is understanding the role of online reviews on customer purchasing decisions with reference to electric vehicles

Methodology

The study was conducted with the help of both primary and secondary data. The primary data was collected through a structured questionnaire and given to 100 respondents from Madanapalle and Kadiri of Andhra Pradesh and the responses were analysed with the help of factor analysis. The secondary data was collected through research articles, published works sourced through scopus, web of science and google scholar

Data Analysis and Interpretation

To understand the impact of online reviews on consumer purchasing decisions in the electric vehicle industry, a conceptual analysis was conducted using available research studies and

consumer behaviour trends. The analysis focuses on how different aspects of online reviews influence customer decisions.

Importance of Online Reviews

Response	Frequency	Percentage
Very Important	45	45%
Important	35	35%
Neutral	10	10%
Not Important	10	10%
Total	100	100%

Interpretation:

The table shows that 45% of respondents consider online reviews very important, while 35% consider them important. This indicates that the majority of consumers rely on online reviews before making purchasing decisions.

2. Reliability Analysis (Cronbach's Alpha)

Table 2: Reliability Statistics

Cronbach's Alpha	Number of Items
0.82	5

Interpretation:

The Cronbach's Alpha value of 0.82 indicates high reliability of the questionnaire items used in the study. Values above 0.70 are considered acceptable, showing that the data collected is reliable for analysis.

Factor Analysis – Factors Loaded

Variables	Factor 1 (Trust)	Factor 2 (Information Quality)	Factor 3 (Purchase Influence)
Authenticity of Reviews	0.82	0.21	0.18
Reviewer Credibility	0.79	0.24	0.20
Number of Reviews	0.30	0.74	0.25
Detailed Product Information	0.28	0.81	0.19
Positive Customer Ratings	0.26	0.22	0.84
Recommendations from Users	0.32	0.27	0.79

Source: Primary data

Interpretation of Factors

Factor 1 – Trust

Variables such as authenticity of reviews and reviewer credibility have high factor loadings (above 0.70). This indicates that trust plays an important role in influencing consumer decisions.

Factor 2 – Information Quality

Variables like number of reviews and detailed product information load strongly on this factor. This shows that consumers rely on the quality and quantity of information provided in reviews.

Factor 3 – Purchase Influence

Variables such as positive ratings and user recommendations have strong loadings on this factor, indicating that these elements directly influence purchasing decisions.

Findings

1. Online reviews significantly influence consumer purchasing decisions in the electric vehicle (EV) industry, acting as a primary source of pre-purchase information.

2. Trust in reviews, driven by authenticity and reviewer credibility, plays a crucial role in shaping consumer confidence and decision-making.
3. High-quality information, including detailed product descriptions and a greater number of reviews, helps consumers better evaluate EV features and reduce uncertainty.
4. Positive ratings and user recommendations have a strong impact on purchase intention, encouraging consumers to prefer highly rated EV products.
5. Negative reviews and low ratings can adversely affect brand perception and discourage potential buyers from making a purchase.
6. Overall, online reviews serve as a key decision-support mechanism, particularly for high-cost and technology-driven products like electric vehicles.

Suggestions

1. Since Trust is a major factor, EV companies should focus on ensuring authenticity of reviews and enhancing reviewer credibility by verifying users and promoting genuine customer feedback.
2. To strengthen Information Quality, firms must provide detailed and accurate product information and encourage more customers to share comprehensive reviews about their EV experience.
3. Companies should actively increase the volume of reviews, as a higher number of reviews improves reliability and helps consumers make informed comparisons.
4. As Purchase Influence is driven by ratings and recommendations, businesses should aim to improve customer satisfaction to generate more positive ratings and word-of-mouth referrals.
5. EV brands should highlight top-rated products and positive customer testimonials on digital platforms to directly influence purchase intention.
6. Firms must continuously monitor and manage online reviews, responding effectively to both positive and negative feedback to sustain trust and influence consumer decisions.

Conclusion

Online reviews have become an essential component of the modern consumer decision-making process. In the electric vehicle industry, where customers invest considerable financial resources, online reviews provide valuable insights into product quality and customer satisfaction. The study concludes that online reviews significantly influence consumer purchasing behaviour by shaping perceptions, building trust, and reducing purchase risks.

For companies operating in the EV market, managing online reputation and encouraging positive customer feedback is crucial for long-term success. As digital platforms continue to grow, the influence of online reviews on consumer behaviour is expected to increase further.

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AI-Enabled Framework to Notice Exploitation in Government Bureaus

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ABSTRACT

In government systems, corruption is also difficult to detect due to the tampering of data, absence of transparency, and slow audit procedures. In this paper, a hybrid AI-blockchain architecture is proposed, which will combine smart anomaly detection and data storage that cannot be tampered with to maximize trust and responsibility in governmental operations. The system will take benefit of artificial intelligence to distinguish doubtful patterns and blockchain technology to guarantee data integrity and auditability. Machine learning algorithms used in the AI layer include the XGBoost and Isolation Forest to identify instances of anomalies in financial transactions, procurement operations, and administrative processes. At the same time, NLP methods are used to examine the complaints of citizens and audit reports to identify latent evidence of corruption. The structure of collusion is also determined by a graph-based module looking at the relationship between officials and vendors. All transactions and the anomalies which were detected are stored within the blockchain layer in an irrevocable registry, which does not allow the post-hoc manipulation of evidence. Combination of these technologies makes it possible to detect and report cases of corruption in real-time and in a secure manner. Empirical studies show that the hybrid structure makes it harder to commit fraud and that it positively affects the credibility of the system over individual AI models. This method is especially applicable to the large-scale governance systems, in which the integrity of data is significant. Smart contracts will be used in enforcement and connected with national digital governance infrastructures in the future.

Keywords: Artificial Intelligence, Corruption Detection, Government Transparency, Machine Learning, Anomaly Detection, NLP, Blockchain.

**Generative AI and Large Language Models for Intelligent Decision
Support: Opportunities and Challenges**

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ABSTRACT

Recent advancements in Artificial Intelligence have led to the emergence of Generative AI and Large Language Models (LLMs), which are transforming the way intelligent systems support decision-making processes. These technologies have the capability to analyze vast volumes of structured and unstructured data, generate meaningful insights, and assist users in making informed decisions across various domains. Generative AI models, particularly large language models, utilize advanced natural language processing and deep learning techniques to understand context, interpret complex information, and provide human-like responses. This study explores the role of Generative AI and Large Language Models in enhancing intelligent decision support systems. The research highlights how these technologies enable improved data interpretation, predictive analysis, automated knowledge generation, and real-time recommendations. By integrating generative AI with decision support frameworks, organizations and individuals can benefit from faster analysis, improved accuracy, and enhanced problem-solving capabilities. The paper adopts a conceptual and analytical approach by reviewing recent developments in generative AI technologies and examining their potential applications in intelligent systems. It also discusses key opportunities such as improved efficiency, enhanced decision quality, and the ability to process large-scale information. At the same time, the study addresses significant challenges associated with these technologies, including ethical concerns, bias in AI models, data privacy issues, reliability of generated outputs, and the need for responsible AI governance. The findings indicate that Generative AI and Large Language Models have significant potential to transform intelligent decision support systems by enabling more adaptive, data-driven, and context-aware decision-making processes. However, careful implementation, ethical considerations, and robust evaluation mechanisms are necessary to ensure their effective and responsible adoption. Future research should focus on developing transparent, reliable, and human-centered AI systems that can support complex decision-making in diverse environments.

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Keywords: Generative AI, LLM, NLP

**AI-Enabled Real-Time Predictive Framework for Enhancing Road Safety
in Smart Transportation Systems**

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ABSTRACT

Road safety has emerged as a critical global challenge, with increasing vehicular density, urbanization, and human behavioural factors contributing to a significant rise in traffic accidents and fatalities. Traditional traffic management systems remain largely reactive, lacking real-time intelligence and predictive capabilities. This paper proposes an AI-enabled, data-driven framework for proactive road safety management that integrates Internet of Things (IoT) sensors, computer vision, and machine learning algorithms to monitor, analyze, and mitigate road risks in real time. The system leverages heterogeneous data sources, including CCTV feeds, vehicular telemetry, weather conditions, and driver behavior patterns, to identify high-risk scenarios such as over-speeding, lane violations, distracted driving, and accident-prone zones. The proposed framework employs advanced deep learning models, such as convolutional neural networks (CNNs) for visual data analysis and gradient boosting techniques for risk prediction, enabling accurate detection and classification of hazardous events. Furthermore, a real-time alerting mechanism is incorporated to notify drivers and traffic authorities, thereby facilitating immediate intervention. The integration of edge computing ensures low-latency processing, making the system suitable for deployment in smart city environments. Experimental evaluation using simulated and real-world datasets demonstrates improved accuracy, reduced response time, and enhanced predictive capability compared to conventional systems. The novelty of this work lies in its unified architecture that combines predictive analytics with real-time enforcement, offering a scalable and efficient solution to current road safety challenges. Future work includes the integration of blockchain technology for secure data sharing among stakeholders, incorporation of vehicle-to-everything (V2X) communication, and adaptation of reinforcement learning models for dynamic traffic control optimization. The proposed system aims to significantly reduce accident rates and improve overall transportation safety in rapidly evolving urban ecosystems.

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Keywords: Road Safety, Artificial Intelligence (AI), Machine Learning, Computer Vision, Internet of Things (IoT), Smart Transportation, Accident Prediction, Real-Time Monitoring



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GC³I²–25	Dr. N. Revathi Adaptive Learning Systems Using Generative AI for Personalized Education
GC³I²–26	Suriya S Modelling Socio-economic Drivers of Peak Electricity Demand in Tamil Nadu Using Statistical and Machine Learning Approaches
GC³I²–27	Dr. S. Andar Recent AI in Education and Skills
GC³I²–28	Dr. Pushpalatha H Green, HRM Practices Towards Environmental Affairs in Organisation
GC³I²–29	Dr. S. Usharani, Nashath Rukhaiah Shaik AI-driven Blockchain Framework for Access Control, Anomaly Detection, and Secure Data Sharing
GC³I²–30	Dr. S. Chithra Smart Campus IoT Infrastructure for Enhanced Academic and Operational Efficiency.

Adaptive Learning Systems Using Generative AI for Personalized Education

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ABSTRACT

The rapid advancement of artificial intelligence has significantly transformed modern education systems, enabling the development of intelligent and personalized learning environments. This paper proposes an adaptive learning system powered by generative AI techniques to deliver customized educational content tailored to individual student needs. The system analyzes student performance, learning pace, preferences, and behavioral patterns using machine learning algorithms to dynamically generate and recommend appropriate learning materials.

Unlike traditional e-learning platforms that provide uniform content to all learners, the proposed framework continuously adapts the difficulty level and presentation style of content in real time. Generative AI models are utilized to create quizzes, summaries, explanations, and practice exercises that align with the learner's current knowledge level. The methodology involves data preprocessing, feature extraction, and the implementation of supervised and deep learning models to predict learner outcomes and optimize content delivery. Experimental evaluation is conducted using student datasets, and performance is measured based on accuracy, engagement rate, and learning improvement metrics. The results demonstrate that the adaptive system significantly enhances learning efficiency, retention, and student satisfaction compared to conventional e-learning methods.

This research highlights the potential of generative AI in revolutionizing education by promoting personalized learning experiences, reducing learning gaps, and supporting educators in delivering more effective instruction. The proposed system can be widely applied in schools, higher education institutions, and online learning platforms to improve overall educational outcomes.

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Keywords: Adaptive Learning, Personalized Education, Machine Learning, Data-Driven Learning, Artificial Intelligence in Education (AIED).

Modelling Socio-Economic Drivers of Peak Electricity Demand in Tamil Nadu Using Statistical and Machine Learning Approaches

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ABSTRACT

The demand for electricity at peak hours in Tamil Nadu has been increasing steadily over the years. This is largely because of rapid economic growth, urbanization, and changing patterns of consumption in the state. To improve demand forecasting and planning in the electricity sector, it is important that we understand the socio-economic factors that influence the demand. In this study, we aim to examine the influence of socio-economic factors such as income, family sizes, urban and rural locations, educational levels, appliance ownership, tariff structures, and subsidy policies on the demand for electricity in the state of Tamil Nadu. To achieve this objective, we have used recent survey data and demand records in the state. The socioeconomic factors like Urbanization, Population growth, Industrial development affects the Tamil Nadu's electricity demand. Because of peoples, Air conditioner, Machinery and Lighting are simultaneously used during hot afternoons. The Machinery feature which influences and ranked no 2 apart from the weather effect, by the regression analysis It is identified that quantifying demand increases per 1% economic growth. Based on the socioeconomic dataset the factory influencing results in 15-20% of evening spikes daily peaks matches the quarterly economic data. These features and factors improve the heat wave of Chennai by 32% and for 42% of the additional demand that weather model miss. In the analysis, we have attempted to estimate regression equations that explain the elasticity of demand with respect to socio-economic factors. From the analysis, we are expected to find that an increase in income, appliance ownership (especially cooling and lighting), and urbanization are important factors that influence the demand for electricity in the state.

Keywords: Socio economic factor, regression, demand forecasting, rural area.

Recent AI in Education and Skills

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Abstract

Recent developments in Artificial Intelligence (AI) are significantly transforming the field of education by introducing innovative tools and methods that enhance teaching and learning processes. AI in education refers to the use of intelligent systems, machine learning, and generative technologies to support personalized learning, automated assessment, content creation, and data-driven decision making. These technologies enable educators to design adaptive learning environments where instructional materials and learning activities can be tailored to the individual needs, abilities, and learning pace of students. For example, platforms such as Khan Academy use AI to provide personalized practice and feedback, while AI assistants like ChatGPT help learners understand complex concepts through instant explanations and interactive dialogue.

The integration of AI has also led to the development of new skills required for both teachers and learners. These include AI literacy, critical thinking, data interpretation, prompt engineering, ethical awareness, and the ability to collaborate with intelligent systems. Such competencies allow students to effectively interact with AI-based tools such as chatbots, intelligent tutoring systems, and generative AI platforms. For instance, language learning applications like Duolingo use AI to adapt lessons to the learner's progress and provide personalized feedback.

Furthermore, AI applications help automate administrative tasks like grading, feedback generation, and learning analytics, enabling teachers to focus more on creative instruction and student engagement. AI technologies also promote inclusive education by supporting diverse learning needs through translation tools, accessibility features, and adaptive learning platforms. Tools such as Canva also integrate AI features that assist teachers in creating engaging educational content such as presentations, worksheets, and visual learning materials.

Overall, the recent integration of AI in education is reshaping traditional educational practices by fostering personalized, efficient, and technology-enhanced learning environments.

Developing AI-related skills is therefore essential for preparing students and educators to adapt to the rapidly evolving digital and knowledge-based society.

Keywords

AI Literacy, Personalized Learning, Educational Technology, Intelligent Tutoring Systems, Digital Skills, and Adaptive Learning

Green HRM Practices Towards Environmental Affairs in Organisation

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Abstract

The role that green human resource Management (GHRM) practices play in the sustainable performance of an organization within environmental, social, and economic frameworks. The primary objective is to consolidate and structure disjointed discoveries in order to present both theoretical and practical knowledge. The review was conducted under the methodology of systematic literature. Findings identify three core GHRM practices, green training and development, green performance and management of sustainability outcomes. A combination of most of these practices also leads to the management of the environment, as well as the health of both employees and their financial performance.

INTRODUCTION:

In the present scenario, green HRM is an emerging topic. Green HRM was developed from the green movement around the world. The concept of green HRM has become popular worldwide. It differs from person to person. It attempts to boost energy savings or reduce the pollution created by industries. The primary aim of freeing is to minimize the possible effects of pollution on the environment and energy consumption. Today the topic green HRM not only includes awareness toward environmental affairs, but also stands for the social as well as economic well-being of both the organization and the employees within a broader prospect.

Green human resources management (GHRM) can be defined as a set of policies, practices, and systems that stimulate the green behaviour of a company's employees in order to create an environmentally sensitive, resource-efficient, and socially responsible organization. The word green HRM applies mainly to the contribution of policies and activities in the field of HRM. It is commonly known as green human resource management or GHRM. Green HRM is important and green HR helps in achieving greater efficiency and lower costs within a process, reducing and eliminating ecological wastage and refurbishing HR products, tools, and procedures. Being environmentally conscious, organizations are starting to integrate green attempts into their regular work environment with society.

Going “green” means to pursue knowledge and practice that can lead to more environmentally friendly and ecologically responsible decisions and lifestyles, which can help protect the environment and its natural resources for current and future generations.

Going “green” has four major implications regarding human resource management.

1. Protection of the natural environment
2. Taking care of the surroundings.
3. Creating green fields and natural looking places.

4. Minimization of green fields and natural looking places.
5. Maintain the natural ecological balance on earth so that all living things can survive and be in their habit.

BENEFITS OF GREEN HRM

Companies need to carry out an environmental audit that can focus on recycling and help society and its people. It will help the workers and members of organizations to understand the use of natural resources and promote eco-friendly goods. The benefits of green HRM are:

- a) It helps to achieve greater employee job satisfaction and dedication towards the work, and it leads to increased productivity
- b) Small companies can also reduce their electricity costs by using technologies that are energy-efficient and cost less.
- c) If an organisation adds a green initiative to its workplace, it will create some good public relations.
- d) It can support organizations to minimize expenses without losing talent.

FACTORS AFFECTING GREEN HRM PRACTICES

- Cost
- Employee perception and mindset
- Organization Culture
- Top management support
- Lack of organizational infrastructure and resources
- Government rules and regulations

GREEN HRM INITIATIVES

Some of the techniques adopted by organizations are:

- Plant trees and conduct green awareness for employees. Recycled water can also be used in gardens.
- Construct buildings, and workspaces that support a green architecture by allowing natural lighting, heating, and cooling and get it certified.
- Awareness and training about electronic waste, bio-medical waste for employees.
- Rewarding employees for coming up with sustainable green ideas and strategies

- Implementing a smoking free office.
- Discourage the use of paper or if required use recycled paper for filing of records and other pertinent information through the use of technology like electronic / digital filing instead of physical documents and file.
- Carpooling/ Bike pooling or sometimes provide transportation to employees for their daily commute to the workplace so as to save fuel, lessen traffic thereby ensuring faster commute and reduction in carbon emissions. Sometimes, even encourage employees to cycle to work, wherever possible.
- Minimum usage or nil usage of lights and allowing sunlight during the day wherever possible.
- Using energy efficient LED (Light Emitting Diode) bulbs in the night so as to save electricity.
- Use of energy efficient air conditioners and heaters Using recycled and refurbished products and ensuring products are disposed of for recycling.

OBJECTIVES:

1. To have an in-sight into the concept of green HRM
2. To overview the current practices GHRM practices in the organizations towards environmental affairs.
3. To suggest the organization and add a green initiative.

REVIEW OF LITERATURE:

Prasad (2013) defines green HRM as the contribution of hr policies to protect and preserve the natural resources. The prime aim of green HRM is sustainable development through human resource management. It has extended the boundaries of conventional HRM practices towards more sustainable and environmental strategies.

Sarode et. al.,(2016) today the topic green HRM not only includes awareness toward environmental affairs but also stands for the social as well as the economic well-being of both the organization and the employees within a broader.

Ahmad, (2015) it can be used to reduce costs, better efficiencies, to reduce carbon footprints, to make green awareness among the employees and initiate green work-life balance programmes

(nijhawan,2014). It has been stressed upon the need for developing the correlation between the existing HR policies and practices with green HRM principles for sustainable development. He suggests that green HRM can become a key business strategy for an organization with active participation by her department.

Makeover and pyke (2009), on the other hand, say that the green business demands a dedication towards financial gain, sustainability and humankind. The business dictionary states that the green business is “a business working in a way that has no harmful effects on the environment, the community, or the economy”, and also that “green business partake in creating approaches that include the pro-environmental issues and policies that affect the human rights” (business dictionary,).

Croston (2009) writes that the green organizations integrate more sustainable business activities that the rivals, enhance the natural environment and improve the living conditions, while profiting and adding to the economy.

FUNCTIONS OF GREEN HRM:

The main objective of green HRM is to make the employees aware of the intricacies of environment management i.e. what action is needed, how it functions, and how it helps the environment. The exercise really motivates the employees and develops a sense of pride in them for being a part of the going green program.

Green training and Development

Green training and development is defined as the process of equipping employees with working approaches that ensure adequate resource utilisation, reduce waste, energy conservation, and environmental degradation cause reduction. Training and development is a practice that focuses on development of employees' skills, knowledge, and attitudes, preventing deterioration of em-related knowledge, skills, and attitudes (zoogah, 2011 p. 17). Green training and development educate employees about the value of training them in working methods that conserve energy, reduce waste, diffuse environmental awareness within the organization, and provide opportunities to engage employees in environmental problem-solving. Training is a field where the part of human resource management in ecological management it's been recognized for quite a while. Besides, Fernandez, Junquera and Ortiz (2003) stated that establishing an ecological methodology needs increased employees' awareness and learning of procedures and abilities – which means representatives need “integrated training” and they need to demonstrate a contribution to achieve goals.

Green compensation:The term green compensation can be described as reward and compensation packages utilised to influence employees' green practices. Rewards and compensation are the major HRM processes through which employees are rewarded for their performance. These practices are the most powerful method which links together an individual's interest to that of the organization's. It is also asserted that incentives and rewards can influence employees' attention to the maximum at work and motivate them to exert

maximum effort on their part to achieve organizational goals. Employee compensation programs can be modified to give bonuses based in part on the employee's appraisal ratings on the behavioural and technical competencies. In addition, employees could be awarded bonuses for their outstanding work on special projects (liebowitz, 2010,).

Green rewards can include the use of workplace and lifestyle benefits, ranging from carbon credit offsets to free bicycles, to engage people in the green agenda while continuing to recognize their contribution (pillai & sivathanu, 2014).

Green employee relations

Employee relations are that aspect of HRM which is concerned with establishing an amicable employer–employee relationship. The relationship facilitates motivation and morale of the employees as well as increases productivity. Basically, employee relations involve employee participation and empowerment activities. It also helps prevent and resolve problems that arise at the workplace that may affect the work. In fact, positive employee relations are an intangible and enduring asset and a source of competitive advantage for any organization.

Green employee participation

Employee participation in green initiatives increases the chances of better green management as it aligns employees' goals, capabilities, motivations, and perceptions with green management practices and systems. Involving employees in emphasis has been reported as improving em systems such as efficient resource usage (florida & davison, 2001); reducing waste (may & flannery, 1995) and reducing pollution from workplaces (kitazawa & sarkis,2000). Several workers in their study concluded that individual empowerment positively influences productivity and performance, and facilitates self-control, individual thinking, and problem-solving skills (renwick, 2008; wee & quazi, 2005).

An important way in which employee involvement and participation can be encouraged within the organization is to seek entrepreneurs within the company who are socially or ecologically oriented known as eco-entrepreneurs (Mendip, 2012 p. 246). Eco-friendly ideas should be welcomed from all employees irrespective of their designation which will encourage their interest in environmental issues and make best use of applying their skills.

GREEN INITIATIVES FOR HR

lado and wilson (1994) defines HRM system as a set of distinct but interrelated activities, functions, and processes that aims to attract, develop, and maintain a firm's human resource. Organizations generally organize their practices into systems that are consistent with their culture and business strategy (boselie et al., 2001).we can say that green initiatives included in the HRM manifesto are a part of corporate social responsibility in the long run. Today,

organizations are implementing and integrating green initiatives in their agenda with the help of their human resource. Managers make sure that their HR is utilizing green human resource practices in an appropriate manner. As an addendum to the statement, several authors have suggested that it is important to promote a great deal of technical and management skills among all employees of the organization in order to implement an effective corporate green management system in companies (daily, bishop, & govindarajulu, 2009, unnikrishnan & hegde, 2007)

CONCLUSION:

The human resource is the most important asset of an organization that plays an important role in managing the employee in an Organization. GHRM contributes to the HRM policies and practices towards corporate environmental affairs. It includes sustainable practices and commitments on sustainability issues. Green HR is the usage of HRM approaches to realize the feasible utilization of assets inside a business unit and develops the cause for natural sustainability. Contemporary Corporate can link the workforce with a mission to save a stable environment and improve the service sector towards a green society. As we look to the future, the importance of Green HRM will only continue to grow, making it an essential consideration for all businesses.

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Websites:

<https://www.iedunote.com/green-hrm>

**AI-Driven Blockchain Framework for Access Control, Anomaly Detection,
and Secure Data Sharing**

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¹ Ph.D., Professor / Head, Viswam Engineering College, India

² Viswam Engineering College, India
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ABSTRACT

The exponential growth of distributed digital ecosystems has intensified concerns around data privacy, access control, and anomaly detection. Traditional centralized security models suffer from limited transparency, scalability issues, and vulnerability to single points of failure. This paper introduces an AI-driven blockchain framework that synergizes the predictive intelligence of Artificial Intelligence (AI) with the immutability and decentralization of Blockchain technology.

The proposed architecture leverages smart contracts for autonomous access control and integrates machine learning models for real-time anomaly detection and adaptive policy enforcement. Explainable AI (XAI) techniques ensure interpretability of decisions, enhancing trust and compliance. Experimental results demonstrate 95.2% anomaly detection accuracy, a 42% improvement in unauthorized access prevention, and access latency reduced to ~250 ms per transaction.

Unlike conventional systems, this framework offers a unified, interpretable, and decentralized solution for secure data sharing across heterogeneous environments. It ensures scalability, transparency, and resilience against evolving cyber threats. The integration of AI and blockchain presents a promising paradigm for building intelligent, tamper-proof, and self-regulating security infrastructures in modern data-driven applications.

Keywords: AI, Blockchain, Access Control, Anomaly Detection, Smart Contracts, Explainable AI, Secure Data Sharing.

Smart Campus IoT Infrastructure for Enhanced Academic and Operational Efficiency

Dr. S. Chithra

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chithras@srmist.edu.in

ABSTRACT

Universities and colleges often face challenges in managing energy consumption, resource allocation, and real-time communication between administration, faculty, and students. Internet of Things (IoT) technologies presents an opportunity to create a smart campus environment that can optimize facilities management, improve learning experiences, and enhance safety. This project proposes an IoT-enabled smart campus ecosystem that integrates smart energy management, environmental monitoring, classroom utilization tracking, and connected campus services. By leveraging IoT sensors, edge computing, and centralized dashboards, the project will deliver operational efficiency and improved academic support.

Keywords: IoT, Sensor, Edge Computing, Dash Board

Multi-Class Fruit Image Classification Using Cnn

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ABSTRACT

Image classification is an important task in computer vision that involves identifying and categorizing objects in images. Fruit classification is an important task in agriculture, food processing, and automated quality inspection systems. This research proposes a **multi-class fruit image classification system using CNN** to automatically identify different types of fruits based on their visual characteristics. The goal of the system is to automatically identify different types of fruits from images. The model learns visual patterns such as color, shape, and texture from fruit images and classifies them into multiple categories. The dataset used consists of labeled images of various fruits such as apples, bananas, oranges, and grapes. CNN models are highly effective for image classification because they automatically extract features from images without manual intervention. A CNN architecture is designed with convolutional layers, pooling layers, and fully connected layers to extract meaningful features from the fruit images. The model is trained and evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score. This paper demonstrates the effectiveness of deep learning in solving real-world image classification problems and can be applied in agriculture, supermarkets, and automated fruit sorting systems.

Keywords: Convolutional Neural Networks (CNN), Image Classification System, deep learning

Smart E-Commerce Price Comparison Platform

Shaik Salma Anjum¹, M. Gowthami²

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² Assistant Professor, Department of MCA, Viswam Engineering College, Madanapalle, India
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ABSTRACT

The exponential growth of online retail platforms has transformed consumer purchasing behaviour, creating a demand for efficient tools to compare product prices across multiple vendors. However, manually navigating different websites to identify the most economical option is both time-consuming and inefficient. This paper presents a Smart E-Commerce Price Comparison Platform that aggregates product data and provides a unified interface for comparing prices from various sellers. The system is designed using a multi-layered architecture incorporating frontend technologies such as HTML, CSS, Bootstrap, and JavaScript, along with a backend developed using Python and Django. SQLite3 is used for data storage. The proposed platform enhances user convenience, reduces search time, and improves decision-making efficiency. Experimental evaluation indicates improved usability and faster response times compared to traditional manual comparison methods.

Keywords: E-commerce, Price Comparison System, Web Application, Django Framework, Product Search, Full Stack Development

BEST PAPER AWARD WINNERS GC³I² – 2026

Recognizing Excellence in Research & Innovation

S.No	Author(s)	Affiliation	Paper Title
1	Amulya Gollavilli & Dr. M. Karthikeyan	SRM University, Vadapalani Campus, Chennai	Enhancing Investor Trust in Robo-Advisory Systems Through Explainable AI
2	Dr. Gyaneshwari Vyas	Principal, Gurukul College, Bhopal, M.P., India	Role of Green Technologies in Achieving Sustainable Development Goals (SDGs): A Study with Special Reference to D.El.Ed and B.Ed Students
3	Ms. Deepthi M & Dr. Niviya Feston	Department of Psychology, Garden City University, Bangalore	Influence of Socio-Demographic Factors on Depression and Sleep Quality among Working Women: A Cross-Sectional Study
4	Mithun Meenakshi S & Dr. Shanmugam DB	Department of Data Science, SRM Institute of Science and Technology	Comparing Physics-Informed and Physics-Enforced Neural Networks for the Chaotic Three-Body Problem
5	Prof. (Dr.) Bharat Raj Singh & Dr. Asha Kulshrestha	School of Management Sciences, Lucknow	Intelligent Sustainable Technologies and Green Energy Systems for Environmental Resilience and Climate-Smart Development



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GLOBAL EXCELLENCE AWARDS – 2026

GC³I² – 2026

The Global Excellence Awards – 2026, hosted by Edinz Tech Private Limited, an Inspire Softech Group Company, are presented as a distinguished recognition platform under the Global Conference & Excellence Awards on Innovation, Intelligence & Impact (GC³I² – 2026).

These awards honour individuals and institutions that have demonstrated exceptional contributions across diverse domains including healthcare, education, research, technology, and industry, reflecting excellence, innovation, leadership, and societal impact.

The 2026 edition witnessed an overwhelming response, with over 320+ nominations received from across the globe. Each profile underwent a rigorous double-blinded evaluation process, complemented by comprehensive background verification to ensure authenticity, merit, and credibility.

The selection process was conducted by an expert jury panel using a structured evaluation framework that assessed domain expertise, research contributions, innovation, professional achievements, and societal impact.

In addition to recognition, a unique feature of the evaluation process was the provision of detailed review reports, offering constructive feedback and scope for enhancement, thereby encouraging continuous growth and excellence among nominees.

The awardees presented in this section represent a distinguished cohort of professionals and institutions whose work exemplifies the spirit of the conference theme — Innovation, Intelligence & Impact.

Through the Global Excellence Awards – 2026, Edinz Tech Private Limited and Inspire Softech Group Companies reaffirm their commitment to fostering a culture of recognition and strengthening the synergy between academia, industry, and society.

We extend our heartfelt congratulations to all awardees for their outstanding accomplishments and contributions.

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**GC³I² – 2026
Awardees Index**

S. No.	Name	Qualification	Designation	Affiliation	Award Name	Page No.
1	Dr. Swetha S	M.D. (Respiratory Medicine)	Director & Consultant Pulmonologist	Ramani Respiratory Clinic, Chennai, India	Excellence in Clinical & Academic Pulmonology	
2	Dr. M. Sharmila Devi	M.S. (Ophthalmology), FMRF	Consultant Ophthalmologist	Vetrichelvi Anbazhagan Eye Hospital, Chennai, India	Excellence in Compassionate Vision Care	
3	Dr. K. Shanmugapriya	M.B.B.S., M.D.	Associate Professor – Pulmonology	Sri Lalithambigai Medical College and Hospital, Chennai, India	Excellence in Medical Education & Research	
4	Dr. Nandini	MBBS, MD	Assistant Professor	Shri Sathya Sai Medical College and Hospital, Chennai, India	Excellence in Medical Education & Research	
5	Dr. Sravani Chithra Ch	MD	Associate Professor	Sri Muthukumaran Medical College Hospital, Chennai, India	Excellence in Medical Research	
6	Dr. Vijaiyalakshmi Praveen	MBBS, MD	Assistant Professor & Psychotherapist	Makkal Nalam Clinic, Chennai, India	Innovation Leader in Public Health	
7	Dr. Saranya Chithra	MS	Associate Professor, ENT	Sri Muthukumaran Medical College Hospital, Chennai, India	Innovation in Medical Technology	
8	Dr. Prasanth G	MD, DNB	Consultant Pulmonologist	Prana ENT and Chest Clinic, Chennai, India	Young Medical Professional of the Year	
9	Dr. R. Ravanan	M.Sc., M.Phil., Ph.D.	Vice President / Former Principal	Presidency College, Chennai, India	Lifetime Contribution to Higher Education	
10	Dr. S. Uma	Ph.D.	Professor & Head	Dr. N.G.P. Arts and Science College, Coimbatore, India	Best Academician of the Year	
11	Dr. Thailambal G	Ph.D.	Professor & Head	VISTAS, Chennai, India	Excellence in Teaching & Learning	
12	Dr. A. Akila	Ph.D.	Associate Professor & Associate Dean	VISTAS, Chennai, India	Outstanding Teaching Excellence Award	

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13	Dr. S. Andal	Ph.D.	Assistant Professor	Tamil Nadu Teachers Education University, Chennai, India	Excellence in Teacher Education & Academic Service
14	Mr. Brahmam Odugu	M.Sc., MCA, M.Ed.	Mathematics Faculty Coordinator	New Era High School, Maharashtra, India	Excellence in Mathematics Education
15	Dr. S. Uma Shankari	Ph.D.	Associate Professor	SRM Institute of Science and Technology, Chennai, India	Excellence in Research & Academic Innovation
16	Dr. K. Sony	Ph.D.	Assistant Director & Assistant Professor	KL University, India	Emerging Researcher Excellence
17	Dr. Manikandan S	Ph.D.	Assistant Professor	Sree Balaji Institute of Science and Technology, Chennai, India	Young Researcher Excellence Award
18	Dr. Dhaval Pravinbhai Soneri	Ph.D.	Principal	Shri Vivekananda Pre School, Gujarat, India	Outstanding Contribution to Educational Research
19	Dr. S. Perumal	Ph.D.	Professor	VISTAS, Chennai, India	Excellence in Computer Science Research & Teaching
20	Dr. K. Sharmila	Ph.D.	Professor	VISTAS, Chennai, India	Excellence in Artificial Intelligence & Data Science Research
21	Dr. R. Padma	Ph.D.	Assistant Professor	VISTAS, Chennai, India	Excellence in Artificial Intelligence & Data Science Research
22	Dr. R. Priscilla	Ph.D.	Professor & Head	St. Joseph's Institute of Technology, Chennai, India	Excellence in Artificial Intelligence & Data Science Research
23	Dr. S. Vidhushavarshini	Ph.D.	Associate Professor	Rajalakshmi Institute of Technology, Chennai, India	Excellence in AI & Healthcare Data Science Research
24	Mr. Yashwanth A S	M.Com.	CEO & Founder Director	Y17J Cyber Security LLP, Bangalore, India	Excellence in Cyber Security & Legal Innovation
25	Dr. S. Umarani	MCA., Ph.D.	Professor	SRM Institute of Science and Technology, Chennai, India	Technical Advisory Excellence Award
26	Dr. K. Amutha	Ph.D.	Associate Professor	PSG College of Arts & Science,	Excellence in Commerce &

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				Coimbatore, India	Financial Research
27	Dr. Pushpalatha H	Ph.D.	Assistant Professor	Anna Adarsh College for Women, Chennai, India	Excellence in Commerce Education & Academic Contribution
28	Dr. T. Deepa Augustine	Ph.D.	Assistant Professor & Head	Annai Violet Arts & Science College, Chennai, India	Excellence in Commerce & Management Education
29	Dr. Baby Niviya Feston	Ph.D.	Dean – Research & Innovation	Garden City University, Bangalore, India	Excellence in Management Education & Research
30	V. Venkatasubramanian	—	CEO & Managing Director	Devconsol Pvt Ltd, Chennai, India	Emerging Leader in AI & Data- Driven Enterprise Innovation
31	Mr. Sashikumar VS	—	Managing Director	Praanha Marcomm Solutions, Chennai, India	Excellence in Brand Strategy & Integrated Marketing Communications
32	Dr. C. Sundar	Ph.D.	Principal	Dhanraj Baid Jain College, Chennai, India	Distinguished Academic Leadership Award
33	Dr. V. Jayalakshmi	Ph.D.	Director – Digital Infrastructure & Professor	VISTAS, Chennai, India	Lifetime Achievement in Academic Leadership & Research Excellence
34	Dr. A. Antony Prakash	Ph.D.	Head & Assistant Professor	St. Joseph's College, Trichy, India	Excellence in Academic Leadership & Computer Science Education
35	Prof. Dr. J. Indumathi	Ph.D., D.Sc.	Dean – Artificial Intelligence	Sree Balaji Institute of Science and Technology, Chennai, India	Distinguished Academic Leadership & Research Excellence in AI
36	Dr. Aneeshkumar A.S	Ph.D.	Associate Professor & Head	AJK College of Arts and Science, Coimbatore, India	Excellence in Research & Academic Leadership
37	Dr. S. Usharani	Ph.D.	Professor & Head	Viswam Engineering College, Andhra Pradesh, India	Best Faculty & Academic Leadership Award
38	Dr. R. Parameswari	Ph.D.	Professor & Head	VISTAS, Chennai, India	Best Faculty & Excellence in Academic Leadership Award

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39	Dr. G. Fathima	Ph.D.	Professor & Head	Adhiyamaan College of Engineering, Hosur, India	EXCELLENCE IN INDUSTRY-ORIENTED EDUCATION
39	Sri Balaji Institute of Science and Technology	—	Institution	Chennai, India	Global Emerging Leader in Engineering & Technology Education Award
40	Mr. S. Aravind	—	Head – Learning & Development	Edinz Tech Private Limited, Chennai, India	Best Employee of the Year



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**EXCELLENCE IN CLINICAL & ACADEMIC
PULMONOLOGY**



Dr. Swetha S, M.D. (Respiratory Medicine),
Director and Consultant Pulmonologist,
Ramani Respiratory Clinic,
Chennai, India.

Dr. Swetha S is a distinguished pulmonologist and healthcare leader serving as Director and Consultant Pulmonologist at Ramani Respiratory Clinic, Chennai, with 7 years of clinical experience. She has made valuable contributions to the field through various research publications that have advanced understanding in respiratory medicine. Her commitment to healthcare extends beyond the clinic, as she actively organizes free healthcare camps and engages in social activities aimed at promoting public health awareness. Her dual role as a clinical expert and community health advocate reflects her holistic approach to healthcare, combining scientific rigor with compassionate service to patients and society.

EXCELLENCE IN COMPASSIONATE VISION CARE



Dr. M. Sharmila Devi, M.S. (Ophthalmology), FMRF,
Consultant Ophthalmologist,
Vetrichelvi Anbazhagan Eye Hospital,
Chennai, India.

Dr. M. Sharmila Devi is a distinguished Consultant Ophthalmologist born in Chennai. A District First Rank holder in her 12th grade and a Gold Medalist from Sri Ramachandra Medical College, she has consistently demonstrated academic and clinical excellence. She completed her Retina specialty training at the prestigious Sankara Nethralaya and has over 11 years of experience delivering advanced, compassionate, and patient-centered eye care. Her commitment to accessible healthcare is demonstrated through the numerous free cataract surgery camps she has organized, bringing quality vision care to underserved communities. Her combination of surgical precision, academic excellence, and compassionate patient care makes her a role model in ophthalmology.

EXCELLENCE IN MEDICAL EDUCATION & RESEARCH



Dr. ShanmugaPriya M.B.B.S., M.D.

Associate Professor -pulmonology

Sri Lalithambigai Medical College and Hospital

Chennai,India

Dr. K. Shanmugapriya, M.B.B.S., M.D. (TB & Pulmonary Medicine) is a dedicated medical professional and academician specializing in Respiratory Medicine, with extensive clinical, teaching, and research experience. She currently serves as Associate Professor in the Department of Respiratory Medicine, contributing actively to medical education and patient care. Her expertise includes pulmonary diseases, tuberculosis management, chronic respiratory disorders, and clinical research. She has published multiple research papers in reputed medical journals, focusing on areas such as COPD, pulmonary tuberculosis, lung diseases, and respiratory diagnostics. Dr. Shanmugapriya has actively participated in national and international conferences, including NAPCON, where she has contributed as a presenter, faculty member, and chairperson. She has also served as an external examiner in reputed medical institutions, reflecting her academic credibility. With a strong commitment to healthcare excellence, teaching, and research, she continues to contribute to advancements in respiratory medicine while mentoring future medical professionals and enhancing patient outcomes.

**EXCELLENCE IN MEDICAL EDUCATION &
RESEARCH**



Dr. Nandini, MBBS, MD,

Assistant Professor, Department of Microbiology,
Shri Sathya Sai Medical College and Hospital and Research Institute,
Chennai, India.

Dr. Nandini is an Assistant Professor of Microbiology with over 7 years of teaching experience. She completed her MBBS from Saratov State Medical University and her MD in Microbiology from SRM Medical College Hospital and Research Centre. Her areas of interest include mycology, infectious diseases, and diabetic-associated infections. She has multiple national and international publications, including studies on *Candida* in immunocompromised patients, invasive fungal infection genomics, disinfectant efficacy in operation theatres, and a tuberculosis case report. She is the recipient of an institutional funded research project on platelet-based antimicrobial therapy and is currently working on antimicrobial resistance in diabetic uropathogens. She is trained in medical education, biomedical research, hospital infection control, and ISO 15189:2022 Quality Management Systems. She has served as Guest Speaker in Hospital Infection Control and as a Judge for Medstar Summit 2024 and 2025.

EXCELLENCE IN MEDICAL RESEARCH



Dr. Sravani Chithra Ch, MD,

Associate Professor, Department of Obstetrics and Gynaecology,
Sri Muthukumaran Medical College Hospital and Research Institute,
Chennai, India.

Dr. Sravani Chithra Ch is an Associate Professor of Obstetrics and Gynaecology with 14 years of clinical and academic experience. She has completed Fellowships in Reproductive Medicine and Minimal Access Surgery. She has authored a chapter in 'Mastering the Techniques in Hysteroscopy' by Osama Shawki, contributed to the 13th edition of Mudaliar and Menon's Clinical Obstetrics textbook, and authored a chapter in the FOGSI Handbook on Labour Management. She has made 25 presentations at national and international conferences and has 24 publications in reputed national and international journals with 4 ongoing research papers. She is a member of FICS, FOGSI, WALIS, IFS, IAGE, and ISAR. She has received the ISAR Youth Icon Award by the Indian Society of Assisted Reproduction and the ISSN International Outstanding Researcher Award.

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INNOVATION LEADER IN PUBLIC HEALTH



Dr. Vijaiyalakshmi Praveen, MBBS, MD,

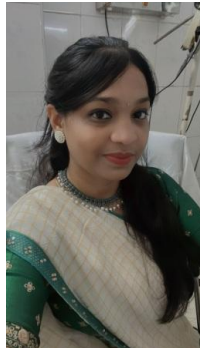
Assistant Professor of Community Medicine & Certified Psychotherapist,

Makkal Nalam Clinic, CIT Colony, Mylapore,

Chennai, India.

Dr. Vijaiyalakshmi Praveen is a Medical Doctor, Assistant Professor of Community Medicine, and Certified Psychotherapist with a unique and impactful approach to mental health and community wellness. She brings a compassionate and innovative lens to healthcare, combining medical expertise with psychotherapy to address a wide spectrum of patient needs. Through 3 years of dedicated talk therapy practice, she has made a meaningful difference in the lives of diverse individuals including school students overcoming academic anxiety, medical professionals dealing with workplace challenges, new mothers healing from childhood trauma, and young adults navigating mental health difficulties. Her work bridges the gap between community medicine and mental health, reflecting a forward-thinking approach to holistic patient care.

INNOVATION IN MEDICAL TECHNOLOGY



Dr. Saranya Chithra, MS,

Associate Professor, Department of ENT,

Sri Muthukumaran Medical College Hospital and Research Institute,

Chennai, India.

Dr. Saranya Chithra is an ENT surgeon, Associate Professor, researcher, and healthcare innovator with over 10 years of dedicated service in advancing medical education and patient care through technology-driven solutions. She was honored with the International Best Researcher Award in March 2025. She has authored 10 national publications and currently guides multiple research projects, including serving as a guide for an ICMR national project on biofilms. She is developing a Virtual Reality-based surgical training model for postgraduate ear surgery to enhance precision and skill acquisition through immersive technology. She has also designed a SmartBox medication dispenser aimed at improving medication adherence and safety among elderly patients. She holds a postgraduate qualification in Healthcare Management and is actively involved in NAAC-related student progression initiatives and the Medical Education Unit. She has mentored over 2,000 medical students in her academic career.

YOUNG MEDICAL PROFESSIONAL OF THE YEAR



Dr. Prasanth G, MD, DNB (Respiratory Medicine),
Consultant & Interventional Pulmonologist,
Prana ENT and Chest Clinic, Kolathur,
Chennai, India.

Dr. Prasanth G is a dedicated medical professional with advanced training and 8 years of clinical experience in Pulmonology. He specializes in managing complex respiratory disorders, interstitial lung diseases, drug-resistant infections, and critically ill patients in high-dependency and ICU settings. He has been actively involved in postgraduate teaching for the past 5 years and served as Assistant Head of Department in the Respiratory Medicine department of a tertiary care medical college hospital. He has over 20 publications in national and international journals, reflecting his commitment to evidence-based clinical practice and academic contribution. His strong analytical skills in clinical decision-making and patient-centered care philosophy make him an exemplary medical professional in the field of pulmonology.

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LIFETIME CONTRIBUTION TO HIGHER EDUCATION



Dr. R. Ravanan, M.Sc., M.Phil., Ph.D., FSMS.,

Vice President, World University Service (WUS) Centre; Joint Director (P&D) (Retd.),

Directorate of Collegiate Education,

Former Principal, Presidency College,

Chennai, India.

Dr. R. Ravanan is a distinguished statistician and educational administrator who began his academic journey in 1990 with the Tamil Nadu Government Collegiate Service. He served as Principal of Presidency College, Chennai — one of India's premier institutions — under whose leadership the college secured the 3rd Rank in the All India NIRF Rankings 2019. He has guided 19 M.Phil. and 15 Ph.D. scholars and authored 6 books, including the Tamil Nadu Higher Secondary Statistics textbooks. He has delivered over 410 invited lectures on statistical applications using SPSS and AMOS across institutions in medicine, engineering, and social sciences. He was honoured with the Tamil Nadu Scientists Award (TANSA) 2019 in Mathematical Sciences by the Tamil Nadu State Council for Science and Technology. His other prestigious awards include the Dr. A.P.J. Abdul Kalam Life Time Achievement National Award (2020), the Dr. Radhakrishnan Teaching Excellence Award (Lions Club, 2018 & 2019), the Pillars of India Award (2022), and the Life Time Achievement Award in Statistics Education. He has served as Secretary of the Indian Society for Probability and Statistics (ISPS) and General Secretary of the Tamil Nadu Statistical Association. He retired as Joint Director (Planning and Development), Directorate of Collegiate Education, Chennai.

BEST ACADEMICIAN OF THE YEAR



Dr. S. Uma, Ph.D.,

Professor and Head, Department of Computer Science,

Dr. N.G.P. Arts and Science College,

Coimbatore, India.

Dr. S. Uma is an accomplished educator and researcher with over 21 years of teaching experience and 16 years of sustained research engagement. Her vision is to transform classrooms into inclusive, research-driven learning ecosystems that cultivate innovation, leadership, and academic excellence. By integrating outcome-based pedagogy with structured research mentorship, she has fostered a strong postgraduate research culture, reflected in consecutive TNSTS-funded projects and the consistent submission of more than 32 PG research proposals annually. She has guided students to secure prestigious internships at IITs, achieve Top 2% national rankings in NPTEL, and participate in DBT-supported interdisciplinary research initiatives. Her scholarly contributions include publications in Scopus, Web of Science, and IEEE-indexed journals, along with national and international patents. She is a recipient of the Best Senior Faculty and Excellence in Innovative Teaching awards and serves as an active mentor for women students pursuing higher education.

EXCELLENCE IN TEACHING & LEARNING



Dr. Thailambal G, Ph.D.,

Professor and Head, Department of Advanced Computing and Analytics,
Vels Institute of Science, Technology and Advanced Studies (VISTAS),
Chennai, India.

Dr. Thailambal G is a seasoned academician with over 21 years of service at VISTAS. She completed her UG and PG studies at Bharathidasan University, her M.Phil. at Periyar University, and her Ph.D. at Sankara University, Kancheepuram. Since joining VISTAS in 2005, she has been a pillar of the institution's academic community, consistently motivating and guiding students to excel in the field of computer science. Her passion lies in classroom teaching and mentoring students to achieve their full potential. Her two decades of dedicated service stand as a testimony to her commitment to education, skill development, and student empowerment in advanced computing disciplines.

OUTSTANDING TEACHING EXCELLENCE AWARD



Dr. A. Akila, Ph.D.,

Associate Professor & Associate Dean – Digital Infrastructure,
Department of Computer Science and Information Technology,
Vels Institute of Science, Technology and Advanced Studies (VISTAS),
Chennai, India.

Dr. A. Akila is an accomplished academician and researcher with 16 years of academic experience and 3 years of industry experience. She completed her Ph.D. in Computer Science on Tamil-based Speech Recognition from Bharathiar University in 2014. Her research has resulted in over 65 published articles in international journals including Web of Science, Scopus, and UGC-approved journals. She holds 1 granted patent and has filed 2 more. She has guided 5 Ph.D. scholars and is currently guiding 4. She has delivered 10 E-content Edutech lecture videos and established a Faculty Club to coach teachers through external expertise. She has received several awards including the Best Women Faculty Award from DK International Research Foundation, the Young Educator & Scholar Award from NFED, and an Appreciation Award for Article Publication from VISTAS. She has visited institutions in Thailand and Dubai, reflecting her global academic engagement.

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**EXCELLENCE IN TEACHER EDUCATION &
ACADEMIC SERVICE**



Dr. S. Andal, Ph.D.,

Assistant Professor, Department of Educational Planning and Administration,
Tamil Nadu Teachers Education University, Karapakkam,
Chennai, India.

Dr. S. Andal is a dedicated academician with 21 years of teaching experience, including 8 years as a Matriculation School Teacher, 1 year as a Resource Person, 18 years at Annamalai University, and over 2 years at Tamil Nadu Teachers Education University. She holds a Ph.D. and has 20 years of research experience. She serves as a Question Setter for TNPSC examinations, reflecting her recognition as a subject matter expert. She is a member of the UWA-Chidambaram Chapter and the IIPian group in EUDOXIA, and is actively engaged in activities related to Apex Sustainable Development. Her diverse academic roles across school and university levels, combined with sustained research engagement and community involvement, exemplify a multifaceted and impactful academic career.

EXCELLENCE IN MATHEMATICS EDUCATION

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Mr. Brahman Odugu, M.Sc. (Maths), MCA, M.Sc. (Psychology), M.Ed., (Ph.D. Scholar),

Mathematics Faculty Coordinator,

New Era High School, Panchgani, Satara District,

Maharashtra, India.

Mr. Brahman Odugu is a Mathematics Educator, Research Mentor, and Ph.D. Scholar with over 18 years of experience in teaching, research, and academic mentorship. His multidisciplinary academic background enables him to integrate mathematics, psychology, and technology into innovative teaching practices. He is widely recognized for promoting research-based learning at the school level, mentoring students to conduct original research and publish in international journals. He has introduced innovative mathematical concepts including the 'Brahman Mirror Number' and 'Brahman Equilibrium Theorem.' He founded PRIME LAB (Programme for Research, Innovation, and Mathematical Excellence) to cultivate a research culture among school students. He is the recipient of the EducationWorld India TREE Awards 2026-27 (Progressive Researcher) and the India Elite Education & Institutional Excellence Award 2026 for Excellence in Mathematics Education & Academics.



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**EXCELLENCE IN RESEARCH & ACADEMIC
INNOVATION**



Dr. S. Uma Shankari, Ph.D.,

Associate Professor & Program Coordinator (BCA), Department of Computer
Applications,
SRM Institute of Science and Technology, Ramapuram Campus,
Chennai, Tamil Nadu, India.

Dr. S. Uma Shankari is a dedicated academician and researcher with 22 years of teaching experience and 10 years of focused research engagement. She holds a Ph.D. in Computer Science from Bharathiar University (2020), an M.Phil. from Periyar University, and an MCA from the University of Madras. She has published over 30 research articles in Scopus, Web of Science, and UGC-CARE indexed journals covering recommendation systems, machine learning analytics, image processing, and biomedical applications, and is currently guiding 6 research scholars. She received the NPTEL Star Discipline Award in 2025 for completing 50+ credits in computer science. She was a Grand Finale Winner for the Smart India Hackathon (December 2025), and received the Mentorship Excellence Award from ICTACT for mentoring 500+ students to complete MongoDB Certification. She is also NET and SET qualified and holds an M.Sc. in Yoga.

EMERGING RESEARCHER EXCELLENCE

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Dr. K. Sony, Ph.D.,

Assistant Director, Alumni Relations; Assistant Professor, ECE Department,
Koneru Lakshmaiah Education Foundation (K L University),
India.

Dr. K. Sony is a researcher and academician specializing in Free Space Optics (FSO) with over 13 years of academic experience. Her research interests encompass optical and wireless networks, atmospheric effects on FSO links, system modeling, and hybrid communication networks. She has published 20+ research papers in Scopus and Web of Science indexed journals and has presented at various national and international conferences. Beyond teaching, she actively contributes to institutional initiatives through the Women Development Cell and as Assistant Director of Alumni Relations. She is deeply committed to mentoring students in research and innovation and nurturing critical thinking to prepare them for professional and academic pursuits. Her vision is to broaden collaborative research in optical communications and integrate emerging wireless technologies to address real-world communication challenges.

YOUNG RESEARCHER EXCELLENCE AWARD



Dr. Manikandan S, Ph.D.,

Assistant Professor, Department of Chemistry,
Sree Balaji Institute of Science and Technology,
Chennai, India.

Dr. Manikandan S is an early-career researcher specializing in advanced functional materials for energy and electronic applications, with particular focus on thermoelectric, ferroelectric, and polymer-based materials. His expertise lies in the design, synthesis, and characterization of inorganic and polymeric materials, including perovskite oxides, polymer composites, and flexible functional films. He has active research involvement in thermoelectric energy harvesting, smart materials, and sustainable energy technologies, aiming to develop flexible and low-carbon energy solutions. He has contributed to interdisciplinary research spanning physics, chemistry, and materials science, with applications in sensors, smart windows, and flexible electronics. His published work covers polymer-based thermoelectric materials, ferroelectric and piezoelectric PVDF-based composite films, and double perovskite oxide materials, contributing to next-generation electronic and energy device applications.

**OUTSTANDING CONTRIBUTION TO EDUCATIONAL
RESEARCH**



Dr. Dhaval Pravinbhai Soneri, Ph.D.,

Principal,

Shri Vivekananda Pre School,

Matar, Gujarat, India.

Dr. Dhaval Pravinbhai Soneri is an educationist and researcher with 16 years of teaching experience and 6 years of dedicated research engagement. He holds a Ph.D. degree and has made meaningful contributions to educational research, with findings that were reported to and positively received by the Gujarat Government for potential policy implementation. He has provided honorary teaching service for 2 years in Government schools during a teacher shortage period, reflecting his deep commitment to the cause of education. He has received the Best International Paper Presentation Award, recognizing his contributions to scholarly discourse. His work bridges academic research and real-world educational policy, demonstrating the impactful role that education researchers can play in shaping institutional and governmental frameworks.



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**EXCELLENCE IN COMPUTER SCIENCE RESEARCH &
TEACHING**



Dr. S. Perumal, Ph.D.,

Professor, Department of Computer Science & IT,
Vels Institute of Science, Technology and Advanced Studies (VISTAS),
Chennai, India.

Dr. S. Perumal is an eminent researcher and academician in the field of Computer Science with 32 years of experience. He has produced 12 Ph.D. scholars and 30 M.Phil. scholars, and is currently guiding 8 Ph.D. researchers. His research output includes approximately 75 published articles in reputed international journals, reflecting a sustained commitment to advancing knowledge in computer science and related disciplines. He holds 5 patents to his credit, underscoring his contributions to innovation and applied research. His academic journey is marked by mentoring excellence, prolific publication, and significant contributions to building a robust research culture within his institution.

**EXCELLENCE IN ARTIFICIAL INTELLIGENCE &
DATA SCIENCE RESEARCH**



Dr. K. Sharmila, Ph.D.,

Professor, School of Applied Computing and Emerging Technologies,
Vels Institute of Science, Technology and Advanced Studies (VISTAS),
Chennai, India.

Dr. K. Sharmila is an accomplished academician with 17 years of teaching experience at both UG and PG levels. She received her M.Sc. from Alagappa University, her M.Phil. from Bharathidasan University, and her Ph.D. in Computer Science from VISTAS University. Her research interests encompass Big Data Analytics, Cloud Computing, Machine Learning, and Blockchain Technology, areas in which she has made substantial contributions. She has guided 8 research scholars and continues to inspire the next generation of researchers and technology professionals. Her dedication to quality education, innovative research, and student mentorship has been a hallmark of her distinguished academic career.

**EXCELLENCE IN ARTIFICIAL INTELLIGENCE &
DATA SCIENCE RESEARCH**



Dr. R. Padma, Ph.D.

Assistant Professor

VELS Institute of Science, Technology & Advanced Studies

Chennai, India

Dr. R. Padma, Ph.D. is a dedicated academician and accomplished researcher in the field of Computer Science with over 15 years of rich teaching and research experience. Her areas of expertise include Artificial Intelligence, Machine Learning, Data Science, Internet of Things (IoT), and Cybersecurity. She has made significant contributions to academia through numerous research publications in reputed national and international journals, including Scopus-indexed journals. She is also the author of several academic books such as *Deep Learning*, *Web Technology*, and *Applied Cryptography*, reflecting her strong commitment to knowledge dissemination. Dr. Padma holds multiple patents in emerging technologies, particularly in IoT, machine learning, and blockchain security, showcasing her innovation-driven approach to research. She has actively participated in more than 200 Faculty Development Programs, workshops, and conferences, continuously upgrading her academic and technical expertise. She has received several prestigious awards and recognitions, including the NPTEL Domain Scholar Award, Best Educator Award, Faculty Excellence Award, and multiple mentoring recognitions from NPTEL-SWAYAM. She also serves as a mentor, editor, and academic contributor in various capacities.

**EXCELLENCE IN ARTIFICIAL INTELLIGENCE &
DATA SCIENCE RESEARCH**



Dr. R. Priscilla, Ph.D.

Professor & Head

St. Joseph's Institute of Technology

Chennai, India

Dr. R. Priscilla, Ph.D. is a distinguished academician and research leader in the field of Artificial Intelligence and Data Science, with over two decades of extensive teaching and academic experience. She currently serves as Professor and Head of the Department of Artificial Intelligence and Data Science, contributing significantly to academic excellence and institutional development. Her research expertise lies in Data Mining, Bioinformatics, Machine Learning, and AI-driven healthcare analytics, with a strong focus on disease prediction and classification using advanced computational techniques. She has published numerous research papers in reputed international journals and IEEE conferences, demonstrating her impactful contributions to the research community. Dr. Priscilla is a recognized Ph.D. supervisor under Anna University and has successfully guided multiple research scholars. She has also received research funding from organizations such as AICTE and TNSCST for innovative student projects. She actively contributes to academia through various leadership roles, editorial responsibilities, and professional memberships. With a strong commitment to innovation, research, and student mentoring, Dr. Priscilla continues to inspire future technologists and researchers in the domain of Artificial Intelligence.

**EXCELLENCE IN ARTIFICIAL INTELLIGENCE &
HEALTHCARE DATA SCIENCE RESEARCH**



Dr. S. Vidhushavarshini, Ph.D.,

Associate Professor, Department of Computer Science and Engineering,
Rajalakshmi Institute of Technology,
Chennai, India.

Dr. S. Vidhushavarshini is a results-driven academic professional with over 13 years of experience in teaching, research, and mentorship in Computer Science. She completed her Ph.D. in Information and Communication Engineering from Sona College of Technology, Anna University in 2022. Her research specializations include Bioinformatics, Machine Learning, Deep Learning, and Data Science, with a particular focus on healthcare analytics. She has published 18+ research papers in peer-reviewed international journals indexed in SCI, SCIE, and Scopus, including journals such as BMC Cardiovascular Disorders, Diagnostics, Expert Systems, Fractal and Fractional, and the Journal of Personalized Medicine. She has served as a reviewer for prestigious journals including Scientific Reports, BMC Cancer, and Journal of Medical Systems. She holds the Red Hat Certified System Administrator (RHCSA) certification and an IITD Outreach qualification. She has conducted over 15 Faculty Development Programs and technical workshops on AI and deep learning. She is also an ISTE Life Member and has served in roles including Infosys SPOC, ICT Academy Coordinator, MoU In-charge, and EDC Coordinator across her academic career at SRM Institute of Technology and Sona College of Technology.

**EXCELLENCE IN CYBER SECURITY & LEGAL
INNOVATION**



Mr. Yashwanth A S, M.Com.,

CEO & Founder Director,

Y17J Cyber Security and Management Consultant LLP,

Bangalore, India.

Mr. Yashwanth A S is a cybersecurity expert and entrepreneur with 5 years of impactful experience in cyber law, digital security, and legal advisory services. He serves as CEO and Founder Director of Y17J Cyber Security and Management Consultant LLP, and holds certifications including CCIO and CISA. He is a Training Partner with ISAC India and has an agreement with Vaikunta Baliga College of Law, Udupi. He works as a Visiting Faculty for Cyber Law and conducts cyber awareness programmes. He has published 20+ articles in national and international law journals and presented 25+ papers at conferences. He has received prestigious recognitions including the Swami Vivekananda Innovation and Start Up Excellence Award 2026, the Subhash Chandra Bose National Icon Award, the Shrest Bharat Puraskar, and the Men of the Icon Award 2026.

TECHNICAL ADVISORY EXCELLENCE AWARD



Dr. S. Umarani, MCA., Ph.D.,

Professor, Department of Computer Applications

SRM Institute of Science and Technology, Ramapuram

Chennai, Tamil Nadu, India

Dr. S. Umarani is a distinguished academician serving as Professor , Department of Computer Applications at SRM Institute of Science and Technology. She holds a Ph.D. in Computer Science from Anna University and has extensive teaching and research experience in areas such as Big Data, Internet of Things (IoT), and Data Mining. Her research contributions include work in computer networks, wireless sensor networks, and topology optimization, with 42 publications,16 patents in reputed journals and conferences. She has received recognition such as the Best Paper Award and has actively guided students in innovative projects, including participation in events like the Microsoft Windows AppFest 2012. Along with her academic excellence, she is also known for promoting value-based education, mentoring students in life skills, and encouraging holistic development through various knowledge-sharing sessions.



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**EXCELLENCE IN COMMERCE & FINANCIAL
RESEARCH**



Dr. K. Amutha, Ph.D.,

Associate Professor, Department of Commerce,
PSG College of Arts & Science,
Coimbatore, Tamil Nadu, India.

Dr. K. Amutha is an accomplished academician in the field of Commerce with 22 years of teaching and research experience at PSG College of Arts & Science, Coimbatore. Her academic contributions reflect a deep commitment to excellence in commerce education, research, and mentorship. She has contributed significantly to the scholarly advancement of commerce studies through her publications, research guidance, and active participation in academic activities. Her sustained dedication to quality education and her role in shaping future commerce professionals stand as a testament to her academic leadership and institutional commitment.

**EXCELLENCE IN COMMERCE EDUCATION &
ACADEMIC CONTRIBUTION AWARD**



Dr. Pushpalatha H, Ph.D.,

Assistant Professor, Department of Commerce,
Anna Adarsh College for Women,
Chennai, India.

Dr. Pushpalatha H is a dedicated Commerce educator with 15 years of teaching experience across reputed institutions in Chennai. She completed her Ph.D. in Banking from Vels University, her M.Phil. from Priest University, and her M.Com. from Annamalai University. Her academic specialization spans Banking, Insurance, Financial Accounting, Cost Accounting, Income Tax, Management Accounting, and Business Law. She has served in key institutional roles including NIRF Chief Coordinator, Examination Committee Coordinator, FDP Coordinator, Placement Coordinator, and Club Coordinator. She has published 10 research papers in national and international journals and conferences on topics including banking rebranding, digitalization in banking, green banking initiatives, and the impact of demonetization. She received the Best Teacher Award (2009) and the Kaviarasu Award for Tamil Poetry (2017). She has served as an External Examiner and Resource Person, presenting the seminar topic 'Fundamentals of Investment' at Annai Violet College. She has worked at Bhaktavatsalam College for Women, Guru Shree Shantivijai Jain College for Women, and currently at Anna Adarsh College for Women, demonstrating a sustained commitment to women's education and empowerment.

**EXCELLENCE IN COMMERCE & MANAGEMENT
EDUCATION**



Dr. T. Deepa Augustine, Ph.D.

Assistant Professor & Head
Annai Violet Arts & Science College
Chennai, India

Dr. T. Deepa Augustine, Ph.D. is a dedicated academician and accomplished professional in the field of Commerce and Management, with extensive experience in teaching, research, and academic administration. She currently serves as Assistant Professor and Head, contributing significantly to student development and institutional growth. Her areas of expertise include Marketing, Human Resource Management, Business Analytics, and Organizational Behavior. She has published research papers in reputed national and international journals, including Scopus-indexed publications, and has authored academic books and book chapters focusing on modern business practices and emerging technologies such as Artificial Intelligence in education. Dr. Deepa is NET and SET qualified and holds a Ph.D. in Management. She has received the IRSD International Academic Excellence Award in recognition of her contributions to academia and research. She also holds additional responsibilities such as NAAC coordination, placement coordination, academic auditing, and institutional development initiatives. With a strong passion for teaching and research, she is committed to empowering students with practical knowledge, fostering critical thinking, and promoting innovation in the field of commerce and management.

**EXCELLENCE IN MANAGEMENT EDUCATION &
RESEARCH**



Dr. Baby Niviya Feston, Ph.D.,

Dean – Research & Innovation,

Garden City University,

Bangalore, India.

Dr. Niviya Feston is a distinguished academic leader, researcher, and institutional strategist with over 22 years of experience in higher education, research governance, and academic administration. She currently leads the doctoral research ecosystem across multidisciplinary domains at Garden City University. She holds a Ph.D. and multiple post-graduate qualifications including M.Sc. (Mathematics), M.Com. (Banking & Insurance), MBA (Finance & Marketing), and a Master's in Data Science from Purdue University, USA. She is UGC NET qualified in Management. She has published more than 30 peer-reviewed research papers in Scopus and ABDC indexed journals, holds 2 national patents granted by the Government of India in Artificial Intelligence and e-Financial Governance, and has guided 7 Ph.D. scholars. She has coordinated multiple KSCST-funded research projects and has played a pivotal role in institutional rankings including NAAC, NIRF, NBA, and QS. She is co-author of 'Marketing Research' and co-editor of 'Recent Advances in Commerce, Management, and Tourism.'

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**EMERGING LEADER IN AI & DATA-DRIVEN
ENTERPRISE INNOVATION AWARD**



V. Venkatasubramanian

CEO & Managing Director

Devcansol Pvt Ltd

Chennai, India

V. Venkatasubramanian is a dynamic entrepreneur and technology leader, currently serving as the CEO and Managing Director of Devcansol Private Limited. With a strong foundation in finance, data analytics, and software development, he specializes in building AI-driven enterprise solutions that enable business transformation and sustainable growth. He has extensive industry experience working with reputed organizations such as Dun & Bradstreet and Verumex, where he contributed to data integration, financial analysis, and business intelligence. His technical expertise includes Artificial Intelligence, Generative AI, data modeling, Python, SQL, and modern web technologies. As a visionary leader, he drives innovation through the development of scalable AI systems, including Gen-AI and Agentic AI applications, enhancing operational efficiency and strategic decision-making. In addition to his corporate role, he actively engages in knowledge sharing by delivering guest lectures, training programs, and workshops across academic and professional platforms. With a passion for innovation and a results-driven approach, Venkatasubramanian continues to bridge the gap between technology and business, empowering organizations through data-driven solutions and cutting-edge advancements in AI engineering.

EXCELLENCE IN BRAND STRATEGY & INTEGRATED MARKETING COMMUNICATIONS



Mr. Sashikumar VS

Managing Director

Praanha Marcomm Solutions

Anna Nagar West Extension, Chennai-600050, Tamil Nadu, India.

Praanha Marcomm Solutions is a dynamic and innovative branding and marketing agency dedicated to creating impactful brand identities and driving business growth. With a strong focus on creativity, strategy, and technology, Praanha delivers comprehensive solutions tailored to modern business needs.

The organization specializes in corporate branding, digital marketing, web design, social media management, and advertising strategies. By combining creative storytelling with data-driven insights, Praanha helps businesses enhance their market presence, engage target audiences, and achieve measurable results.

Known for its agile, bold, and creative approach, Praanha adapts quickly to changing market trends and delivers customized solutions that align with client goals. Their expertise spans across industries, enabling businesses to stand out in competitive environments and build strong, lasting brand value.

With a commitment to innovation and excellence, Praanha continues to empower brands by transforming ideas into impactful visual and digital experiences, ultimately contributing to sustainable business growth and success.



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DISTINGUISHED ACADEMIC LEADERSHIP AWARD



Dr. C. Sundar, Ph.D.,

Principal,

Dhanraj Baid Jain College (Autonomous), Thoraiakkam,

Chennai, India.

Dr. C. Sundar is a distinguished academic leader and management educator with over 26 years of experience across prestigious institutions. He currently serves as Principal of Dhanraj Baid Jain College (Autonomous), Chennai. Prior to this, he served as Dean of Management Studies and Humanities at SRM Institute of Science and Technology, Ramapuram, for six years, and as Vice Principal (Academic) and Professor & Head at the same institution. He holds a Ph.D. from Sathyabama University, an M.Phil. in Entrepreneurship from Madurai Kamaraj University, and an MBA in Marketing and HR. He has received an AICTE research grant of Rs. 3,50,000 for an ATAL FDP on AI-enabled Data Analytics for Business Decision Making. He received the Lifetime Achievement Award in Education from Lions International (2024), Award of Excellence from Tamil Nadu Urdu Academy & TNOU (2024), the Startup & HR Excellence Award, and the Award of Excellence from Education & Research Organisation (2024). He has introduced 7 new academic courses at SRM IST, organized conferences and Faculty Development Programmes, and has served as a resource person for IGNOU, Anna University, University of Madras, Alagappa University, and Madurai Kamaraj University. He is a member of the All India Management Association.

**LIFETIME ACHIEVEMENT IN ACADEMIC
LEADERSHIP & RESEARCH EXCELLENCE AWARD**



Dr. V. Jayalakshmi, MCA, M.Phil., SET, NET, Ph.D.,

Director – Digital Infrastructure; Professor, Department of Computer Applications,
Vels Institute of Science, Technology and Advanced Studies (VISTAS),
Chennai, India.

Dr. V. Jayalakshmi is a highly accomplished academician, researcher, and institutional leader with 28 years of teaching and 9 years of research experience. She serves as Director of Digital Infrastructure and Professor in the Department of Computer Applications at VISTAS, Chennai. Her areas of specialization include Wireless Networks, Data Mining, and Image Processing. She has published 60 international and 2 national journal papers, and has presented at 24 international and 14 national conferences. She has authored 4 books and holds 10 published patents, reflecting her significant contribution to innovation and intellectual property. She has guided 69 MCA students, 22 MSc students, 4 M.Phil. scholars, and 8 Ph.D. scholars, underscoring her dedication to research mentorship. She has received 7 awards for her academic and professional contributions. She has participated in international conferences across India, UAE, Switzerland, China, and has been invited as a Technical Program Committee member for IEEE-supported conferences. Her leadership in digital infrastructure development and her sustained contributions to computing research and education make her a distinguished figure in the academic community.

**EXCELLENCE IN ACADEMIC LEADERSHIP &
COMPUTER SCIENCE EDUCATION AWARD**



Dr. A. Antony Prakash, Ph.D.,

Head & Assistant Professor, Department of Information Technology,
St. Joseph's College,
Trichy, India.

Dr. A. Antony Prakash serves as Head and Assistant Professor in the Department of Information Technology at St. Joseph's College, Trichy, with 15 years of academic experience. He has authored a book and published 31 research articles in national and international journals, reflecting his active engagement in scholarly research. He holds 3 patents and has established 2 MOUs, strengthening industry-academia collaboration. His contributions span research, academic leadership, and institutional development, making him a well-rounded contributor to higher education in the field of information technology.

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**DISTINGUISHED ACADEMIC LEADERSHIP &
RESEARCH EXCELLENCE IN AI & EMERGING
TECHNOLOGIES AWARD**



Prof. Dr. J. Indumathi, PDF, Ph.D., D.Sc.,

Dean, Artificial Intelligence,

Sree Balaji Institute of Science and Technology, Chrompet,

Chennai, India.

Prof. Dr. J. Indumathi is a distinguished academic and higher education leader with over 35 years of innovation in engineering education, research, digital governance, and institutional transformation. She retired as Professor of Information Science and Technology from Anna University, Chennai, and continues her engagement as a Professor of Practice in Digital Governance and Trustworthy Systems. Her scholarly work advances interdisciplinary frameworks in trustworthy digital systems, integrating data privacy, cybersecurity, cyber forensics, explainable artificial intelligence, and resilient cyber-physical infrastructure. As Secretary for Tamil Nadu Engineering Admissions, she led the state's first fully online engineering application system. As Director of the Ramanujan Computing Centre at Anna University, she provided strategic oversight of university-wide digital infrastructure. She has authored over 200+ journal publications, contributed to more than 6 books, 35+ book chapters, edited six volumes, and holds multiple granted patents.

**EXCELLENCE IN RESEARCH & ACADEMIC
LEADERSHIP**



Dr. Aneeshkumar A.S., Ph.D.,

Associate Professor & Head, Departments of Computer Science, Computer Applications
and Artificial Intelligence,
AJK College of Arts and Science,
Coimbatore, India.

Dr. Aneeshkumar A.S. is a dynamic academician, researcher, and institution builder with over 18 years of teaching and 16 years of research experience. He completed his Ph.D. in Computer Science (Full-time) from the University of Madras in 2016, with a dissertation on predictive mining algorithms for liver disorder identification, developing three innovative algorithms recognized for their clinical significance. He serves as Associate Professor and Head of three departments at AJK College of Arts and Science, Coimbatore, while also functioning as IIC Coordinator, Research Centre Coordinator, IQAC Member, and Planning Board Member. He has established 32 MoUs with multinational companies including IBM, for training, placement, and internship opportunities. Through his leadership as IIC Coordinator, he elevated the institution's IIC ranking from one to four stars, the highest among Arts and Science Colleges. He has produced 2 Ph.D. scholars and is currently guiding 4. He has organized 21 national and international conferences, holds 3 patents in big data patterns, and serves as Editor-in-Chief for three impact factor journals and reviewer for over 25 international journals. He has presented research at numerous national and international conferences across India and abroad.

BEST FACULTY & ACADEMIC LEADERSHIP AWARD



Dr. S. Usharani, Ph.D.,
Professor and Head, Department of MCA,
Viswam Engineering College,
Andhra Pradesh, India.

Dr. S. Usharani is an accomplished academic leader with over 18 years of extensive teaching and administrative experience in higher education. She completed her Ph.D. in Computer Science from Mother Teresa Women's University. Her research interests span Artificial Intelligence, Data Mining, Cloud Computing, Machine Learning, Big Data Analytics, IoT, and Software Engineering. She has published numerous research papers in reputed national and international journals and holds an Indian patent focused on Machine Learning-based pre-emptive analytics for IoT security. Dr. Usharani is an NPTEL Domain Scholar and Brand Ambassador, SWAYAM SPOC, reviewer for international conferences, and coordinator of FDPs, workshops, and institutional initiatives. She is deeply committed to fostering a research culture, promoting innovation and digital learning, and contributing to holistic student development.

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**BEST FACULTY & EXCELLENCE IN ACADEMIC
LEADERSHIP AWARD**



Dr. R. Parameswari, Ph.D.,

Professor & Head, Department of Computer Science and Information Technology,
Vels Institute of Science, Technology and Advanced Studies (VISTAS),
Chennai, India.

Dr. R. Parameswari is a distinguished academic with over 20 years of experience in computer science education and research. She earned her Doctorate from St. Peter's University, Chennai in 2016. Her research expertise spans Cloud Computing, Big Data Analytics, and the Internet of Things (IoT). She has published 75 research articles in reputed international journals indexed in Scopus, Web of Science, and UGC-approved platforms. She has presented over 50 research papers at national and international conferences. As a registered Ph.D. Supervisor at VISTAS, she has successfully guided 5 M.Phil. scholars and 10 Ph.D. scholars, with 9 Ph.D. research scholars currently under her supervision. She serves as an Editorial Board Member and Reviewer for several national and international Scopus-indexed journals. She is a Life Member of ISTE, CSI, and NFED, and has received multiple prestigious awards for teaching excellence and professional achievement.

EXCELLENCE IN INDUSTRY-ORIENTED EDUCATION



Dr. G. Fathima, Ph.D.,

Professor and Head, Department of Computer Science and Engineering,

Adhiyamaan College of Engineering, MGR Nagar, Hosur,

Hosur, Tamil Nadu, India.

Dr. G. Fathima is a distinguished academician and innovator with over 27 years of teaching experience and more than 18 years of administrative leadership. She completed her B.E. at Thiagarajar College of Engineering, Madurai, her M.E. at NIT Trichy, and her Ph.D. from Anna University, Chennai in 2012. Her extensive academic and innovation portfolio spans patents, journal publications, conference presentations, and authored books across diverse fields such as artificial intelligence, machine learning, IoT, cybersecurity, healthcare technologies, environmental monitoring, and assistive systems. She has published 38+ journal papers in reputed international journals, many indexed in Scopus, SCIE, and ESCI. She has presented at 32+ national and international conferences, holds 4 published patents with 1 granted, and has co-authored 2 books and 15 book chapters. She has received research grants from UGC, AICTE, IE(I), and TNSCST.



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SCIENCE EDUCATION AWARD**



Sri Balaji Institute of Science and Technology (SBIST),

No.7 Works Road, Chrompet,

Bharath Institute of Higher Education & Research,

Chennai, Tamil Nadu, India.

Sri Balaji Institute of Science and Technology (SBIST) is a progressive institution committed to transforming engineering and technology education through industry-linked learning, emerging technology specialization, research-driven innovation, and global competency development. Operating within the academic ecosystem of Bharath Institute of Higher Education and Research, SBIST is led academically by Dean Dr. J. Indumathi — a Post-Doctoral Fellow in Explainable AI with 35 years of experience and over 100 national and international awards. The institution has achieved 7 granted patents and over 200 scientific publications. It embeds industry exposure throughout the academic lifecycle through 100% mandatory student internships, 80+ active MoUs, and NVIDIA, Microsoft, and IEEE collaborations. SBIST's unique ecosystem includes on-campus integration with Sree Balaji Medical College & Hospital under Chairperson Dr. J. Sri Nisha, enabling live interdisciplinary MedTech innovation. Patron Er. N. Eiamaran leads a defense-grade precision gear manufacturing enterprise, providing students unparalleled real-world engineering exposure. Student achievements in 2026 include Winners at the Youth Health Mela 2026 and Runners-Up at the VIT Hackathon 2026, exemplifying the institution's commitment to empowering tomorrow's innovators through industry-linked learning.



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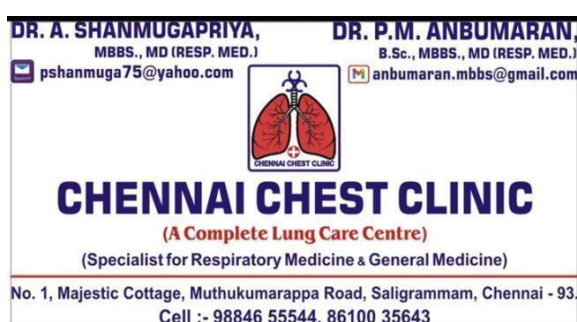
Mr. S. Aravind

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Aravind S is a passionate and dynamic technology professional specializing in Python programming, Data Science, Artificial Intelligence, and Machine Learning. With a strong foundation in Computer Science, he holds a Master's degree and is committed to continuous learning and professional growth. He currently serves as the Head of Learning & Development at Edinz Tech Private Limited, where he has delivered over 200+ technical training sessions and mentored more than 5000 students through internships and skill development programs. His expertise lies in bridging the gap between academic knowledge and industry requirements. Aravind has conducted workshops and training programs across reputed institutions including SRM Institute of Science and Technology, VELS Institute, and other leading colleges. His core competencies include Data Analytics, Power BI, Web Development, Prompt Engineering, and emerging AI technologies such as Generative AI and Agentic AI. With a strong passion for teaching and technology, Aravind S continues to empower students and professionals by equipping them with practical skills and industry-relevant knowledge.

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